

Service Manual

Air Conditioner



Indoor Unit
CS-L72JD1R5

Outdoor Unit
CU-L72JD1R8



WARNING

This service information is designed for experienced repair technicians only and is not designed for use by the general public. It does not contain warnings or cautions to advise non-technical individuals of potential dangers in attempting to service a product. Products powered by electricity should be serviced or repaired only by experienced professional technicians. Any attempt to service or repair the product or products dealt with in this service information by anyone else could result in serious injury or death.

PRECAUTION OF LOW TEMPERATURE

In order to avoid frostbite, be assured of no refrigerant leakage during the installation or repairing of refrigeration circuit.

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1.Safety Precautions

To prevent injury to the user or other people and property damage, the following instructions must be followed.

- Incorrect operation due to ignoring instruction will cause harm or damage. The seriousness is classified by the following indications.

 WARNING	This symbol indicates the possibility of death or serious injury.
 CAUTION	This symbol indicates the possibility of injury or damage to properties only.

- Meanings of symbols used in this manual are as shown below.

	Be sure not to do.
	Be sure to follow the instruction.
	Dangerous Voltage

1.1 Cautions in Repair

 WARNING	
Do not modify the machine,part,material during repairing service. Must not use other parts except original parts describe in catalog and manual.	
Be sure to disconnect the power cable plug from the plug socket before disassembling the equipment for a repair.Internal components and circuit boards are at main potential when the equipment is connected to the power cables. This voltage is extremely dangerous and may cause death or severe injury if come in contact with it.	
Do not touch the discharging refrigerant gas during the repair work. The discharging refrigerant gas.The refrigerant gas can cause frostbite.	
Release the refrigerant gas completely at a well-ventilated place first. Otherwise, when the pipe is disconnected, refrigerant gas or refrigerating machine oil discharges and it Can cause injury.	
When the refrigerant gas leaks during work, execute ventilation. If the refrigerant gas touches to a fire, poisonous gas generates. A case of leakage of the refrigerant and the closed room full with gas is dangerous because a shortage of oxygen occurs. Be sure to execute ventilation.	
When removing the front panel or cabinet, execute short-circuit and discharge between high voltage capacitor terminals. If discharge is not executed, an electric shock is caused by high voltage resulted in a death or injury.	
Do not turn the air-conditioner ON or OFF by plugging or unplugging the power plug. There is risk of fire or electrical shock.	

Do not use a defective or underrated circuit breaker. Use the correctly rated breaker and fuse. Otherwise there is a risk of fire or electric shock.	
Install the panel and the cover of control box securely. Otherwise there is risk of fire or electric shock due to dust, water etc.	
Indoor/outdoor wiring connections must be secured tightly and the cable should be routed properly so that there is no force pulling the cable from the connection terminals. Improper or loose connections can cause heat generation or fire.	
Do not touch, operate, or repair the product with wet hands. Holding the plug by hand when taking out. Otherwise there is risk of electric shock or fire.	

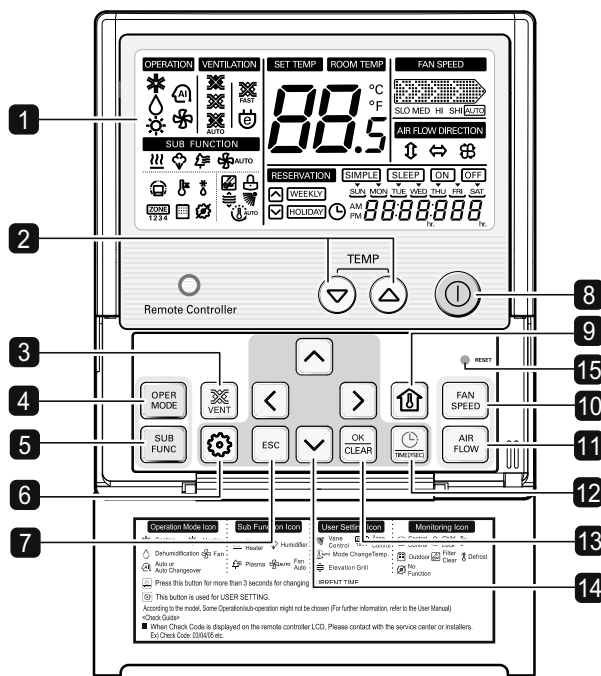
 CAUTION	
Do not turn on the breaker under condition that front panel and cabinet are removed.	
Be sure to earth the air conditioner with an earthing conductor connected to the earthing terminal.	
Conduct repair works after checking that the refrigerating cycle section has cooled down sufficiently. Otherwise, working on the unit, the hot refrigerating cycle section can cause burns.	
Do not tilt the unit when removing panels. Otherwise, the water inside the unit can spill and wet floor.	
Do not use the welder in a well-ventilated place. Using the welder in an enclosed room can cause oxygen deficiency.	
Be sure to turn off power switch before connect or disconnect connector, or parts damage may be occurred.	

1.2 Inspections after Repair

⚠ WARNING	
Check to see if the power cable plug is not dirty or loose. If the plug is dust or loose it can cause an electrical shock or fire.	
Do not use a joined power cable or extension cable, or share the same power outlet with other electrical appliances. otherwise, it can cause an electrical shock, excessive heat generation or fire.	
Do not insert hands or other objects through the air inlet or outlet while the product is operating. There are sharp and moving parts that could cause personal injury.	
Do not block the inlet or outlet of air flow. It may cause product failure	

⚠ CAUTION	
Check to see if the parts are mounted correctly and wires are connected. Improper installation and connections can cause an electric shock or an injury.	
Check the installation platform or frame has corroded. Corroded installation platform or frame can cause the unit to fall, resulting in injury.	
Be sure to check the earth wire is correctly connected.	
After the work has finished, be sure to do an insulation test to check the resistance is 2[Mohm] or more between the charge section and the non-charge metal section (Earth position). If the resistance value is low, a disaster such as a leak or electric shock is caused at user's side.	
Check the drainage of the indoor unit after the repair. If drainage is faulty the water to enter the room and wet floor.	

2. Function of Remote Control



- 1 OPERATION INDICATION SCREEN
- 2 SET TEMPERATURE Button
- 3 VENTILATION Button (Not working)
- 4 OPERATION MODE SELECTION Button
- 5 SUBFUNCTION Button (Not working)
- 6 FUNCTION SETTING Button
- 7 EXIT Button
- 8 ON/ OFF Button
- 9 ROOM TEMPERATURE Button
- 10 FAN SPEED Button
- 11 AIR FLOW Button (Not working)
- 12 RESERVATION/ TIME SETTING Button
- 13 SETTING/ CANCEL Button
- 14 UP, DOWN, LEFT, RIGHT Button
- 15 RESET Button

Display temperature can be different from actual room temperature if the remote controller is installed at the place where sun-rays are falling directly or the place nearby heat source.

3. Specifications

3.1 Indoor Unit

Indoor unit type			Ceiling Concealed Duct
Model			CS-L72JD1R5
Power Supply	Ø / V / Hz		1 / 220~240 / 50
Cooling Capacity	kW		12.6~21.0~23.5
	Btu/h		42 990~71 650~80 180
Heating Capacity	kW		14.1~23.5~26.3
	Btu/h		48 110~80 180~89 740
Current	Nominal Running Current	A	2.8
Fan	Motor Type		BLDC
	Fan Type		Sirocco
	Motor Output(W) x No. of Unit		375 x 2
	Air Flow Rate (H / M / L)	CMM	70 / 65 / 60
		l/s	1 167 / 1 083 / 1 000
	External Static Pressure	Pa	180
	Capacitor	µF / Vac	-
	Drive		Direct Drive
Coil	Row x Column x FPI		4R x 16C x 16FPI
Dimensions (W x H x D)	Body	mm(inch)	1562(+115) x 460 x 688 (61.5(+4.5) x 18.1 x 27.1)
	Decorative Panel	mm(inch)	-
Net Weight	Body	kg (lbs)	90(198.4)
	Decorative Panel	kg (lbs)	-
Air Filter			-
Sound Level (H / M / L)		dB(A)+3	52 / 50 / 49
Piping	Liquid	mm (inch)	9.52(3/8)
Connections	Gas	mm (inch)	22.2(7/8)
	Drain(OD/ID)	mm	25.4 / 19.4
Dehumidification Rate		l/h	5.9
Safety Devices			FUSE, Thermal Protector
Temperature Sensor			Thermistor
Refrigerant			R410A
Refrigerant Control			EEV(Outdoor Unit)
Connectable Outdoor Unit			CU-L72JD1R8
Power and Transmission Interunit Cable	No. x mm ²		4 x 1.0 (Including Earth)

Note :

- Capacities are based on the following conditions:
 Cooling: - Indoor Temperature 27°C(80.6°F) DB/19°C(66.2°F) WB
 - Outdoor Temperature 35°C(95°F) DB/24°C(75.2°F) WB
 Heating: - Indoor Temperature 20°C(68°F) DB/15°C(59°F) WB
 - Outdoor Temperature 7°C(44.6°F) DB/6°C(42.8°F) WB
 Piping Length: - Interconnecting Piping Length 7.5m
 - Level Difference of Zero
- Wiring cable size must comply with the applicable local and national code.
- The specification may be subject to change without notice for purpose of improvement.
- Add 115mm for power supply box.

3.2 Outdoor Unit

Outdoor unit			CU-L72JD1R8
Nominal Capacity	Cooling	kW	12.6~ 21.0~23.5
		Btu/h	42 990~71 650~80 180
	Heating	kW	14.1~23.5~26.3
		Btu/h	48 110~80 180~89 740
Nominal Input	Cooling	kW	3.50~6.56~7.60
	Heating	kW	4.00~7.50~9.20
Testing Combination			-
Running Current (Rating)	Cooling	A	11.0
	Heating	A	12.3
Starting Current	Cooling / Heating	A	12.3
Power Supply		Ø / V / Hz	3 / 380~415 / 50
Power Supply Cable(Outdoor)		No. x mm ²	5 x 5.5 (Including Earth)
Power and Transmission Cable(ODU to IDU or ODU to BD)		No. x mm ²	4 x 1.0 (Including Earth)
Dimensions	W x H x D	mm(inch)	1 280 x 1 509(+96) x 690 (50.4 x 59.4(+3.8) x 27.2)
Net Weight		kg(lbs)	250(550)
Max. Number of Connectable Indoor Units			-
Compressor (#1)	Type		Scroll
	Qty x Model		1 x ANB52FKCMT
	Capacity	kW	17.2 (at 60Hz)
	Motor type		BLDC
	Oil charge volume	cc	2300
	Oil Type		FV50S
Compressor (#2)	Type		-
	Qty x Model		-
	Motor type		-
	Oil charge volume	cc	-
	Oil type		-
Compressor (#3)	Type		-
	Qty x Model		-
	Motor type		-
	Oil charge volume	cc	-
	Oil type		-
Refrigerant	Cycle A Charge (at 7.5m)	g(oz)	8 500(300)
	Cycle B Charge (at 7.5m)	g(oz)	-
	Type		R410A
	Control		EEV
Additional Refrigerant Charge		g/m(oz/ft)	50
Heat Exchanger	(Rows x Column x FPI) x No.		((2R x 17C x 17) x 2)+((2R x 36C x 17) x 2)
	Defrosting Method		Reversing Cycle
Fan	Capacitor	µF/Vac	-
	Drive		Direct Drive
	Discharge Direction(Side/Top)		Top
	Air Flow Rate x No. of Fan	CMM(l/s)	180(3 000)
Sound Level(Cooling/Heating)	Sound Pressure	dB(A)+3	59/59
Piping Connections	Liquid(Cycle A)	mm(inch)	9.52(3/8)
	Gas(Cycle A)	mm(inch)	22.2(7/8)
	Liquid(Cycle B)	mm(inch)	-
	Gas(Cycle B)	mm(inch)	-
	High Pressure Gas	mm(inch)	-
Max. Interunit Piping Length	Total Piping	m	50
	Standard Piping	m	7.5
	Main Piping	m	-
	Total Branch Piping	m	-
	Each Branch Piping	m	-
Max. Elevation Difference	Outdoor Unit~Indoor Unit	m	30
	Indoor Unit~Indoor Unit	m	-
Operation Range (Outdoor)	Cooling	°C(°F)	-10 ~ 46
	Heating	°C(°F)	-10 ~ 24

Note :

1. Capacities are based on the following conditions:

- Cooling: - Indoor Temperature 27°C(80.6°F) DB/19°C(66.2°F) WB
 - Outdoor Temperature 35°C(95°F) DB/24°C(75.2°F) WB
 Heating: - Indoor Temperature 20°C(68°F) DB/15°C(59°F) WB
 - Outdoor Temperature 7°C(44.6°F) DB/6°C(42.8°F) WB

- Piping Length: - Interconnecting Piping Length 7.5m
 - Level Difference of Zero

2. Wiring cable size must comply with the applicable local and national code.

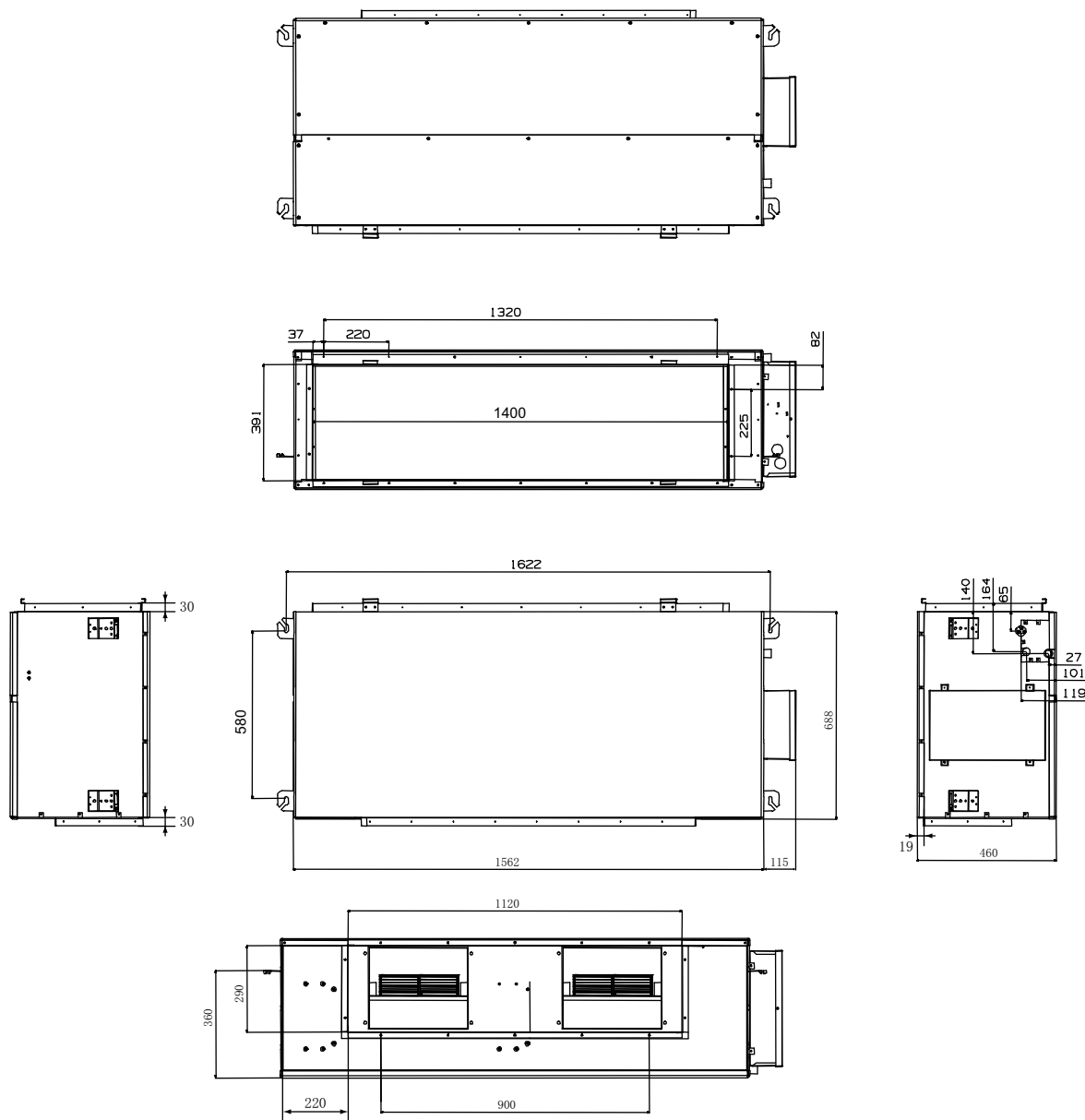
3. The specification may be subject to change without notice for purpose of improvement.

4. Add 96mm for discharge grille.

4. Dimensional Drawings

4.1 Indoor Unit

Model No.: CS-L72JD1R5

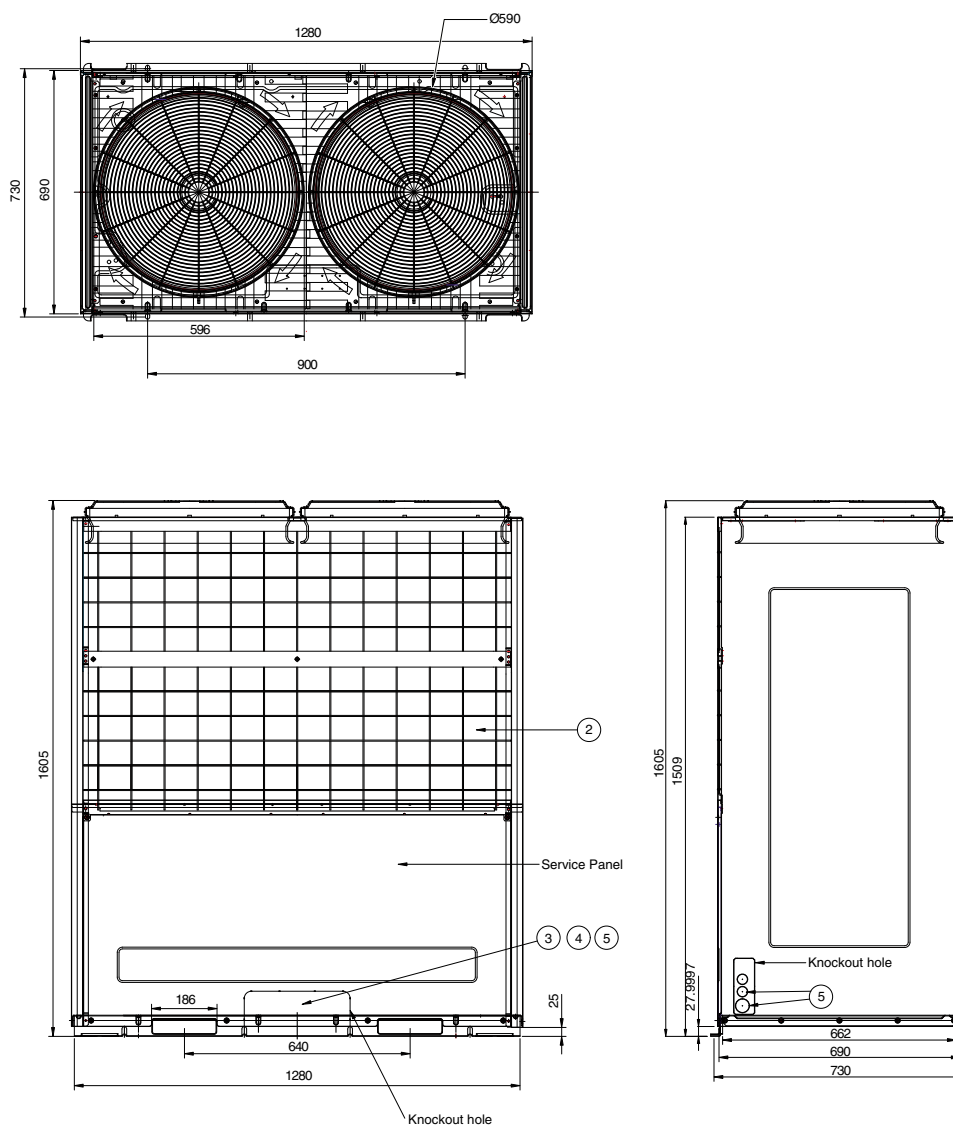


(Unit : mm)

Model	Liquid side service valve	Gas side service valve
CS-L72JD1R5	Ø9.52	Ø22.20

4.2 Outdoor Unit

Model No.: CU-L72JD1R8

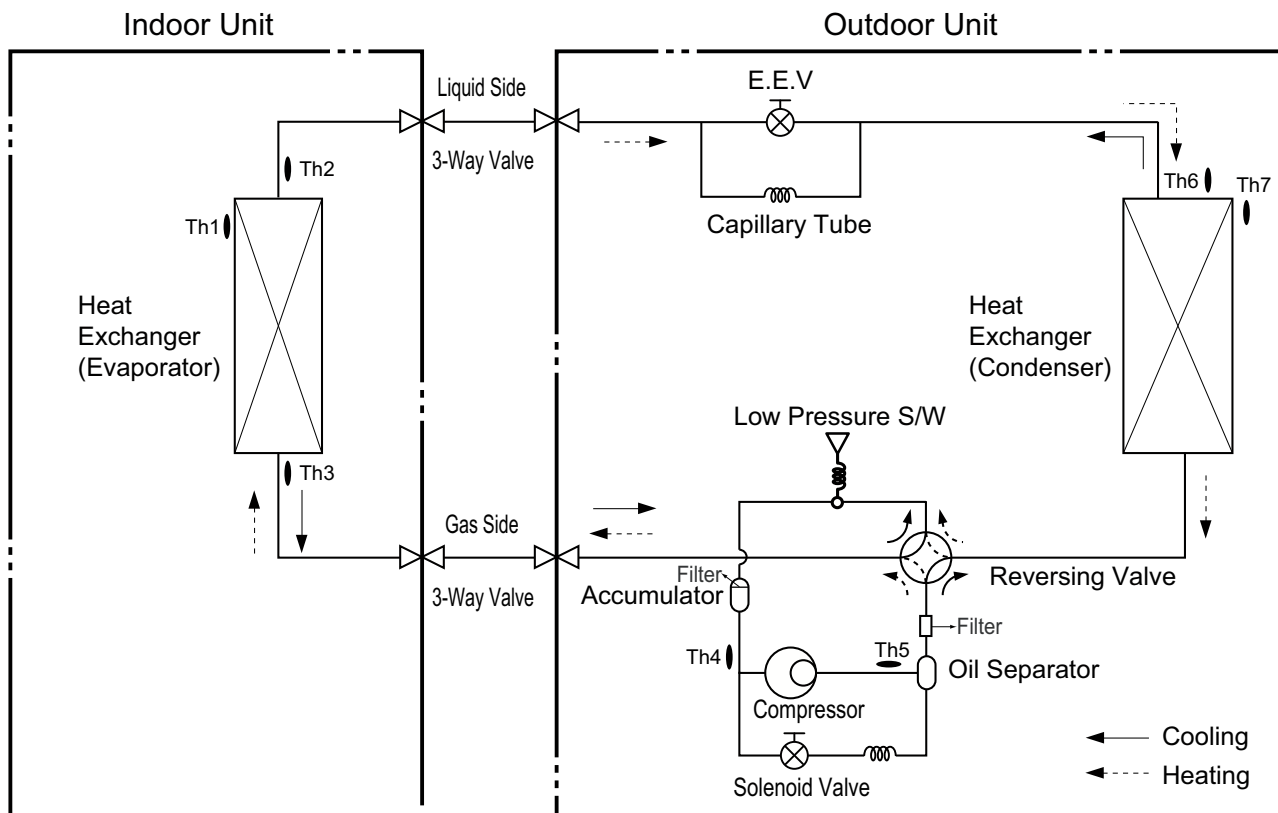


(Unit : mm)

Number	Name	Description
1	Air discharge grill	
2	Air suction grill	
3	Gas side service valve(mm)	Ø22.20
4	Liquid side service valve(mm)	Ø9.52
5	Drain pipe connection	

5. Refrigerant Cycle Diagrams

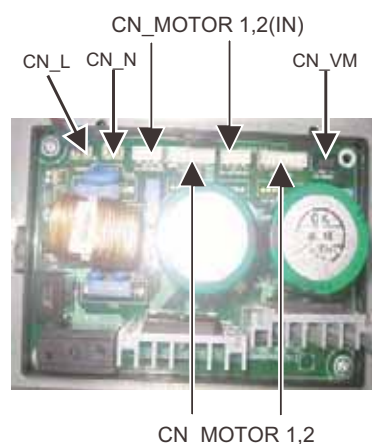
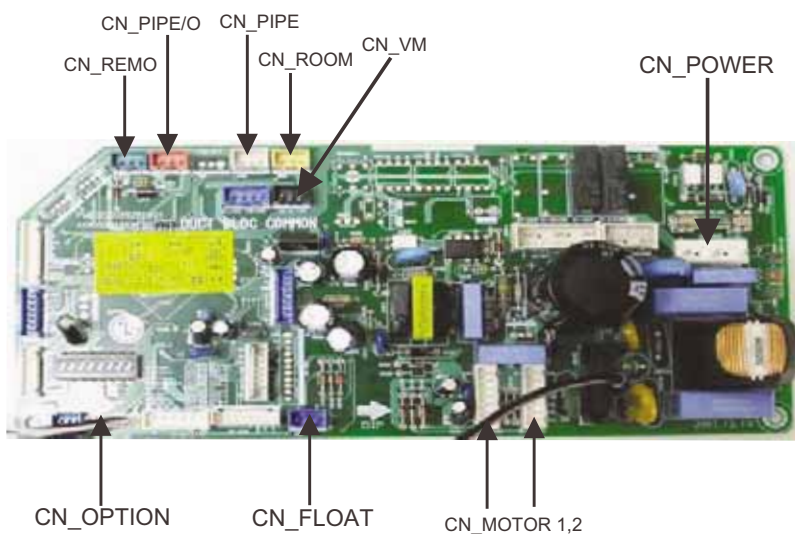
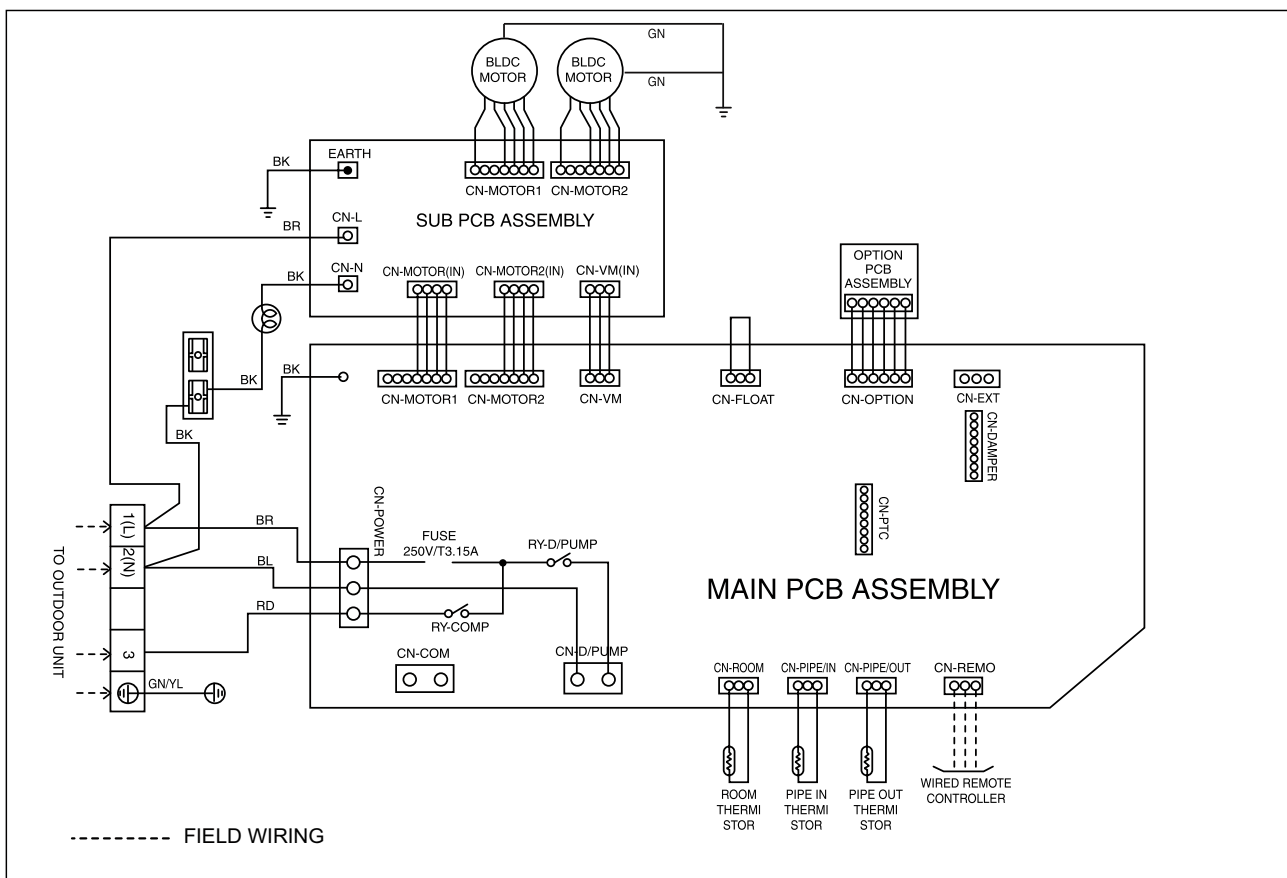
Model No.: CS-L72JD1R5 , CU-L72JD1R8



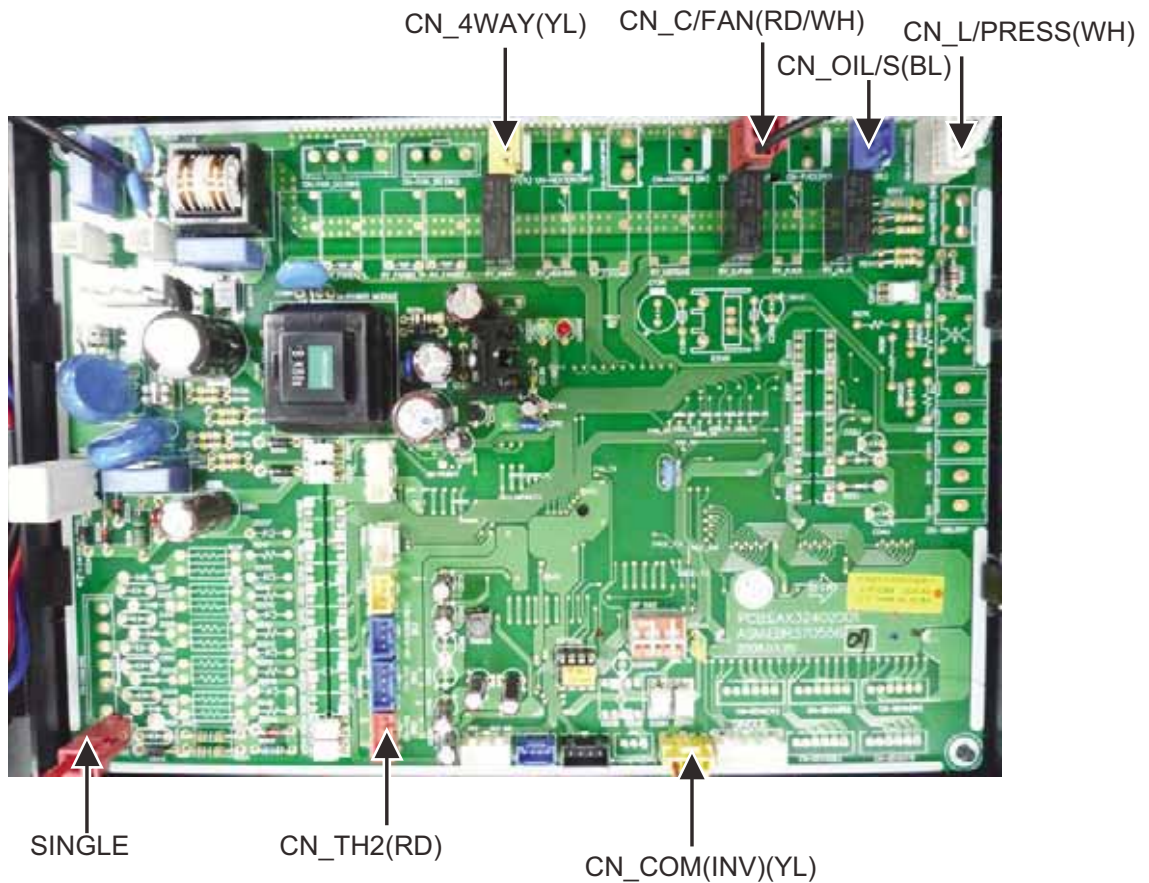
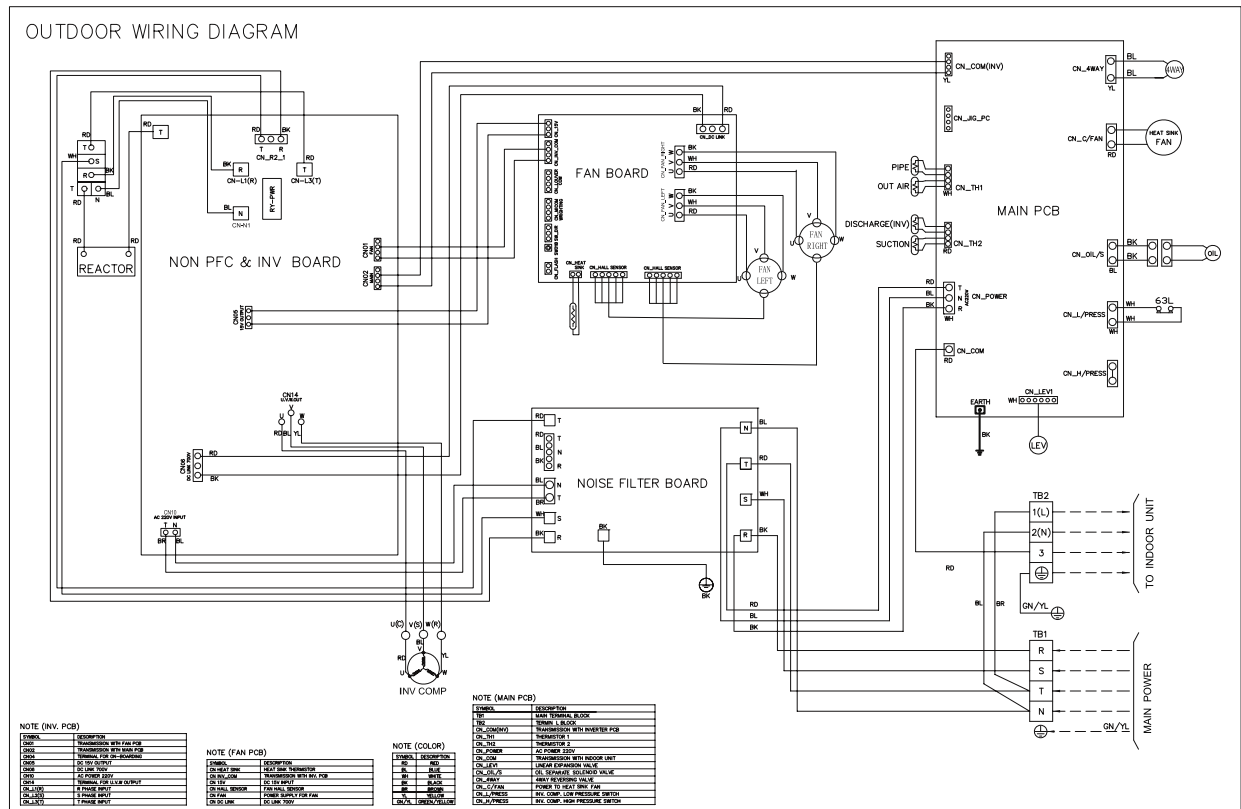
LOC.	Description	PCB Connector
Th1	Thermistor for Inlet Air	CN_ROOM(INDOOR)
Th2	Thermistor for Inlet Evaporator Temp	CN_PIPE(INDOOR)
Th3	Thermistor for Outlet Evaporator Temp	CN-PIPE/O(INDOOR)
Th4	Thermistor for suction pipe temperature	CN_TH2(OUTDOOR)
Th5	Thermistor for discharge pipe temperature	
Th6	Thermistor for condenser pipe temperature	CN_TH1(OUTDOOR)
Th7	Thermistor for outdoor air temperature	
EEV	Electronic Expansion Valve	CN_EEV1(OUTDOOR)

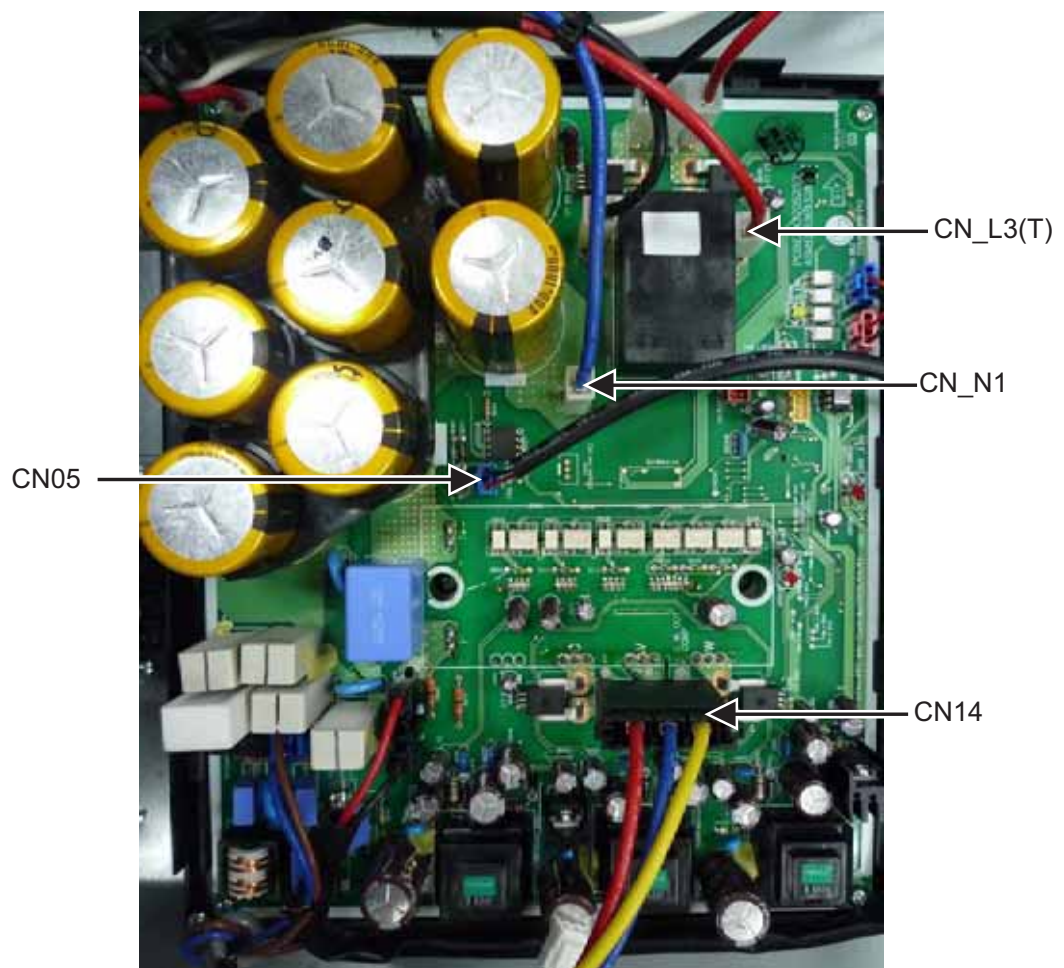
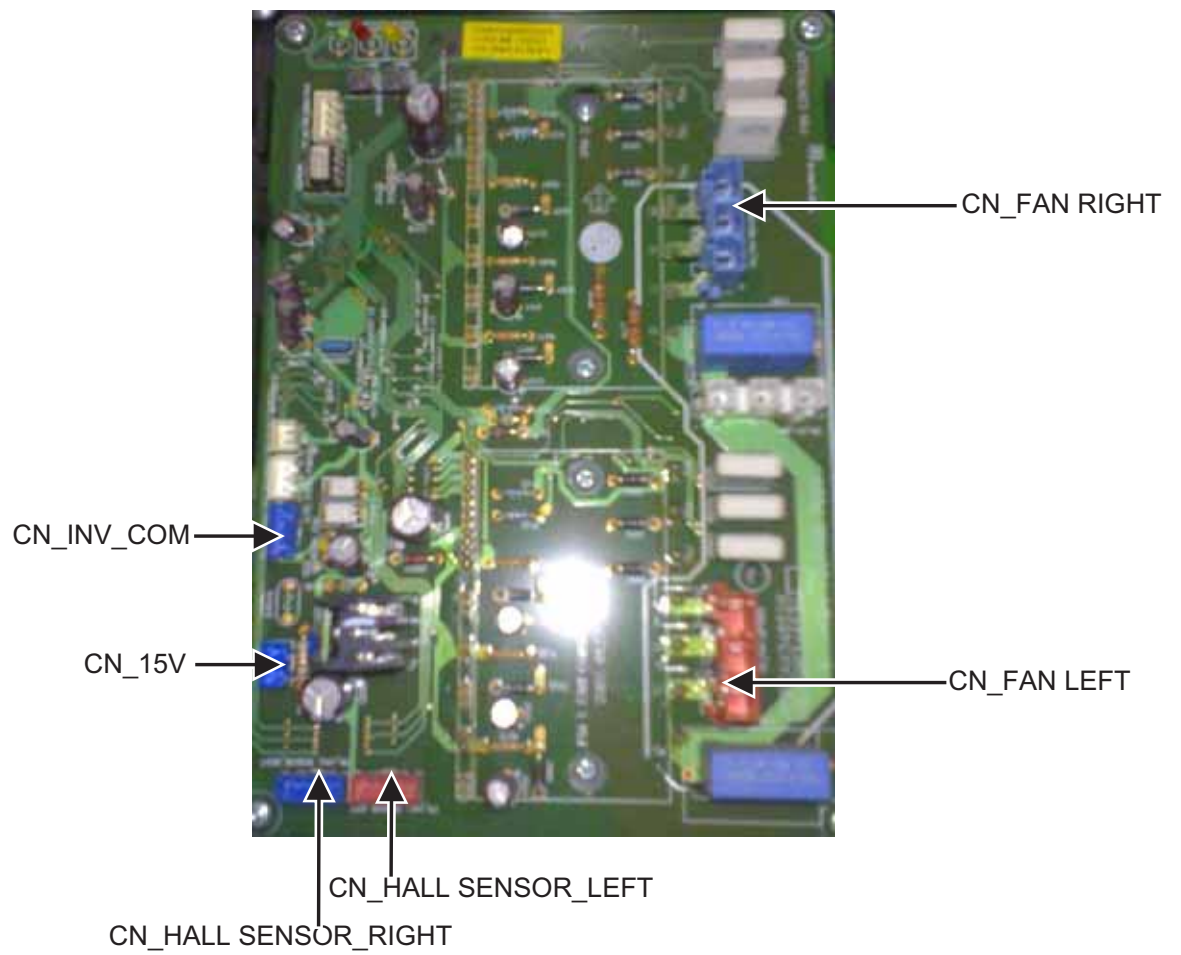
6. Wiring Diagrams

6.1 Indoor Unit



6.2 Outdoor Unit





7. Function Table

7.1 Indoor Unit

Function		Ceiling Concealed Duct
		CS-L72JD1R5
Air flow	Air Supply outlet	1
	Airflow Direction control (left & right)	X
	Airflow Direction control (up & down)	X
	Auto Swing (left & right)	X
	Auto Swing (up & down)	X
	Airflow Steps(Fan / Cool / Heat)	3 / 3 / 3
	CHAOS Swing	X
	CHAOS wind (Auto wind)	X
	Jet Cool (Power Wind)	X
	Swirl Wind	X
Air Purifier	Deodorizing Filter	X
	Plasma Air Purifier	X
	Prefilter(Washable / Anti-fungus)	X
Installation	Drain Pump	X
	E.S.P. Control	O
	Electric Heater(Operation)	X
	High Ceiling Operation	X
Reliability	Hot Start	O
	Self Diagnosis	O
	Soft Dry Operation	O
Convenience	Auto Changeover	O
	Auto Cleaning	X
	Auto Operation (Artificial intelligence)	X
	Auto Restart Operation	O
	Child Lock	O
	Forced Operation	X
	Group Control	O
	Sleep Mode	O
	Timer (On/Off)	O
	Timer (weekly)	O
Individual Control	Two Thermistor Control	O
	Standard Wired Remote Controller	O
	Wired Remote Controller(for hotel use)	X
	Wireless Remote Controller(simple)	X
Others	Wireless LCD Remote Control	X
	Remote Sensor	X
	Fan Auto Operation	O
	Fan Auto Fuction(Wired Remote Controller)	X
	Skip over Evaporator Sensing	X
	Dry Contact Error State Output	X
	RF Controller	X
	Heating Overload	X
	Option Switch	-

Note :

O : Applied, X : Not applied, – : No relation

Option: Model name & price are different according to options, and assembled in factory with main unit.

Accessory: Installed at field, ordered and purchased separately by the corresponding model name, supplied with separate package.
(Refer to model name for accessory above table)

7.2 Outdoor Unit

Function		CU-L72JD1R8
Reliability	Defrost / Deicing	O
	High pressure switch	X
	Low pressure switch	O
	Phase protection	O
	Restart delay (3-minutes)	O
	Self diagnosis	O
	Soft start	O
	Test function	O
Convenience	Auto operation(Artificial intelligence)	X
	Auto restart operation	O
Special function kit	Low ambient operation	-
Others	Thermistor	-

Note :

O : Applied X : Not applied - : No relation

Option : Model name & price are different according to options, and assembled in factory with main unit.

Accessory : Installed at field, ordered and purchased separately by the corresponding model name, supplied with separate package
(Refer to model name for accessory above table)

8. Performance Data

8.1 Cooling Capacity

Model No.: CS-L72JD1R5 , CU-L72JD1R8

AFR	Indoor Air Temperature		Outdoor Air Temperature : °CDB																				
			20			25			32			35			40			43			46		
I/s	°CWB	°CDB	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI	TC	SHC	PI
934	14.0	20.0	19.1	16.4	3.5	18.3	15.9	3.7	17.2	15.2	5.0	16.6	14.7	5.5	15.9	14.3	6.1	15.5	13.8	6.0	15.0	13.3	5.3
	16.0	22.0	20.3	16.6	4.8	19.5	16.1	4.8	18.3	15.5	5.8	17.8	15.0	6.3	17.1	14.6	6.5	16.6	14.1	6.2	16.2	13.7	5.4
	18.0	25.0	21.5	16.8	5.2	20.6	16.3	5.2	19.5	15.7	6.1	19.0	15.3	6.4	18.2	14.9	6.6	17.8	14.4	6.2	17.4	14.0	5.3
	19.0	27.0	22.1	16.9	5.2	21.2	16.5	5.3	20.1	15.9	6.2	19.5	15.4	6.4	18.8	15.1	6.7	18.4	14.6	6.2	17.9	14.2	5.2
	22.0	30.0	23.8	17.3	5.2	23.0	16.8	5.4	21.8	16.3	6.4	21.3	15.9	6.7	20.6	15.6	6.8	20.1	15.1	6.3	19.7	14.7	5.2
	24.0	32.0	25.0	17.6	5.2	24.1	17.2	5.4	23.0	16.7	6.5	22.4	16.3	6.8	21.7	15.9	6.9	21.3	15.5	6.4	20.8	15.1	5.4
1050	14.0	20.0	20.1	17.5	3.6	19.2	16.9	3.7	18.1	16.2	5.0	17.5	15.7	5.6	16.7	15.2	6.1	16.3	14.7	6.0	15.8	14.2	5.4
	16.0	22.0	21.3	17.7	4.8	20.5	17.2	4.9	19.3	16.5	5.9	18.7	16.0	6.3	18.0	15.6	6.6	17.5	15.0	6.3	17.0	14.6	5.4
	18.0	25.0	22.6	17.9	5.2	21.7	17.3	5.3	20.5	16.8	6.2	19.9	16.2	6.5	19.2	15.8	6.7	18.7	15.3	6.3	18.3	14.9	5.3
	19.0	27.0	23.2	18.0	5.3	22.3	17.5	5.3	21.1	17.0	6.3	20.5	16.4	6.5	19.8	16.0	6.7	19.3	15.6	6.3	18.9	15.1	5.3
	22.0	30.0	25.1	18.4	5.3	24.2	17.9	5.4	23.0	17.4	6.4	22.4	16.9	6.8	21.6	16.6	6.8	21.2	16.1	6.3	20.7	15.7	5.3
	24.0	32.0	26.3	18.7	5.3	25.4	18.3	5.5	24.2	17.8	6.5	23.6	17.3	6.9	22.8	17.0	7.0	22.4	16.5	6.5	21.9	16.1	5.4
116	14.0	20.0	20.6	18.8	3.6	19.7	18.2	3.8	18.5	17.4	5.1	17.9	16.9	5.7	17.1	16.3	6.2	16.7	15.8	6.1	16.2	15.3	5.5
	16.0	22.0	21.9	19.0	4.9	21.0	18.5	5.0	19.7	17.8	6.0	19.2	17.2	6.4	18.4	16.7	6.7	17.9	16.2	6.4	17.4	15.7	5.5
	18.0	25.0	23.1	19.2	5.3	22.2	18.7	5.3	21.0	18.0	6.3	20.4	17.5	6.6	19.6	17.0	6.8	19.2	16.5	6.4	18.7	16.0	5.4
	19.0	27.0	23.8	19.4	5.4	22.8	18.8	5.4	21.6	18.2	6.4	21.0	17.6	6.6	20.3	17.3	6.8	19.8	16.7	6.4	19.3	16.3	5.4
	22.0	30.0	25.7	19.8	5.4	24.7	19.3	5.5	23.5	18.7	6.5	22.9	18.2	6.9	22.1	17.8	6.9	21.7	17.3	6.4	21.2	16.9	5.4
	24.0	32.0	26.9	20.1	5.3	26.0	19.7	5.6	24.8	19.1	6.6	24.2	18.6	7.0	23.4	18.3	7.1	22.9	17.8	6.6	22.4	17.3	5.5
1284	14.0	20.0	21.0	19.6	3.6	20.1	19.0	3.8	18.8	18.2	5.1	18.3	17.6	5.7	17.5	17.1	6.2	17.0	16.5	6.1	16.5	15.9	5.5
	16.0	22.0	22.3	19.9	4.9	21.4	19.3	4.9	20.1	18.6	6.0	19.5	18.0	6.4	18.7	17.5	6.7	18.3	16.9	6.4	17.8	16.4	5.5
	18.0	25.0	23.6	20.0	5.3	22.6	19.5	5.3	21.4	18.8	6.3	20.8	18.2	6.6	20.0	17.8	6.8	19.5	17.2	6.4	19.1	16.7	5.4
	19.0	27.0	24.2	20.2	5.4	23.3	19.7	5.4	22.0	19.0	6.4	21.4	18.4	6.6	20.7	18.0	6.8	20.2	17.5	6.4	19.7	17.0	5.4
	22.0	30.0	26.2	20.6	5.4	25.2	20.1	5.5	24.0	19.5	6.5	23.4	19.0	6.9	22.6	18.6	6.9	22.1	18.0	6.4	21.6	17.6	5.4
	24.0	32.0	27.4	21.0	5.3	26.5	20.5	5.6	25.2	20.0	6.6	24.6	19.4	7.0	23.8	19.1	7.1	23.4	18.5	6.6	22.9	18.1	5.5
1400	14.0	20.0	21.2	20.2	3.6	20.3	19.6	3.8	19.0	18.8	5.1	18.4	18.2	5.7	17.6	17.6	6.3	17.1	17.0	6.1	16.7	16.5	5.5
	16.0	22.0	22.5	20.5	4.9	21.6	19.9	5.0	20.3	19.2	6.0	19.7	18.5	6.5	18.9	18.0	6.8	18.4	17.4	6.4	17.9	16.9	5.6
	18.0	25.0	23.8	20.7	5.3	22.9	20.1	5.4	21.6	19.4	6.3	21.0	18.8	6.6	20.2	18.4	6.8	19.7	17.8	6.4	19.2	17.3	5.4
	19.0	27.0	24.4	20.9	5.4	23.5	20.3	5.5	22.2	19.6	6.4	21.6	19.0	6.6	20.8	18.6	6.9	20.4	18.0	6.4	19.9	17.5	5.4
	22.0	30.0	26.4	21.3	5.4	25.4	20.8	5.5	24.2	20.2	6.6	23.6	19.6	6.9	22.8	19.2	7.0	22.3	18.6	6.5	21.8	18.2	5.4
	24.0	32.0	27.7	21.7	5.4	26.7	21.2	5.6	25.5	20.6	6.7	24.9	20.1	7.1	24.1	19.7	7.1	23.6	19.1	6.6	23.1	18.7	5.5

• Symbol

AFR : Air flow rate [l/s]
 DB : Dry bulb temperature [°C]
 WB : Wet bulb temperature [°C]
 TC : Total capacity [kW]
 SHC : Sensible capacity [kW]
 PI : Power Input [kW]
 (Comp.+ indoor fan motor+outdoor fan motor)

• Notes

1. All capacities are net, evaporator fan motor heat is deducted.
2. Indicates rated capacity.
3. Direct interpolation is permissible. Do not extrapolate
4. Capacities are based on the following conditions:
 - Interconnecting Piping Length 7.5m
 - Level Difference of Zero.

8.2 Heating Capacity

Model No.: CS-L72JD1R5 , CU-L72JD1R8

AFR	Indoor Air Temperature	Outdoor Air Temperature : °CWB													
		-15		-10		-5		0		6		10		15	
I/s	°CDB	TC	PI	TC	PI	TC	PI	TC	PI	TC	PI	TC	PI	TC	PI
934	16.0	17.4	6.6	18.3	6.5	19.8	6.8	21.2	7.3	23.5	7.8	24.8	8.1	26.9	8.6
	18.0	17.2	6.7	18.2	6.6	19.8	7.0	21.1	7.5	23.3	8.0	24.5	8.2	26.8	8.7
	20.0	17.1	6.8	18.2	6.8	19.8	7.2	21.1	7.7	23.1	8.1	24.3	8.4	26.9	8.8
	21.0	17.0	6.9	18.2	6.9	19.8	7.3	21.0	7.8	23.0	8.2	24.3	8.4	26.7	8.8
	22.0	17.0	7.0	18.2	7.0	19.8	7.4	21.0	7.9	22.8	8.3	24.3	8.5	26.5	8.8
1167	24.0	16.8	7.2	18.1	7.2	19.6	7.6	20.8	8.0	22.6	8.4	23.9	8.5	26.2	8.8
	16.0	17.7	6.1	18.6	6.0	20.1	6.2	21.5	6.7	23.9	7.2	25.2	7.5	27.4	8.0
	18.0	17.5	6.2	18.6	6.1	20.2	6.4	21.5	6.9	23.7	7.3	24.9	7.6	27.3	8.0
	20.0	17.4	6.3	18.6	6.3	20.2	6.6	21.5	7.1	23.5	7.5	24.8	7.7	27.4	8.1
	21.0	17.3	6.4	18.6	6.4	20.2	6.7	21.4	7.2	23.4	7.6	24.8	7.8	27.2	8.1
1400	22.0	17.3	6.5	18.6	6.4	20.2	6.8	21.3	7.3	23.2	7.6	24.7	7.8	27.0	8.1
	24.0	17.1	6.6	18.4	6.6	19.9	7.0	21.2	7.4	23.0	7.8	24.3	7.9	26.6	8.2
	16.0	18.0	5.8	19.0	5.6	20.6	5.9	22.0	6.3	24.4	6.8	25.8	7.1	28.0	7.5
	18.0	17.9	5.9	19.0	5.8	20.6	6.1	22.0	6.5	24.2	6.9	25.5	7.2	27.9	7.6
	20.0	17.7	6.0	18.9	5.9	20.6	6.2	21.9	6.7	24.0	7.1	25.3	7.3	27.9	7.7
	21.0	17.7	6.0	19.0	6.0	20.6	6.3	21.9	6.8	23.8	7.2	25.3	7.4	27.8	7.7
1400	22.0	17.7	6.1	19.0	6.1	20.6	6.4	21.8	6.9	23.7	7.2	25.3	7.4	27.5	7.7
	24.0	17.5	6.2	18.8	6.3	20.4	6.6	21.6	7.0	23.5	7.3	24.8	7.5	27.2	7.7

• Symbol

AFR : Air flow rate

DB : Dry bulb temperature

WB : Wet bulb temperature

TC : Total capacity

SHC : Sensible capacity

PI : Power Input

(Comp.+ indoor fan motor+outdoor fan motor)

[°C]

[°C]

[kW]

[kW]

[kW]

• Notes

1. All capacities are net, evaporator fan motor heat is deducted.

2. Indicates rating capacity.

3. Direct interpolation is permissible. Do not extrapolate

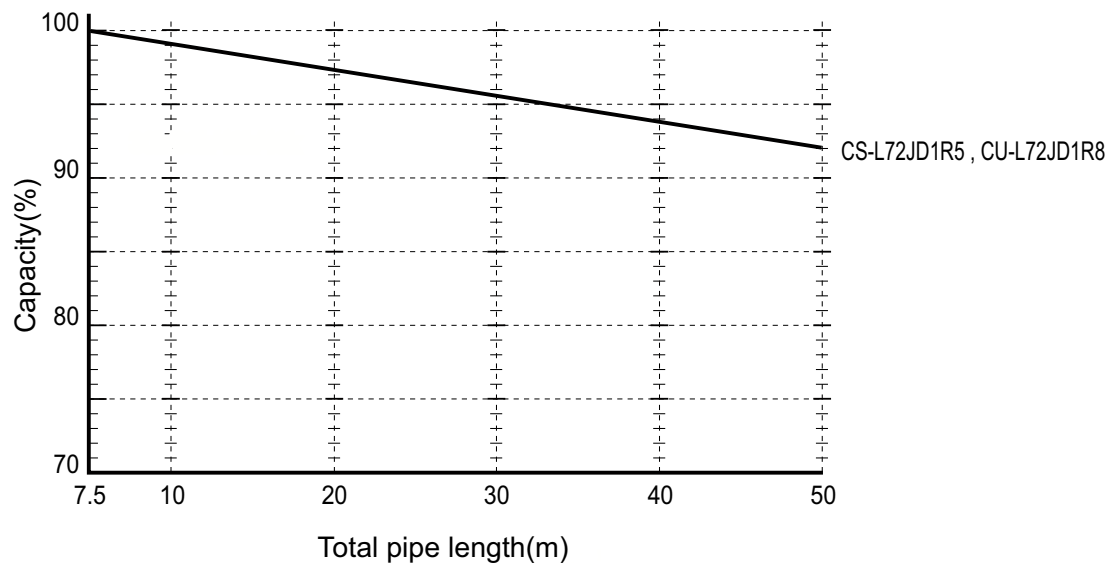
4. Capacities are based on the following conditions:

- Interconnecting Piping Length 7.5m

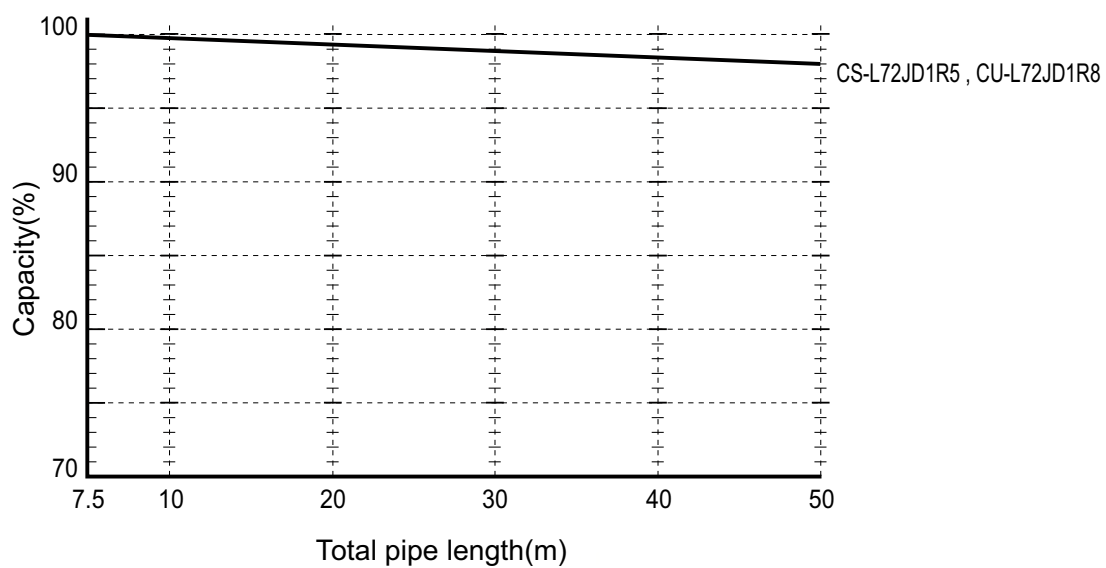
- Level Difference of Zero.

9. The Coefficient of Capacity Change

Cooling



Heating



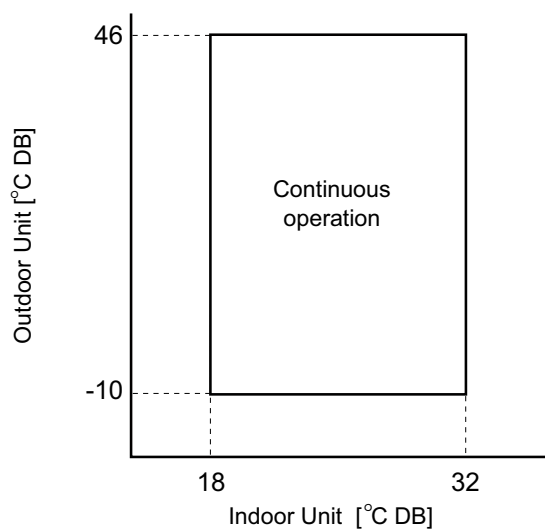
Notes:

* Equivalent pipe length = Actual pipe length + number of bend x 0.3

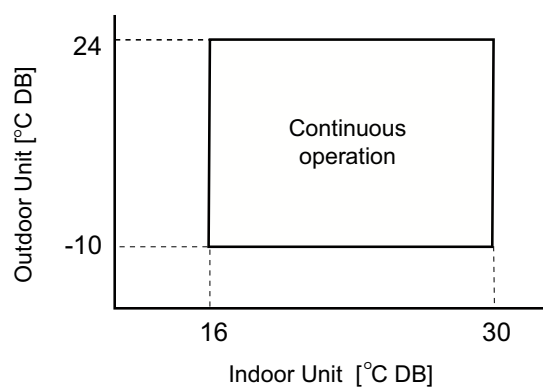
* Refer to the specification for the maximum pipe length of each model.

10. Operation Range

Cooling



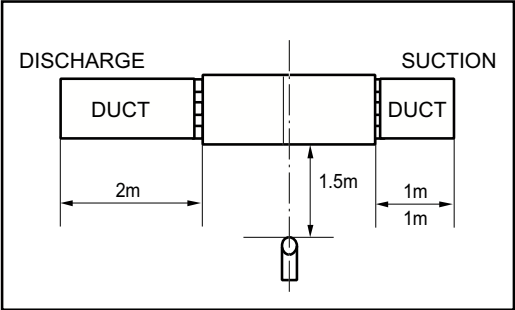
Heating



11. Sound Levels

11.1 Indoor Unit

Overall

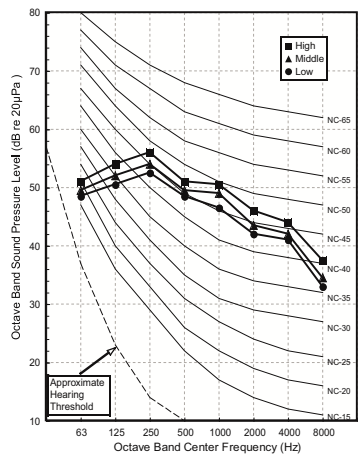


- Notes:
- Sound measured at 1.5m away from the center of the unit.
 - Data is valid at free field condition.
 - Data is valid at nominal operation condition.
 - Reference accoustic pressure $OdB = 20 \text{ Pa}$.
 - Sound level will vary depending on a range of factors such as the construction(acoustic absorption coefficient) of particular room in which the equipment is installed.
 - The operating conditions are assumed to be standard.

Model	Sound Levels(dB(A))		
	H	M	L
CS-L72JD1R5	52	50	49

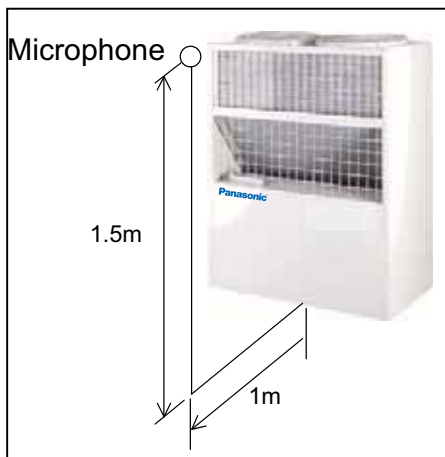
Sound Pressure Level

CS-L72JD1R5



11.2 Outdoor Unit

Overall



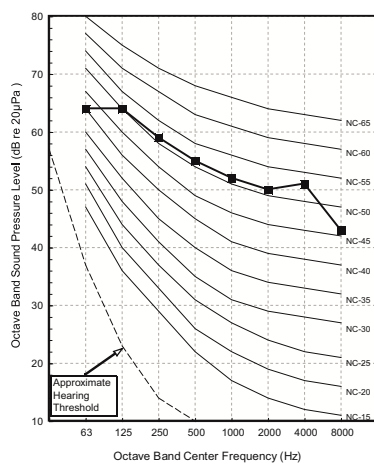
Notes:

- Sound measured at 1m away from the center of the unit.
- Data is valid at free field condition.
- Data is valid at nominal operation condition.
- Reference accoustic pressure 0dB=20 Pa.
- Sound level will vary depending on a range of factors such as the construction(acoustic absorption coefficient) of particular room in which the equipment is installed.
- The operating conditions are assumed to be standard.

Model	Sound Levels[dB(A)]
	Cooling/Heating
CU-L72JD1R8	59/59

Sound Pressure Level

CU-L72JD1R8



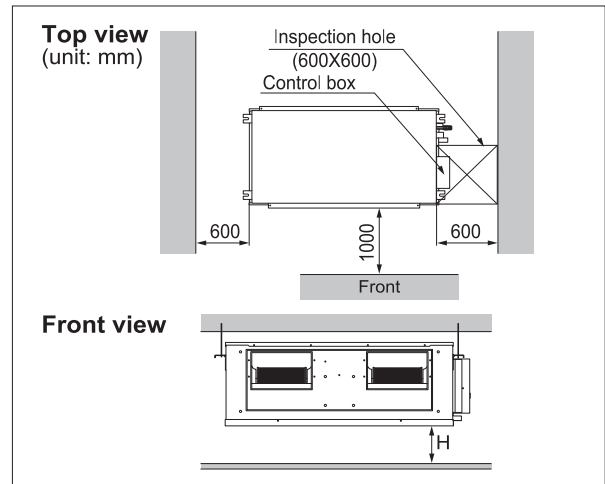
12. Installation

12.1 Select the best location

Indoor unit

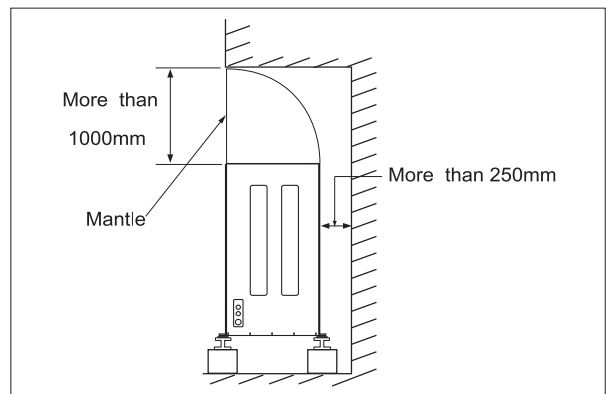
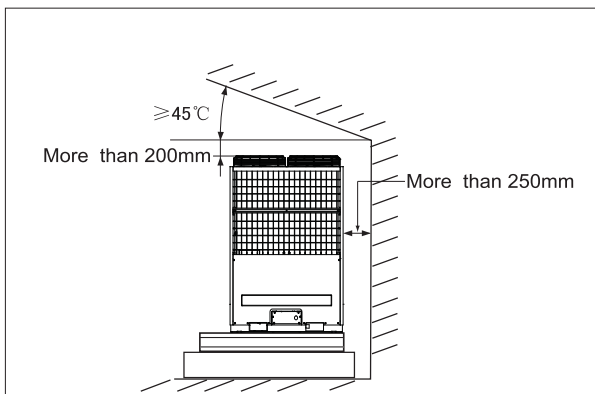
Install the air conditioner in the location that satisfies the following conditions.

- The place shall easily bear a load exceeding four times the indoor unit's weight.
- The place shall be able to inspect the unit as the figure.
- The place where the unit shall be leveled.
- The place shall allow easy water drainage. (Suitable dimension "H" is necessary to get a slope to drain as figure.)
- The place shall easily connect with the outdoor unit.
- The place where the unit is not affected by an electrical noise.
- The place where air circulation in the room will be good.
- There should not be any heat source or steam near the unit.



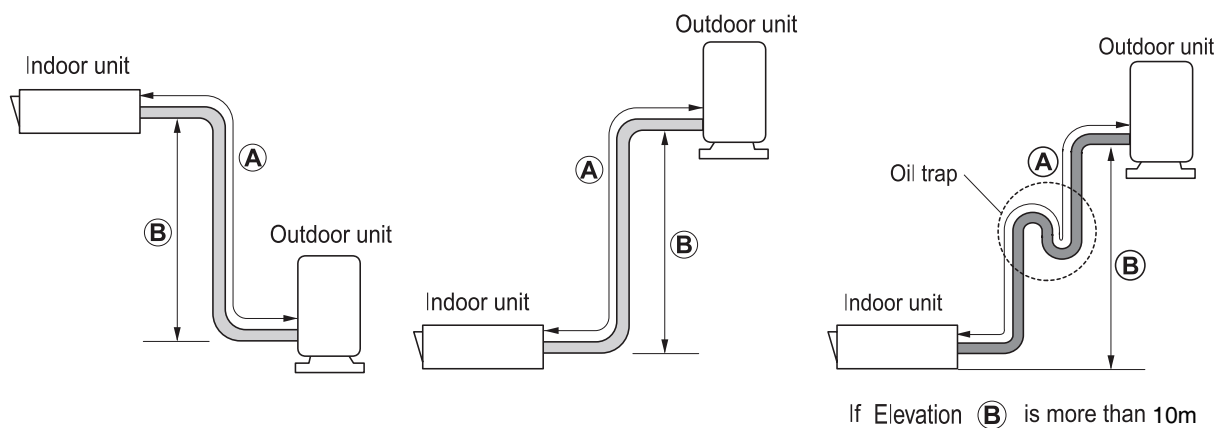
Outdoor unit

- If an awning is built over the unit to prevent direct sunlight or rain exposure, be careful that heat radiation from the condenser is not restricted.
- There should not be any animals or plants which could be affected by hot air discharged.
- Ensure the spaces indicated by arrows from the wall, ceiling, fence or other obstacles.



Piping length and the elevation

Pipe Size (Diameter: Ø)		Length ㉠ (m)			Elevation ㉡ (m)		*Additional refrigerant (g/m)
Gas	Liquid	Min.	Standard	Max.	Standard	Max.	
22.2 (7/8")	9.52 (3/8")	3	7.5	50	0	30	50



CAUTION:

- The Standard pipe length is 7.5 m and no need to additional charge of the refrigerant to max 15 m. If the pipe length exceeds 15 m, need to additional charge of the refrigerant which according to the table.
- If this Model is installed at a distance of 50m, 1750g of refrigerant should be added $(50 - 15) \times 50g = 1750g$
- Capacity is based on standard length and maximum allowance length is on the basis of reliability.
- Improper refrigerant charge may result in abnormal cycle.
- Oil trap should be installed every 5~7 meters.

12.2 Indoor Unit Installation

■ Installation of Unit

Install the unit above the ceiling correctly.

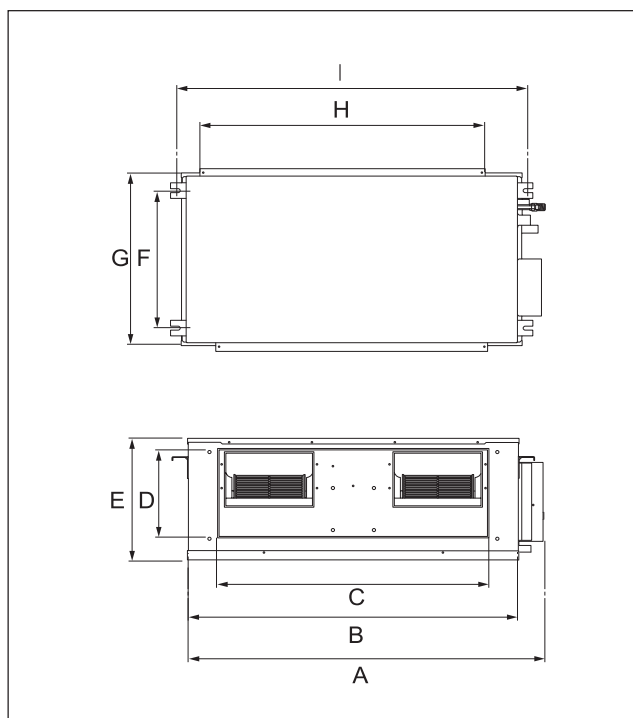
CASE 1

POSITION OF SUSPENSION BOLT

- Apply a joint-canvas between the unit and duct to absorb unnecessary vibration.
- Apply a filter Accessory at air return hole.

(Unit:mm)

Dimension	A	B	C	D	E	F	G	H	I
Model Name									
CS-L72JD1R5	1680	1562	1120	290	460	580	688	1400	1622

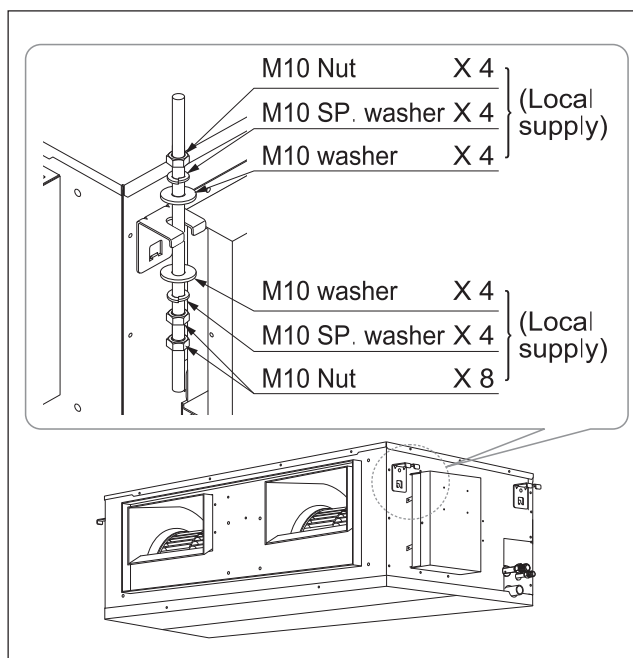


CASE 2

- Install the unit leaning to a drainage hole side as a figure for easy water drainage.

POSITION OF CONSOLE BOLT

- A place where the unit will be leveled and that can support the weight of the unit.
- A place where the unit can withstand its vibration.
- A place where service can be easily performed.

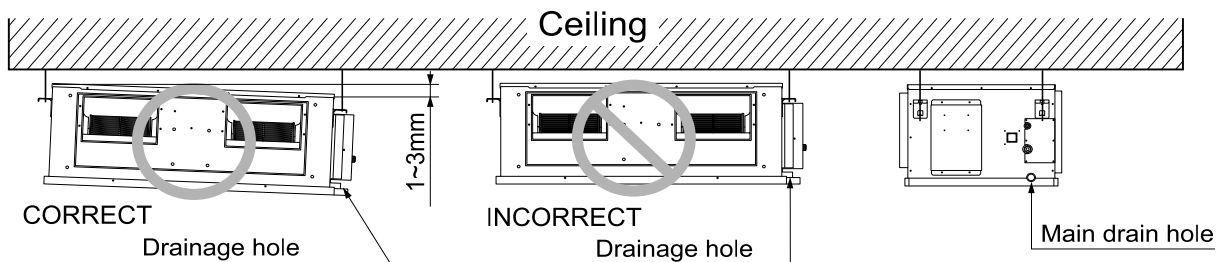


CAUTION

1. Install declination of the indoor unit is very important for the drain of the duct type air conditioner.
2. Minimum thickness of the insulation for the connecting pipe shall be 10 mm.

Front of view

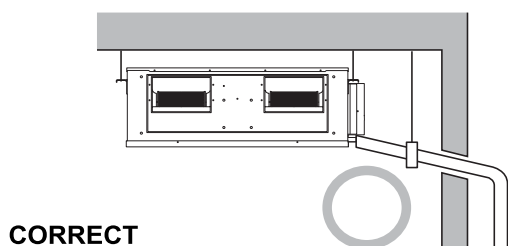
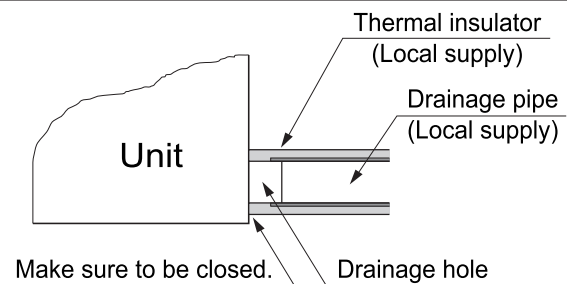
- The unit must be declined to the drain hose connected when finished installation.



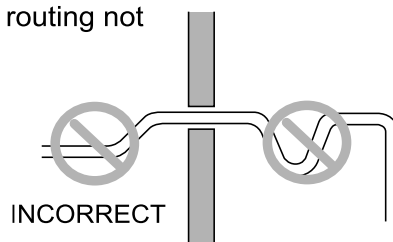
CAUTION FOR GRADIENT OF UNIT AND DRAIN PIPING

Lay the drain hose with a downward inclination so water will drain out.

- Always lay the drain with downward inclination (1/50 to 1/100). Prevent any upward flow or reverse flow in any part.
- 5mm or thicker formed thermal insulator shall always be provided for the drain pipe.



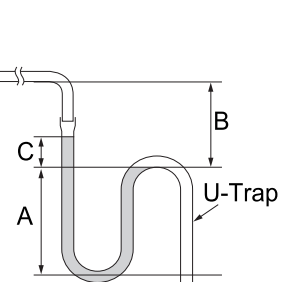
- Upward routing not allowed



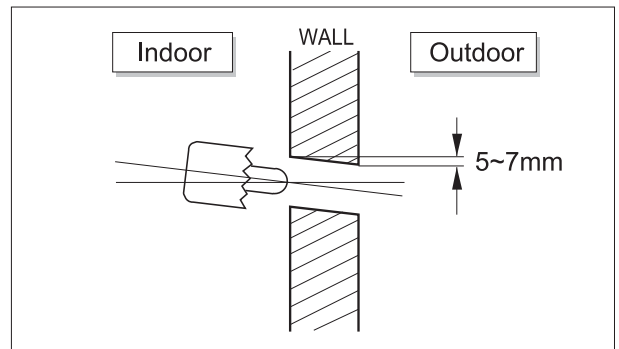
Applied U-Trap Dimension

- Install the P-Trap (or U-Trap) to prevent a water leakage caused by the blocking of intake air filter.

$A \geq 70\text{mm}$
 $B \geq 2C$
 $C \geq 2 \times \text{SP}$
 SP = External Pressure (mmAq)
 Ex) External Pressure = 10mmAq
 $A \geq 70\text{mm}$
 $B \geq 40\text{mm}$
 $C \geq 20\text{mm}$



- Drill the piping hole with 70mm dia, hole core drill.
- Piping hole should be slightly slant to the outdoor side.



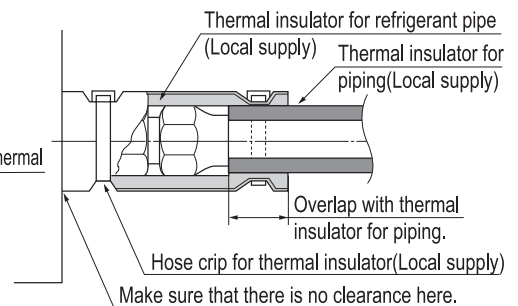
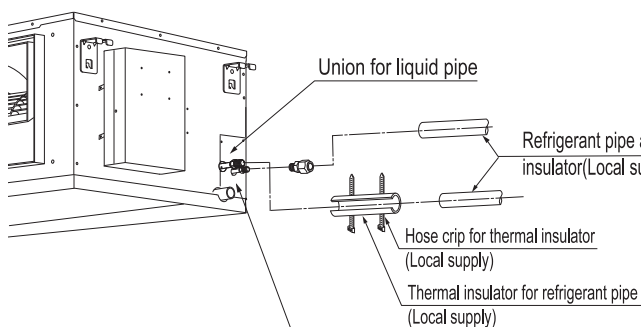
INSULATION, OTHERS

Insulate the joint and tubes completely.

THERMAL INSULATION

All thermal insulation must comply with local requirement.

INDOOR UNIT



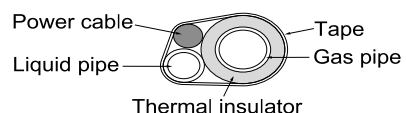
REFRIGERANT PIPE

- Insulate and tape the gas piping.

CAUTION: Cutting line of insulation must look upper direction. Thickness of insulation is 10mm or over.

NOTICE Recommended Insulation material
 Material : FOAM PE
 Thickness : 10mm
 Density : less than $0.032 \pm 0.005(\text{g/cm}^3)$

Thermal conductivity : less than $0.03(\text{kcal/m.hr.}^\circ\text{C})$



TEST AND CHECK

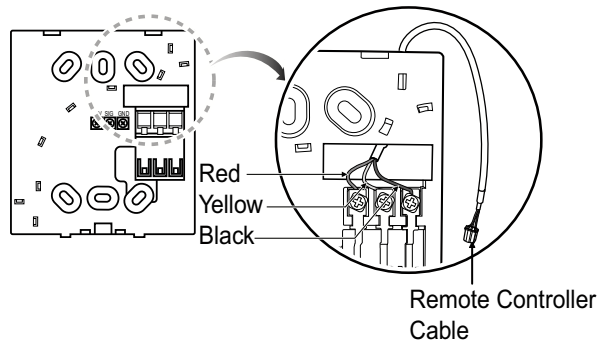
■ After all workings are finished, check the working and operation.

- Air distribution — Is the air circulation good?
- Drain — Is the drainage smoothly and no sweating?
- Gas leakage — Is the piping connection correctly?
- Wiring — Is the wiring connection correctly?
- Lock-bolt — Is the lock-bolt of compressor loosened?

12.3 WIRED REMOTE CONTROLLER INSTALLATION

1. Connect the remote controller cable to the wired remote controller installation board as shown in the right picture.

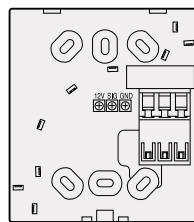
12V	Red wire
SIG	Yellow wire
GND	Black wire



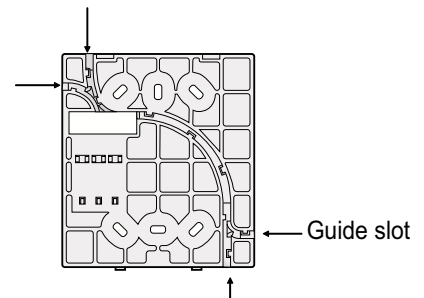
* The remote controller cable is connected as factory default.

2. After fixing the cable to the guide slot, attach the wired remote controller installation board at the desired location.

- Before fixing the remote controller cable to the guide slot, remove any clogged part of the case in the direction to install before the installation.



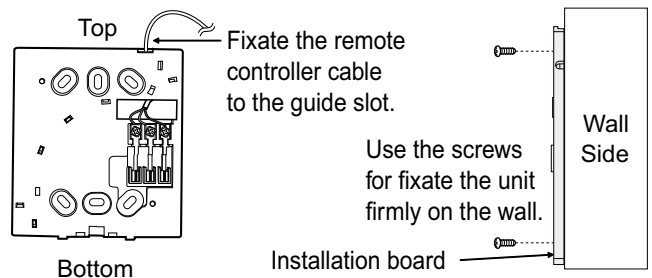
<Front side of installation board>



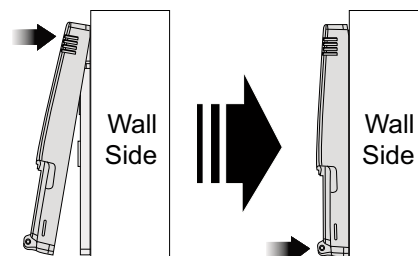
<Rear side of installation board>

3. After locating the wired remote controller installation board at the desired location, screw the unit firmly. (When there is a buried box, install the wired remote controller board to fit the buried box.)

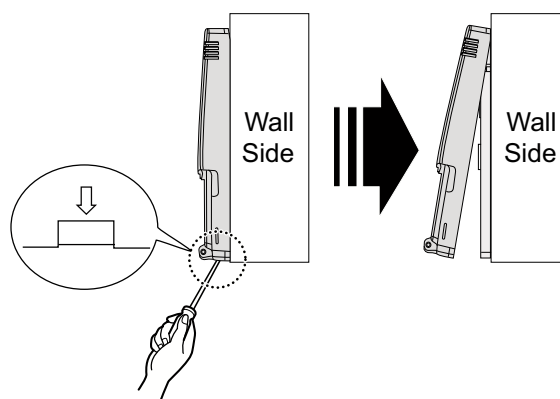
- Use the screws provided.



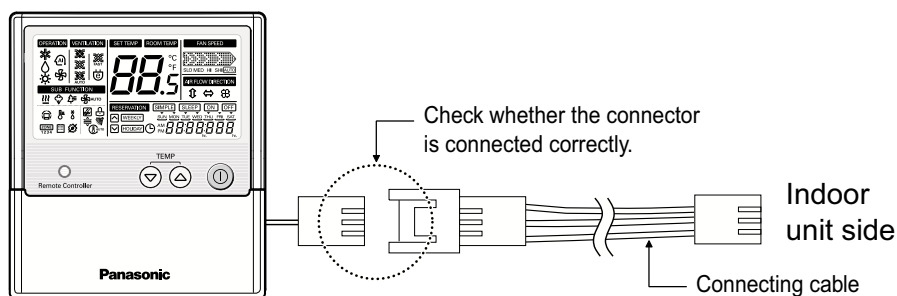
4. After fixing the top part of the remote controller to the installation board as shown in beside picture, press the bottom part to assemble the controller to it's board.



When disassembling the remote controller from the installation board, use the driver as shown in the right picture and insert it into the hole with the arrow. And when you pull the driver in the front direction, the remote controller will be separated.



5. Use the connecting cable to connect the indoor unit and the remote controller.



CAUTION

When installing the wired remote controller, do not bury it in the wall.

(It can cause damage in the temperature sensor.)

Do not install the cable to be 50m or above.

(It can cause communication error.)

- When installing the extension cable, check the connecting direction of the connector of the remote controller side and the product side for correct installation.
- If you install the extension cable in the opposite direction, the connector will not be connected.
- Specification of extension cable: 2547 1007 22# 2 core 3 shield 5 or above.

- Since the room temperature sensor is in the remote controller, the remote controller box should be installed in a place away from direct sunlight, high humidity and direct supply of cold air to maintain proper space temperature.
Install the remote controller about 1.5m above the floor in an area with good air circulation at an average temperature.

Do not install the remote controller where it can be affected by:

- Drafts, or dead spots behind doors and in corners.
- Hot or cold air from ducts.
- Radiant heat from sun or appliances.
- Concealed pipes and chimneys.
- Uncontrolled areas such as an outside wall behind the remote controller.
- This remote controller is equipped with a seven segment LED. display. For proper display of the remote controller LED's, the remote controller should be installed properly as shown in Fig.1.
(The standard height is 1.2~1.5 m from floor level.)

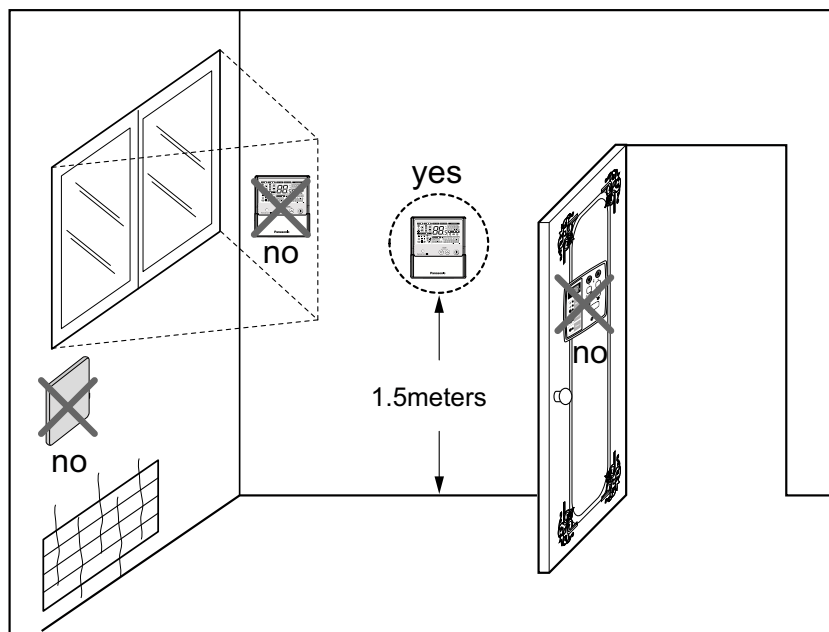
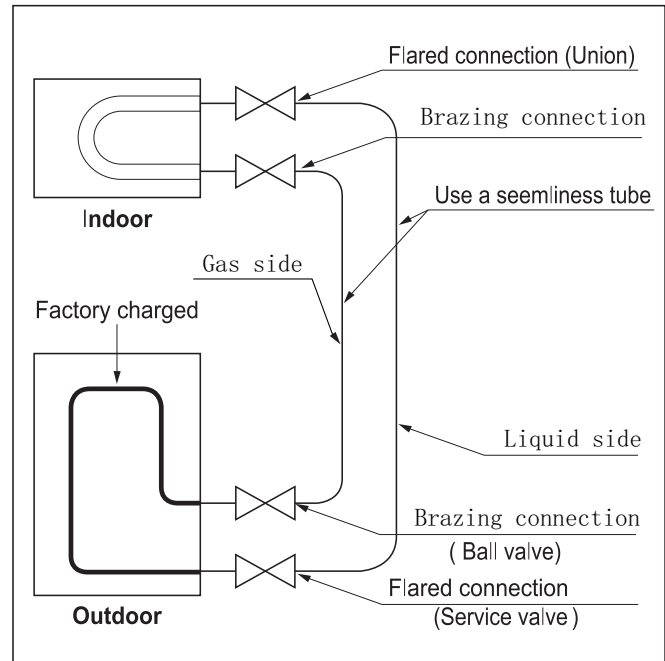


Fig.1 Typical locations for remote controller

12.4 Refrigerant Piping

Perform the work according to the Service Manual or Installation Guide.

- Use two spanners when connecting the refrigerant pipe to the unit.
- Make a bend with a radius as large as possible.
- Perform air purge with vacuum drying.
- When piping work is finished, check all joints.



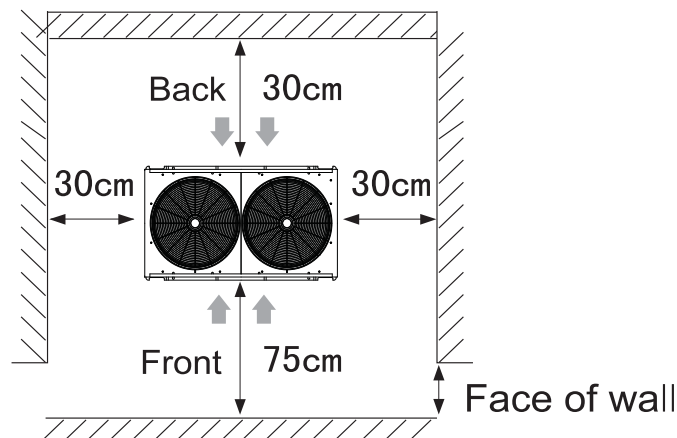
12.5 Installation of outdoor unit

Select a location that satisfies the following conditions. Install the unit firmly in place.

■ Select the following location

- A place where the air conditioner can get good ventilation.
- A place where it shall not annoy the neighbors.
- A place where the unit shall be leveled and that can support the weight of unit and withstand its vibrations.

■ Keep a maintenance space



12.6 Electrical Wiring

Perform the electrical wiring work according to the electrical wiring connection.

- All wiring must comply with local requirements.
- Select a power source that is capable of supplying the current required by the air conditioner.
- Use a recognized circuit breaker between the power source and the unit. A disconnection device to adequately disconnect all supply lines must be fitted.
- Capacity of circuit breaker recommended by authorized personnel only
- Capacity of circuit breaker: 35A

WIRING CONNECTION

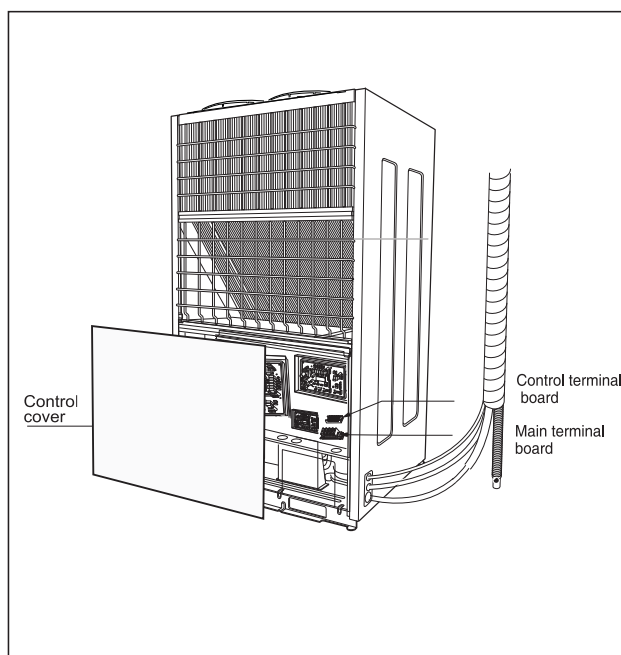
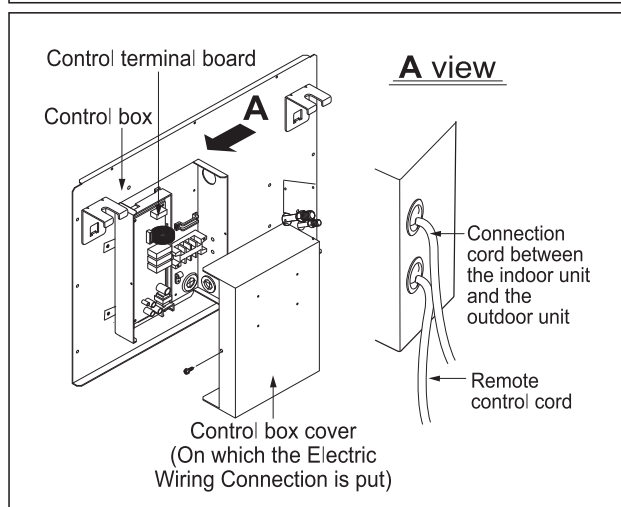
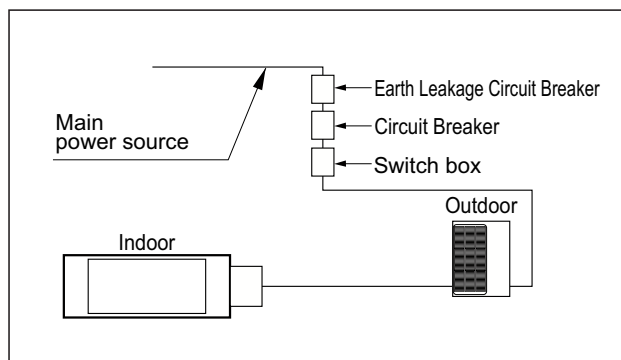
Indoor unit

- Remove the control box cover for electrical connection between the indoor and outdoor unit.
- Use the cord clamber to fix the cord.

Outdoor unit

- Remove the control cover for wiring connection.
- Use the cord clamber to fix the cord.
- Earthing work
 - Case 1 :Terminal block of Outdoor Unit have $\oplus$ mark.
 - Connect the cable of diameter 1.6mm^2 or more to the earthing terminal provided in the control box and do earthing.
 - Case 2 :Terminal block of Outdoor Unit don't have $\oplus$ mark.
 - Connect the cable of diameter 1.6mm^2 or more, to the panel of control box, marked as $\oplus$ and fasten with earth screw.

※ **Please check !!**



12.7 Connecting Pipes to the indoor Unit

Preparation of Piping

Main cause of gas leakage is defect in flaring work. Carry out correct flaring work in the following procedure.

Cut the pipes and the cable.

- Use the accessory piping kit or the pipes purchased locally.
- Measure the distance between the indoor and the outdoor unit.
- Cut the pipes a little longer than measured distance.
- Cut the cable 1.5m longer than the pipe length.

Burrs removal

- Completely remove all burrs from the cut cross section of pipe/tube.
- Put the end of the copper tube/pipe to downward direction as you remove burrs in order to avoid to let burrs drop in the tubing.

Putting nut on

- Remove flare nuts attached to indoor and outdoor units, than put them on pipe/tube having completed burr removal.
(Not possible to put them on after flaring work)

Flaring work

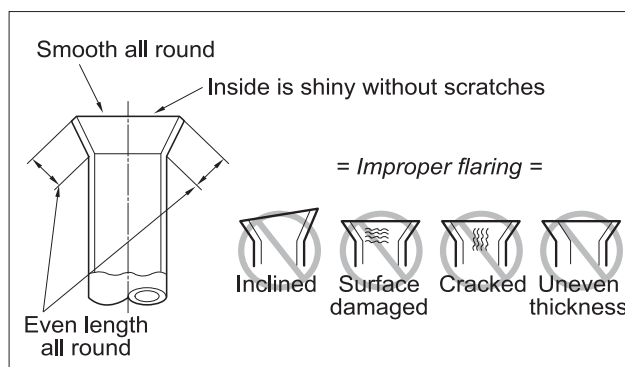
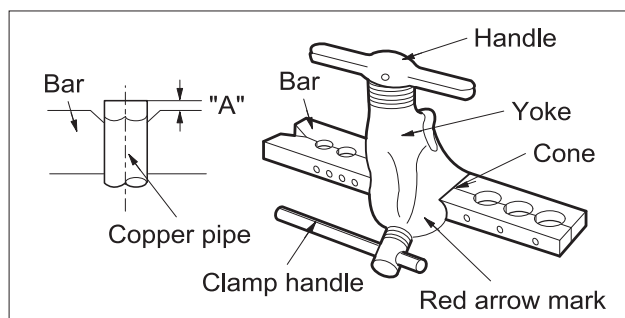
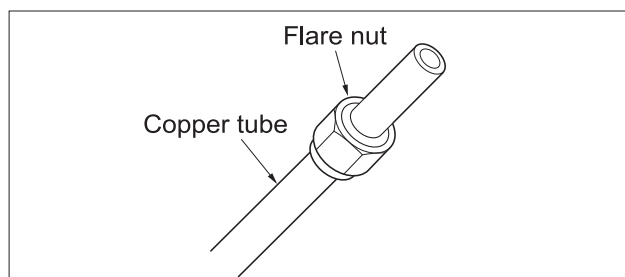
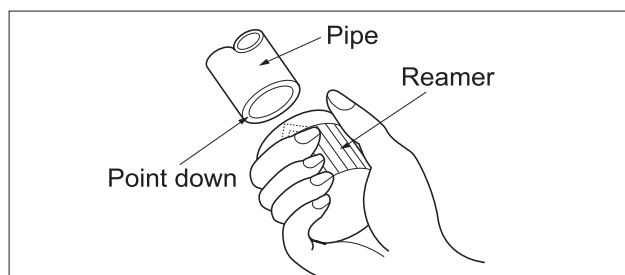
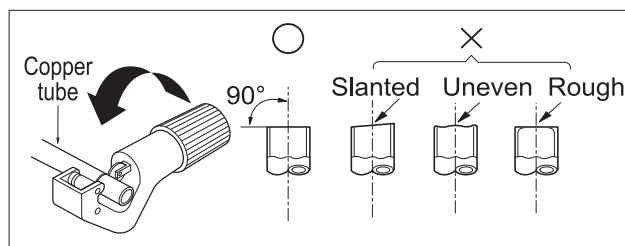
- Carry out flaring work using flaring tool as shown below.

Outside Diameter	"A"
6.35 (1/4")	1.1~1.3
9.52 (3/8")	1.5~1.7
12.70 (1/2")	1.6~1.8
15.88 (5/8")	1.6~1.8
19.05 (3/4")	1.9~2.1

Firmly hold copper tube in a bar(or die) as indicated dimension in the table above.

Check

- Compare the flared work with figure.
- If flare is noted to be defective, cut off the flared section and do flaring work again.



Pipe bending

Annealed copper pipe with small diameter ($\phi 6.35$ or $\phi 9.52$) can be easily bent manually. In this case, secure large R(radius) for the bend section and gradually bend pipe. If annealed copper pipe is large in diameter ($\phi 15.88$ or $\phi 19.05$), bend pipe with bender. Use bender appropriate for the pipe diameter.

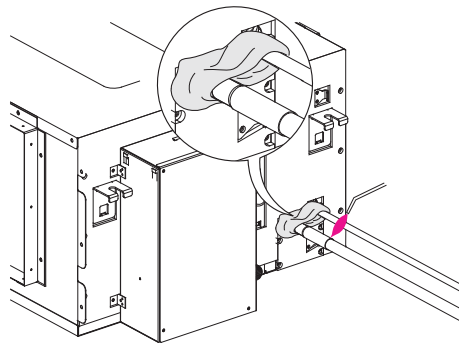
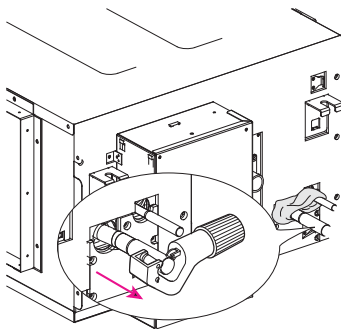
Brazing

In refrigerant piping, bending (in particular, acute bending) must be minimized to reduce piping resistance. Bending is, however, necessary in some places by virtue of the installation position of devices auxiliary to the packaged air conditioner, or of the building structure, piping distance or finishing appearance. If a more acute bend is required than that attainable by pipe bender, perform brazing using ready-made elbow. Aside from this function, brazing also serves to connect straight pipes, generally using ready-made sockets. While brazing, protect piping against heat with wet cloth to avoid damaging valve packing or burning thermal insulator with burner heat. While brazing, blow inert gas (nitrogen gas or carbonic gas) to prevent formation of oxidation film in copper piping; otherwise, the refrigerant circuit will clog. The blowing of nitrogen gas (or carbonic gas) through 3-way valves is described in the following:

Refrigerant piping(Flare piping)

When connecting piping, be sure to keep piping dry(keep piping away from water), clean (keep piping away from dust) and airtight (avoid refrigerant leakage).

When connecting piping on rainy days or making a through-hole in wall, take due care to prevent water or plaster from entering piping.



CAUTION:

- This procedure is designed to prevent formation of oxidation film by filling piping with inert gas. Note that excessive gas pressure will generate pinholes at brazed points.
(Nitrogen gas: Supply pressure 0.005 ~ 0.01MPa G(0.05~0.1kgf/cm²G)
- When supplying inert gas, be sure to open one end of piping.

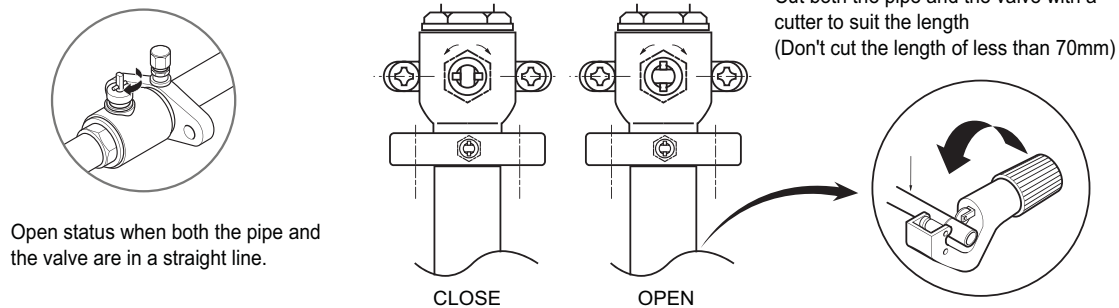
12.8 Connecting Pipes to the Outdoor Unit

The method of connection consists of flare connections at the indoor units, flange connections for the piping of the outdoor unit and flare connections for the gas piping. Note that the branched sections are brazed.

WARNING

Always use extreme care to prevent the refrigerant gas (R410A) from the leakage while using fire or flame. If the refrigerant gas comes in contact with the flame from any source, such as a gas stove, it breaks down and generates a poisonous gas which can cause gas poisoning. Never perform brazing in an unventilated room. Always conduct an inspection for gas leakage after installation of the refrigerant piping has been completed.

Cautions in pipe connection/valve operation

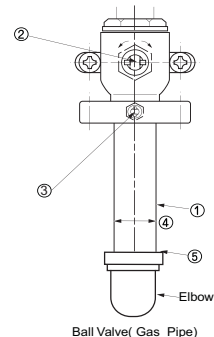


WARNING

After completing work, securely tighten both service ports and caps so that gas does not leak.

Brazing method

- ① Pipe joint (auxiliary parts): Securely perform brazing with a nitrogen blow into the service valve port. (Releasing pressure : 0.02 MPa or less)
- ② Cap: Remove caps and operate valve, etc. After operation, always reattach caps (tightening torque of valve cap: 25Nm (250kg.cm) or more). (Don't remove the internal part of the port)
- ③ Service port: Make the refrigerant pipe vacuum and charge it using the service port. Always reattach caps after completing work (tightening torque of service cap: 14Nm (140kg.cm) or more).
- ④ Gas pipe
- ⑤ Elbow joint (field supply)



Flare nut method

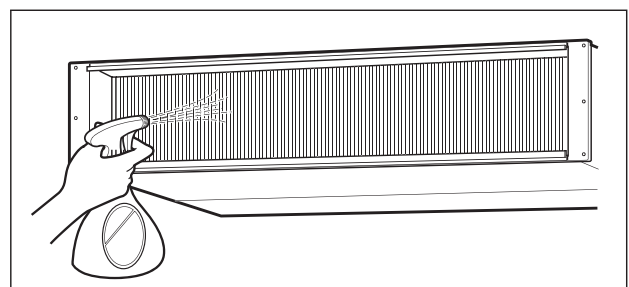
- Align the center of the pipings and sufficiently tighten the flare nut with fingers.
- Finally, tighten the flare nut with torque wrench until the wrench clicks.
- When tightening the flare nut with torque wrench, ensure the direction for tightening follows the arrow on the wrench.

Pipe size	kgf.m	N.m
9.52(3/8")	3.4~4.2	36~42

12.9 Checking the Drainage

Checking the Drainage

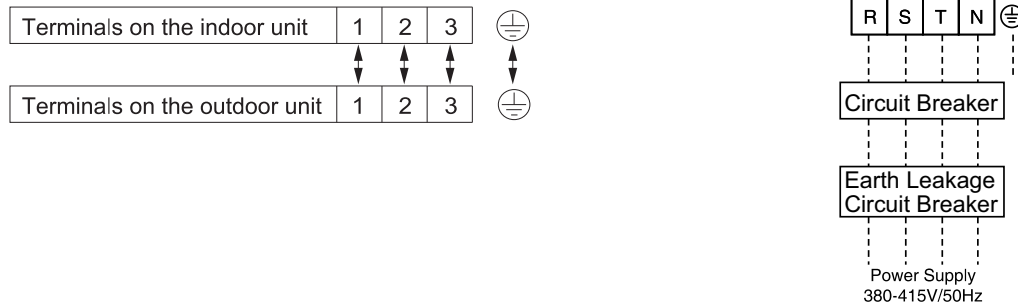
- Check the drainage.
 - Spray one or two glasses of water upon the evaporator.
 - Ensure that water flows drain hose of indoor unit without any leakage.



12.10 Connecting Cables between Indoor Unit and Outdoor Unit

12.10.1 Connecting Cables to the Indoor Unit

- Connect the wires to the terminals on the control board individually according to the outdoor unit connection.
 - Ensure that the color of the wires of outdoor unit and the terminal No. are the same as those of indoor unit respectively
- Heat pump Model



! WARNING	▲ Earth This equipment must be properly earthed. Earth line must not be connected to gas pipe, water pipe, lightning rod and telephone. Otherwise, it may cause electrical shock in case the equipment breakdown or has leakage current.	
	▲ Installation of Earth Leakage Current Breaker This equipment must be installed with earth leakage current breaker. (ELCB:35A) Otherwise, it may cause electrical shock and fire in case the equipment breakdown or has leakage current.	

CAUTION	
The power cord connected to the outdoor unit should be complied with the following specifications (Rubber insulation, type H05RN-F approved by HAR or SAA). NORMAL CROSS-SECTIONAL AREA: 5.5mm²	The connecting cable connected to the indoor and outdoor unit should be complied with the following specifications (Rubber insulation, type H05RN-F approved by HAR or SAA). NORMALCROSS-SECTIONA AREA: 1.0mm²
If the supply cord is damaged, it must be replaced by a special cord or assembly available from the manufacturer of its service agent.	

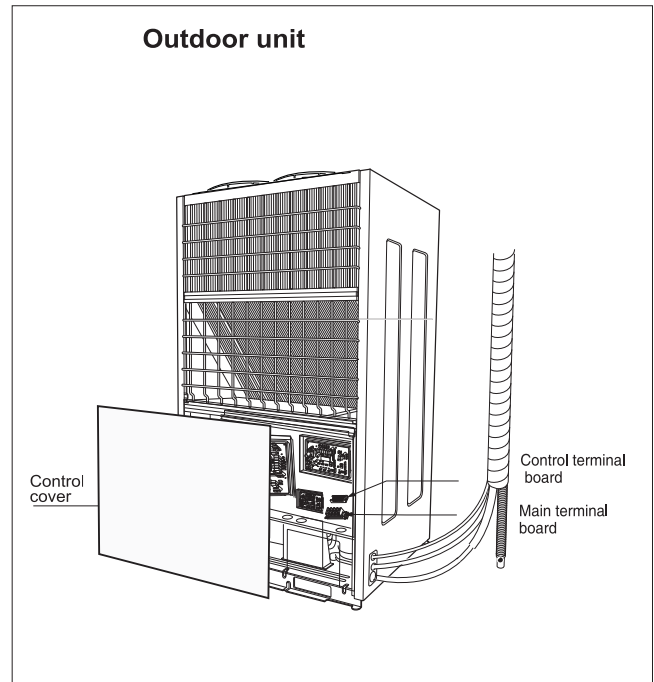
CAUTION: Make sure that the screws of the terminal fixed tightly.
Tightening torque of screws: 2.5kgf·m

12.10.2 Clamping of cables

- Arrange 2 power cables on the control panel.
- First, fasten the steel clamp with a screw to the inner boss of control panel.
- For the cooling model, fix the other side of the clamp with a screw strongly.
For the heat pump model, put the 1.0 mm² cable(thinner cable) on the clamp and tighten it with a plastic clamp to the other boss of the control panel.
- In Australia, the length of power supply cord measured from the entry of the power supply cord to the middle of live pin on the power plug should be over 1.8m.

12.10.3 Connecting the Cable to the Outdoor Unit

- Remove the Cover control from the unit by loosening a screw.
Connect the wires to the terminals on the control board individually as following.
- Secure the cable onto the control board with the holder (clammer).
- Refix the cover control to the original position with the screw.



CAUTION: After the confirmation of the above conditions, prepare the wiring as follows:

- 1) Never fail to have an individual power specialized for the air conditioner. As for the method of wiring, be guided by the circuit diagram pasted on the inside of control box cover.
- 2) Provide a circuit breaker switch between power source and the unit.
- 3) The screw which fasten the wiring in the casing of electrical fittings are liable to come loose from vibrations to which the unit is subjected during the course of transportation. Check them and make sure that they are all tightly fastened. (If they are loose, it could give rise to burn-out of the wires.)
- 4) Specification of power source
- 5) Confirm that electrical capacity is sufficient.
- 6) Be sure that the starting voltage is maintained at more than 90 percent of the rated voltage marked on the name plate.
- 7) Confirm that the cable thickness is as specified in the power sources specification.
(Particularly note the relation between cable length and thickness.)
- 8) Never fail to equip a leakage breaker where it is wet or moist.
- 9) The following troubles would be caused by voltage drop-down.
 - Vibration of a magnetic switch, damage on the contact point there of, fuse breaking, disturbance to the normal function of a overload protection device.
 - Proper starting power is not given to the compressor.

12.11 Forming the piping

Form the piping by wrapping the connecting portion of the indoor unit with insulation material and secure it with two kinds of vinyl tape.

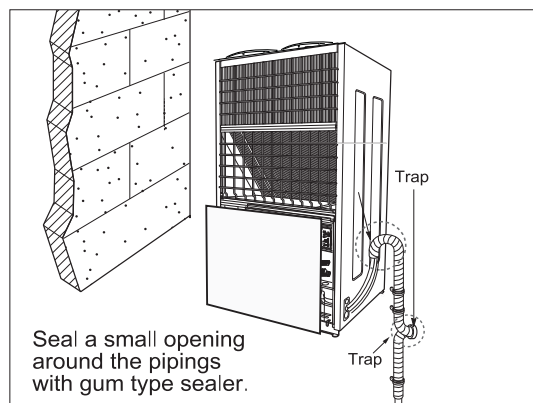
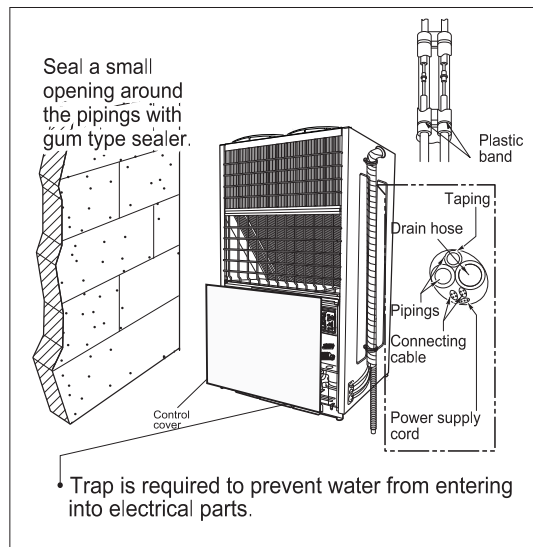
- If you want to connect an additional drain hose, the end of the drain outlet should be routed above the ground. Secure the drain hose appropriately.

In cases where the outdoor unit is installed below the indoor unit perform the following.

1. Tape the piping, drain hose and connecting cable from down to up.
2. Secure the tapped piping along the exterior wall using saddle or equivalent.

In cases where the Outdoor unit is installed above the Indoor unit perform the following.

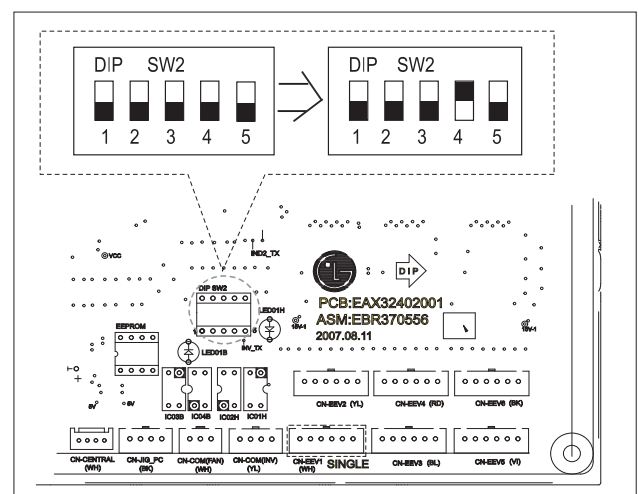
1. Tape the piping and connecting cable from down to up.
2. Secure the taped piping along the exterior wall. Form a trap to prevent water entering the room.
3. Fix the piping onto the wall by saddle or equivalent.



12.12 Long Pipe Setting(more than 15m)

1. Open the control cover of outdoor unit
2. Select 4th slide of the five selectable modes as follows.
3. Move the Slide switch to upside
4. Close the top cover and check whether the product works normally

WARNING: Do not open the control cover or Set the pipe length when operating the product.



12.13 Air Purging and Evacuation

Air and moisture remaining in the refrigerant system have undesirable effects as indicated below.

1. Pressure in the system rises.
2. Operating current rises.
3. Cooling(or heating) efficiency drops.
4. Moisture in the refrigerant circuit may freeze and block capillary tubing.
5. Water may lead to corrosion of parts in the refrigeration system.

Therefore, the indoor/outdoor unit and connecting tube must be checked for leak tight, and vacuumed to remove incondensable gas and moisture in the system.

Checking method

Preparation

- Check that each tube(both liquid and gas side tubes) between the indoor and outdoor units have been properly connected and all wiring for the test run has been completed. Remove the service valve caps from both the gas and the liquid side on the outdoor unit. Check that both the liquid and the gas side service valves on the outdoor unit are kept closed at this stage.

Leakage test

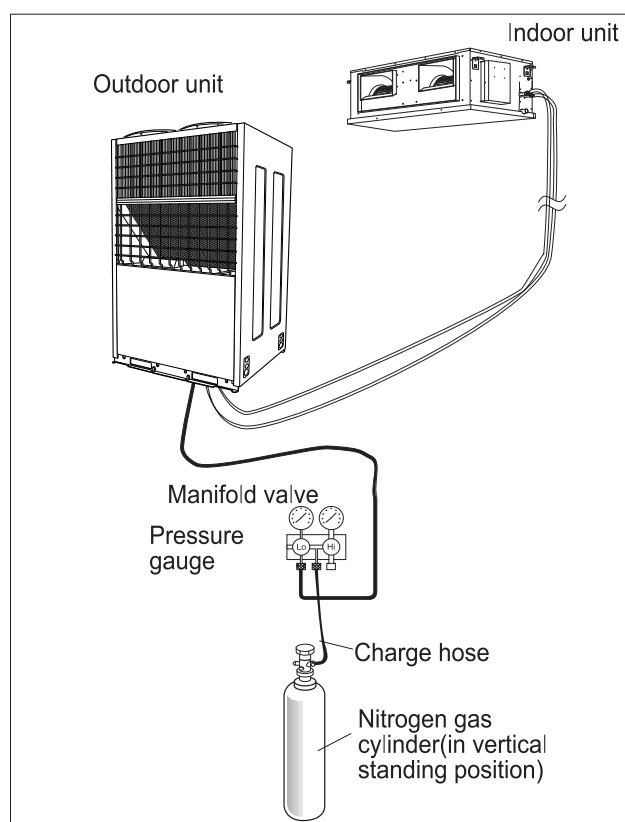
- Connect the manifold valve(with pressure gauges) and dry nitrogen gas cylinder to this service port with charge hoses.

CAUTION: Be sure to use a manifold valve for leakage test. If it is not available, use a stop valve for this purpose. The "Hi" knob of the manifold valve must always be kept close.

- Pressurize the system to no more than 2.94 MpaG. with dry nitrogen gas and close the cylinder valve when the gauge reading reached 2.94 MpaG. Next, test for leaks with liquid soap.

CAUTION: To avoid nitrogen entering the refrigerant system in a liquid state, the top of the cylinder must be higher than its bottom when you pressurize the system. Usually, the cylinder is used in a vertical standing position.

1. Do a leakage test of all joints of the tubing(both indoor and outdoor) and both gas and liquid side service valves. Bubbles indicate a leak. Be sure to wipe off the soap with a clean cloth.
2. After the system is found to be free of leaks, relieve the nitrogen pressure by loosening the charge hose connector at the nitrogen cylinder. When the system pressure is reduced to normal, disconnect the hose from the cylinder.



Evacuation

1. Connect the charge hose end described in the preceding steps to the vacuum pump to evacuate the tubing and indoor unit. Confirm the "Lo and Hi" knob of the manifold valve is open. Then, run the vacuum pump. The operation time for evacuation varies with tubing length and capacity of the pump. The following table shows the time required for evacuation.

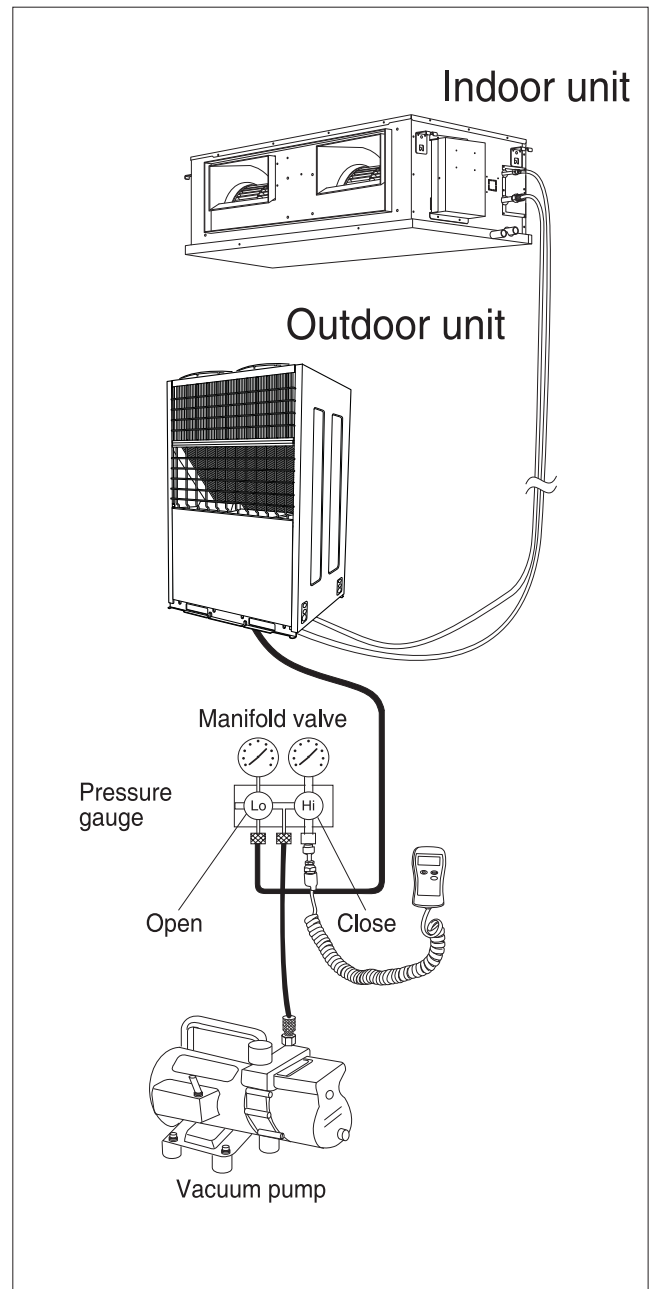
Required time for evacuation when 8.19 m ³ /min. vacuum pump is used	
If tubing length is less than 10 m	If tubing length is longer than 10 m
30 min. or more	1 hour. or more
66.66 Pa or less	

2. When the desired vacuum is reached, close the "Lo and Hi" knob of the manifold valve and stop the vacuum pump.

Finishing the job

1. With a service valve wrench, turn the valve stem of liquid side valve counter-clockwise to fully open the valve.
2. Turn the valve stem of gas side valve counter-clockwise to fully open the valve.
3. Loosen the charge hose connected to the gas side service port slightly to release the pressure, then remove the hose.
4. Replace the flare nut and its bonnet on the gas side service port and fasten the flare nut securely with an adjustable wrench. This process is very important to prevent leakage from the system.
5. Replace the valve caps at both gas and liquid side service valves and fasten them tight.

This completes air purging with a vacuum pump. The air conditioner is now ready to test run.



CAUTION

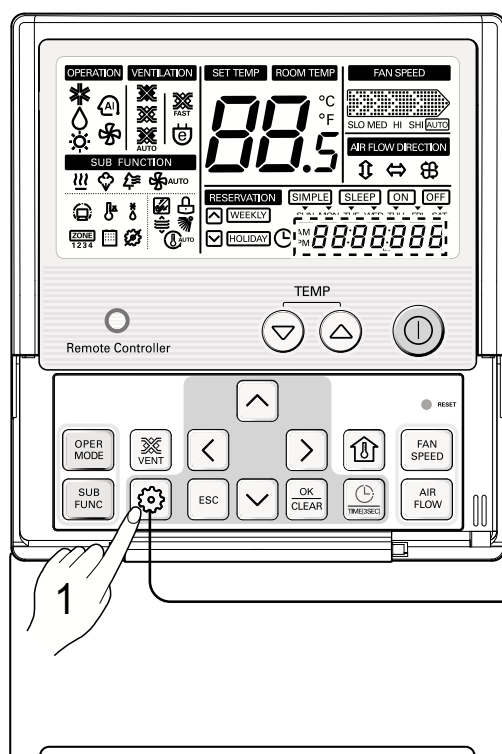
When installing the product, take the necessary measures for typhoon, strong wind such as wind blowing between buildings and earthquake.

12.14 Installer Setting -How to enter installer setting mode

CAUTION

Installer setting mode is to set the detail function of the remote controller.

If the installer setting mode is not set correctly, it can cause problems to the product, user injury or property damage. This must be set by an certificated installer, and any installation or change that is carried out by a non-certificated person should be responsible for the results. In this case, free service cannot be provided.



- 1 Press  button for 4 seconds to enter the installer setting mode. (When you enter the setting mode initially, Function code is displayed on the bottom of the LCD screen.)

0 1 0 1

Function Code Value

- 2 Repeat pressing  button, the menu will change with the corresponding function code.

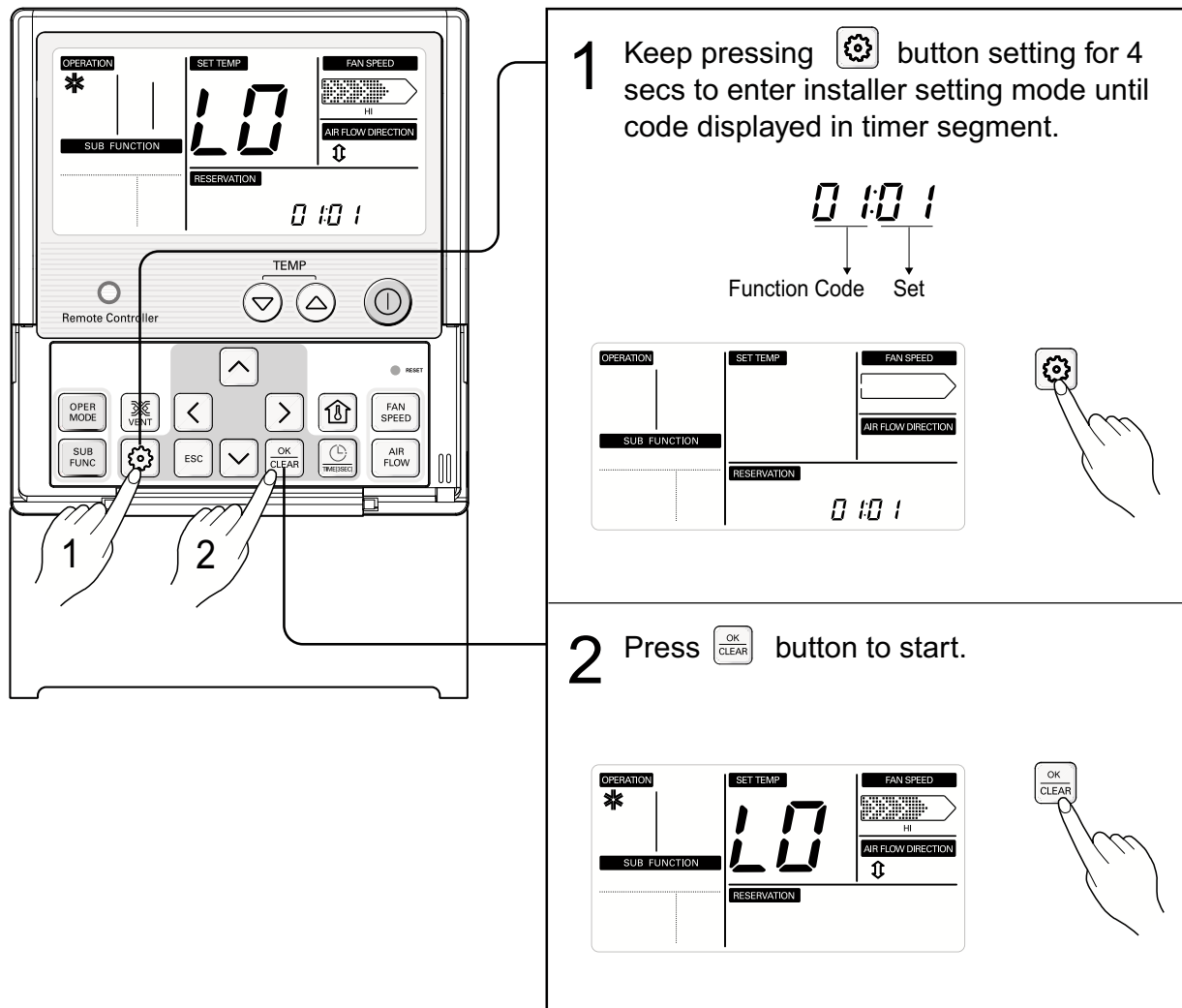
* Please refer to the following page for the function lists.

* Some contents may not be displayed depending on the product function.

12.15 Installer Setting -Test Run Mode

After installing the product, you must run a Test Run mode.

For details related to this operation, refer to the product manual.



1 Keep pressing  button setting for 4 secs to enter installer setting mode until code displayed in timer segment.

0 10 1

Function Code Set

2 Press  button to start.

0 10 1

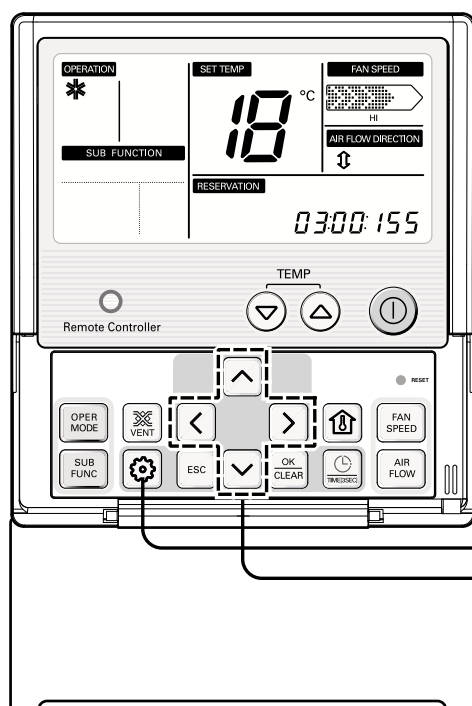
- * 18°C cooling, High Fan Speed, Airflow direction mode will be operated during 18 minutes with ignoring room temperature.
- * After running 18 minutes under test run mode, system will automatically turn OFF.
- * In case of duct type, the Airflow UP/DOWN function is not displayed.

12.16 E.S.P.(External Static Pressure) setting

How to Set E.S.P?

⚠ CAUTION

If you set ESP incorrectly, the air conditioner may malfunction.
This setting must be carried out by a certificated-technician.
This function is used for only Duct product.



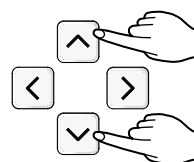
1 Press button for 4 seconds to enter the installer setting mode until timer segment display "01:01".

2 Repeat pressing button to select Function code 03.

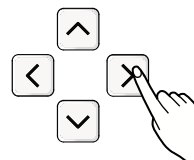
03:00:155

Function Code ESP step ESP value

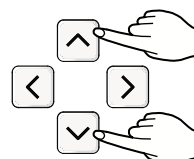
3 Set ESP step by pressing button. (01: very low, 02: low, 03: medium, 04: high. 05: power)

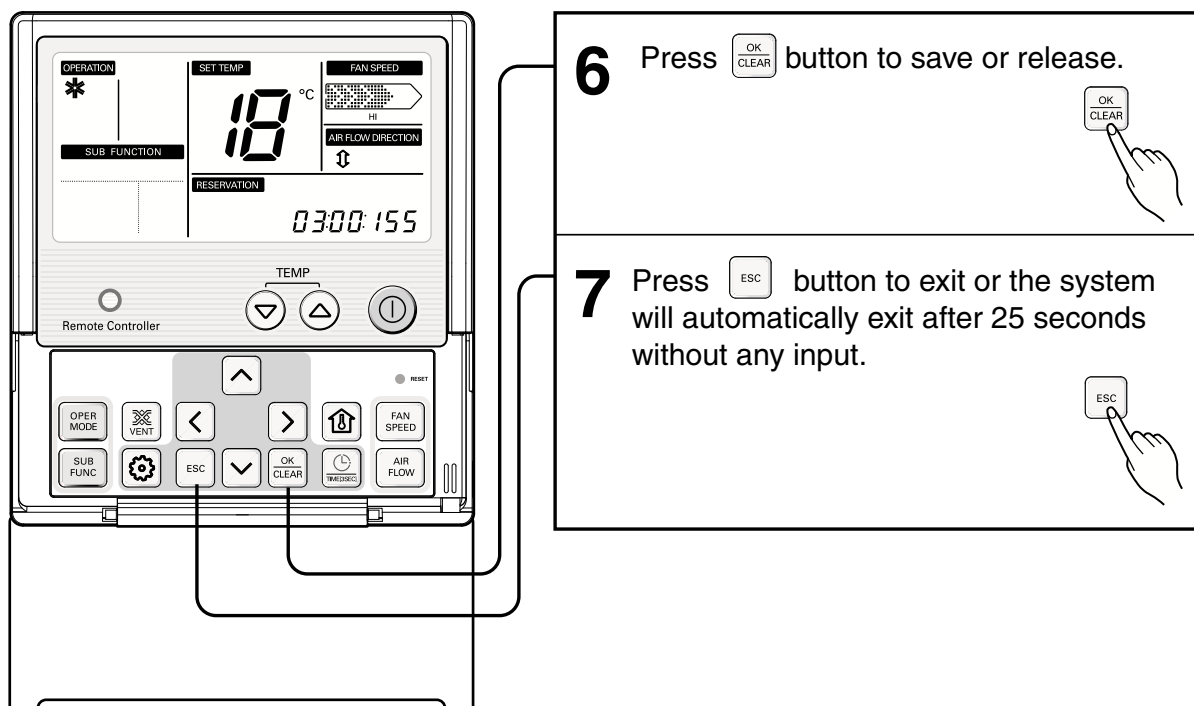


4 Move to ESP setting by pressing button.



5 Press button to select ESP value(0~255).





- * Weak and Power setting is not available for some products.
- * Because the ESP value is already appropriately set when manufactured from the factory, it is recommended that you do not change the ESP value.

12.17. Static Pressure & Air Flow Rate

Static Pressure(Pa)			0	20	40	60	80	100	120	140	160	180	200
MODEL	Step	I/s	Setting Value										
CS-L72JD1R5	HI	1 015	57	62	68	72	77	82	87	91	94	98	103
	Med	972	55	60	66	71	76	81	86	88	93	97	102
	Low	791	44	52	58	65	70	74	81	85	88	93	97

Static Pressure(Pa)		Setting Value	0	50	100	150	200
MODEL	Step		I/s				
CS-L72JD1R5	HI	92	1720.84	1614.84	1420.5	1015.17	789.502
	Med	89	1691.17	1547.67	1364.84	972.002	515.168
	Low	86	1623	1475.5	1260.67	791.168	512.001

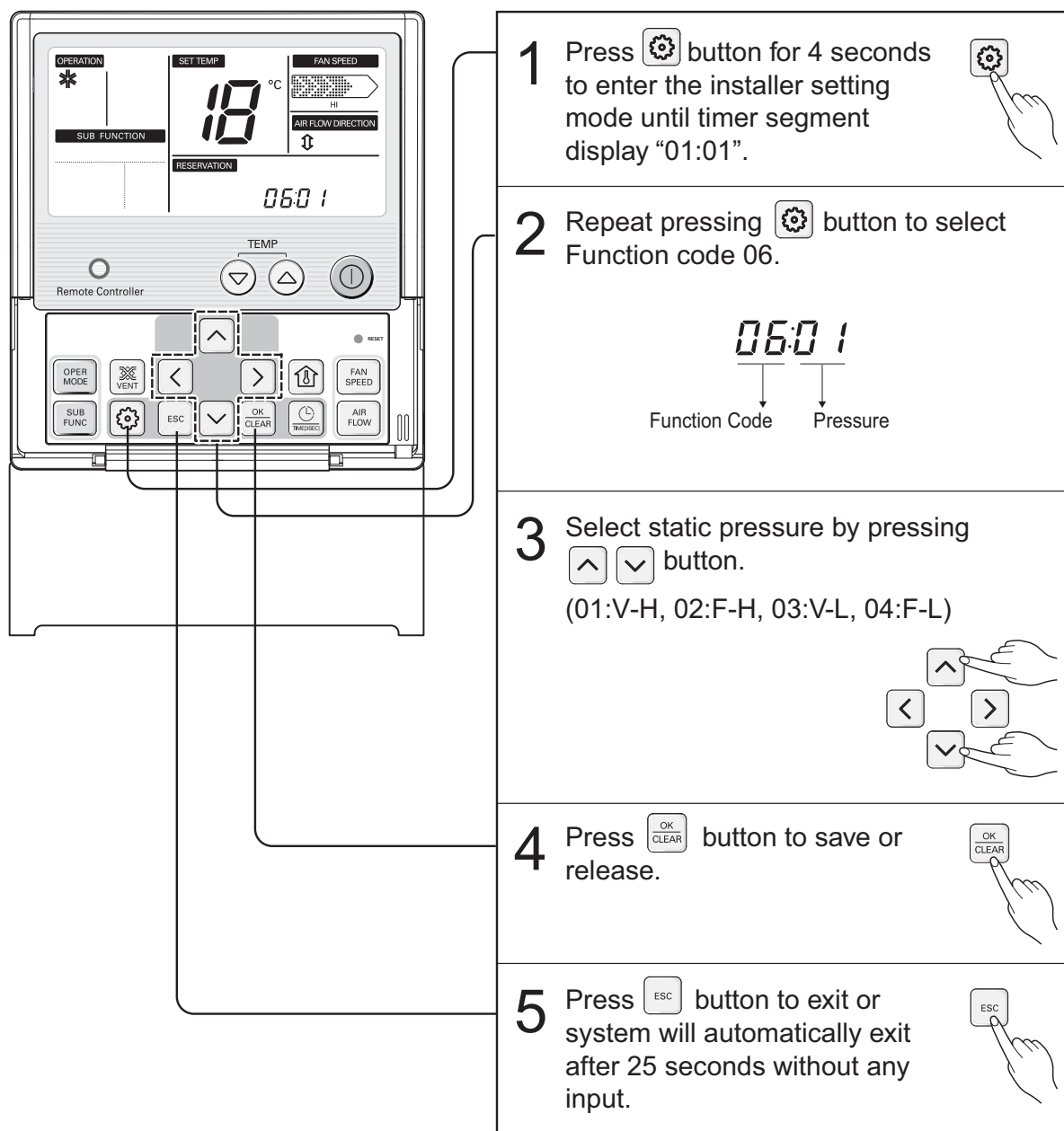
* The above table shows the correlation of External Static Pressure & AIR FLOW.

NOTICE Capacities are tested in the following conditions.

Static Pressure (Pa)	Voltage (Phase/Volt)	Air Flow Step	Piping Length	Temperaure(°C DB/°CWB)			
				Cooling		Heating	
				Indoor	Outdoor	Indoor	Outdoor
130	Ø1/230 V	High	7.5 m	27/19	35/24	20/15	7/6

12.18. Installer Setting -Static Pressure Setting

This function is applied to only duct type. Setting this in other cases will cause malfunction.



- Press  button for 4 seconds to enter the installer setting mode until timer segment display "01:01".
- Repeat pressing  button to select Function code 06.

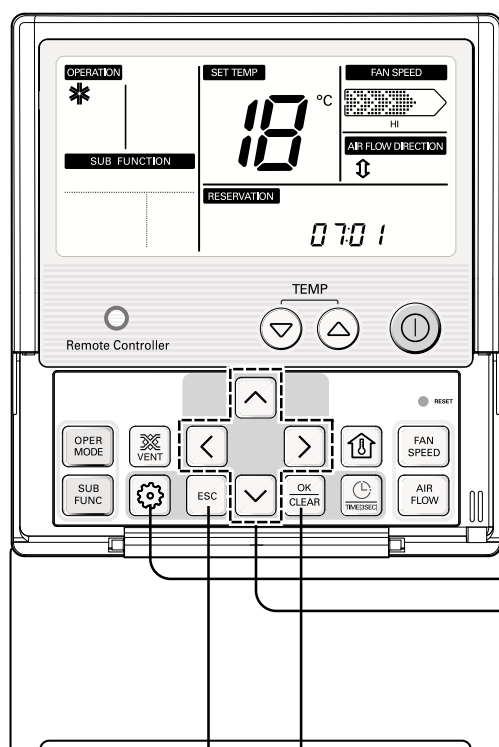

 Function Code Pressure
- Select static pressure by pressing   button.
(01:V-H, 02:F-H, 03:V-L, 04:F-L)
- Press  button to save or release.
- Press  button to exit or system will automatically exit after 25 seconds without any input.

<Static Pressure Setting Table>

Pressure selection		Function	
		Zone state	ESP standard value
01	V-H	Variable	High
02	F-H	Fixed	High
03	V-L	Variable	Low
04	F-L	Fixed	Low

12.19. Installer Setting-Group Setting

This function is only for group control operation. Please don't set this function in case of non-group control.



1 Press button for 4 seconds to enter the installer setting mode until timer segment display "01:01".



2 Repeat pressing button to select Function code 07.

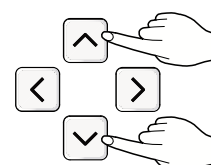
07:01

Function Code Master/Slave value

Ex) 'Master' Setting

3 Select Master/ Slave by pressing button.
(00: Slave, 01: Master)

07:01



4 Press button to save or release.

07:01



5 Press button to exit or system will automatically exit after 25 seconds without any input.



12.20. Continuous Fan Operation

When need to set up the continuous fan operation, chage the option switch 5 in the PCB as below guide line.

ON

12345678

Option Switch



Indoor PCB

Option Switch 5	Function
ON	Continuous Fan Operation - ON
OFF	Continuous Fan Operation - OFF

CAUTION : Continuous fan operation can cause unexpected cold air flow in hot start mode or defrost mode.

It is required to explain to the customer before setting this function.

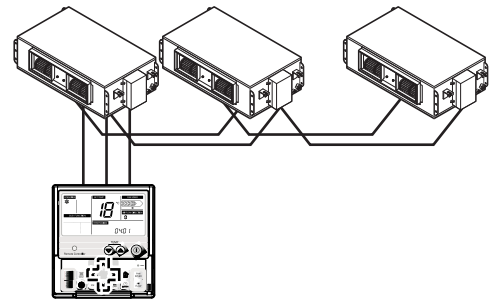
12.21. Optional Operation

12.21.1 Operation Summary

- Where several products are linked, one specific control device can control a specific number of products.

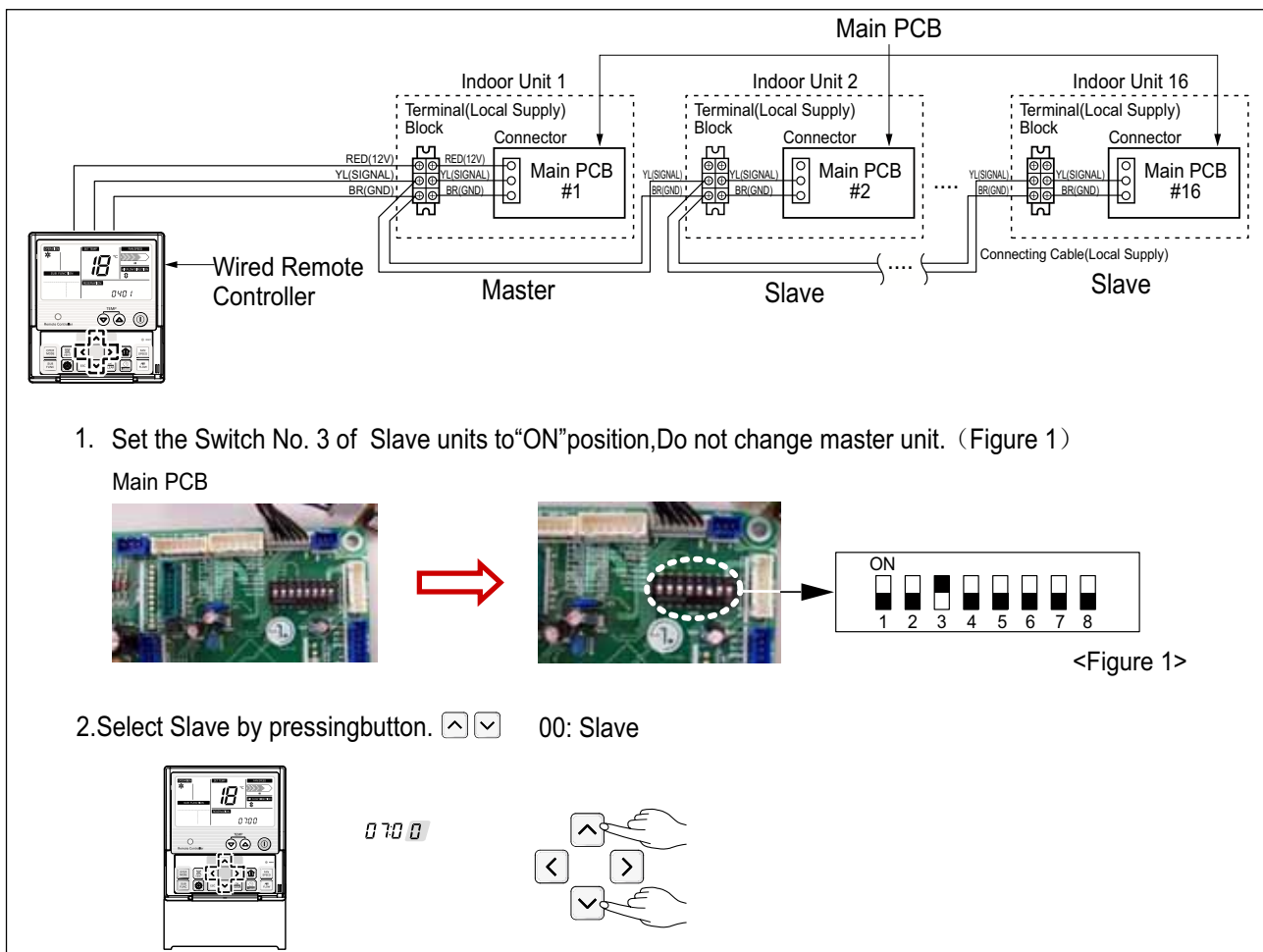
12.21.2 Specific Operation

- Connecting line is linked to each of the indoor equipments for communication.
A specific control device is connected to each of them and this control device can control the same function.
- Group control function is enabled by cutting an optional option switch in the wired remote control. At this time, the main system will not respond in order to prevent data collision.
- While executing group control command, use the random data(0-3minutes) in the main body of indoor equipment for limiting starting current.
- Control device can control up to 16 indoor equipments.



12.21.3 Group Control(Optional Wiring)

It operates maximum 16 Units by only one Wired Remote Controller, and each Unit starts sequentially to prevent overcurrent.



CAUTION

- Be careful not to exchange the color of wires.
- The maximum length of connecting wire should be below 200m(25Ω) on connecting each units.
- Use a wire more than 0.5mm²

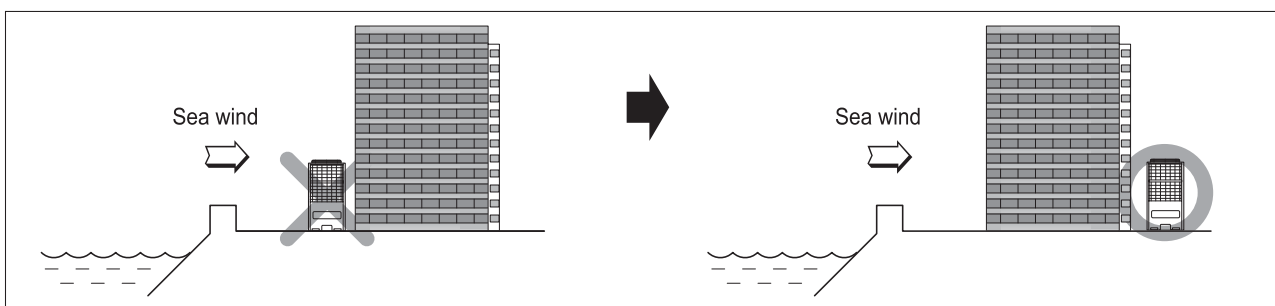
12.22 Installation Guide at the Seaside

CAUTION:

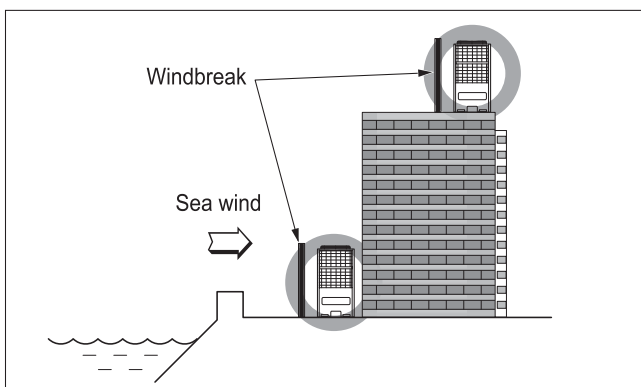
1. Air conditioners should not be installed in areas where corrosive gases, such as acid or alkaline gas, are produced.
2. Do not install the product where it could be exposed to sea wind (salty wind) directly. It can result corrosion on the product. Corrosion, particularly on the condenser and evaporator fins, could cause product malfunction or inefficient performance.
3. If outdoor unit is installed close to the seaside, it should avoid direct exposure to the sea wind. Otherwise it needs additional anticorrosion treatment on the heat exchanger.

Selecting the location(Outdoor Unit)

- 1) If the outdoor unit is to be installed close to the seaside, direct exposure to the sea wind should be avoided. Install the outdoor unit on the opposite side of the sea wind direction.



- 2) In case, to install the outdoor unit on the seaside, set up a windbreak not to be exposed to the sea wind.



- It should be strong enough like concrete to prevent the sea wind from the sea.
- The height and width should be more than 150% of the outdoor unit.
- It should be keep more than 70 cm of space between outdoor unit and the windbreak for easy air flow.

- 3) Select a well-drained place.

1. If you can't meet above guide line in the seaside installation, please contact Panasonic for the additional anticorrosion treatment.
2. Periodic (more than once/year) cleaning of the dust or salt particles stuck on the heat exchanger by using water

13 Control logic

13.1 Hot start

- When heating is started, the indoor fan is stopped or very slow to prevent the cold air carry out
- When the temp. of heat exchanger reach 22°C(model by model), indoor fan is started.

13.2 Self-diagnosis Function

- The air conditioner installed can self-diagnosed its error status and then transmits the result to the central control. Therefore, a rapid countermeasure against failure of the air conditioner allows easy management and increases the usage life of air conditioner.
- Refer to trouble shooting guide.

13.3 Soft dry operation

- When the dehumidification operation input by the remote control is received, the intake air temperature is detected and the setting temp is automatically set according to the intake air temperature.

Intake air Temp.	Setting Temp.
26°C ≤ intake air temp.	25°C
24°C ≤ intake air temp. < 26°C	intake air temp. -1°C
22°C ≤ intake air temp. < 24°C	intake air temp. -0.5°C
18°C ≤ intake air temp. < 22°C	intake air temp.
intake air temp. < 18°C	18°C

- While compressor off, the indoor fan repeats low airflow speed and stop.
- While the intake air temp is between compressor on temp. and compressor off temp., 10-min dehumidification operation and 4-min compressor off repeat.

Compressor ON Temp. → Setting Temp+0.5°C

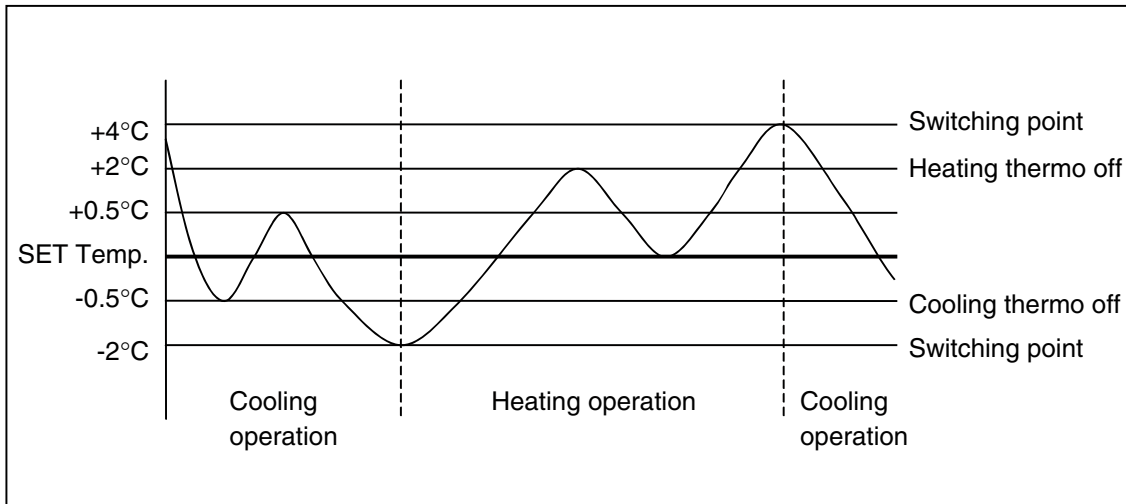
Compressor OFF Temp. → Setting Temp-0.5°C

- The indoor fan speed is automatically set to the low, so the shift of the indoor fan speed is impossible because of already being set to the best speed for Dry Operation by Micom Control.

13.4 Convenience Functions & Controls

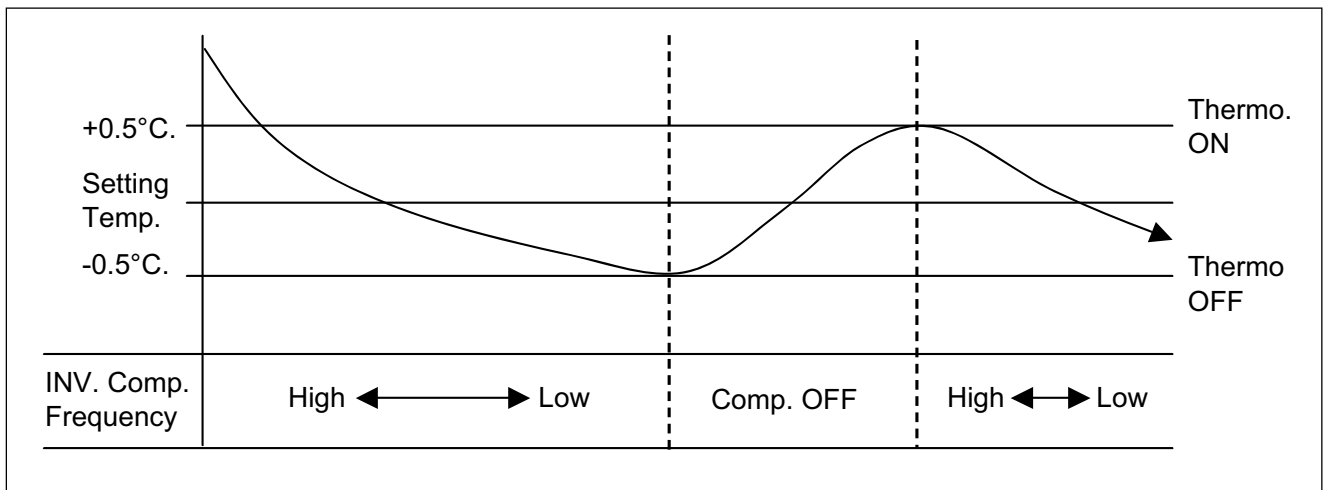
13.4.1 Auto changeover operation

- The air conditioner changes the operation mode automatically to keep indoor temperature.
- When room temperature vary over $\pm 2^{\circ}\text{C}$ with respect to setting temperature, air conditioner keeps the room temperature in $\pm 2^{\circ}\text{C}$ with respect to setting temperature by auto change mode.



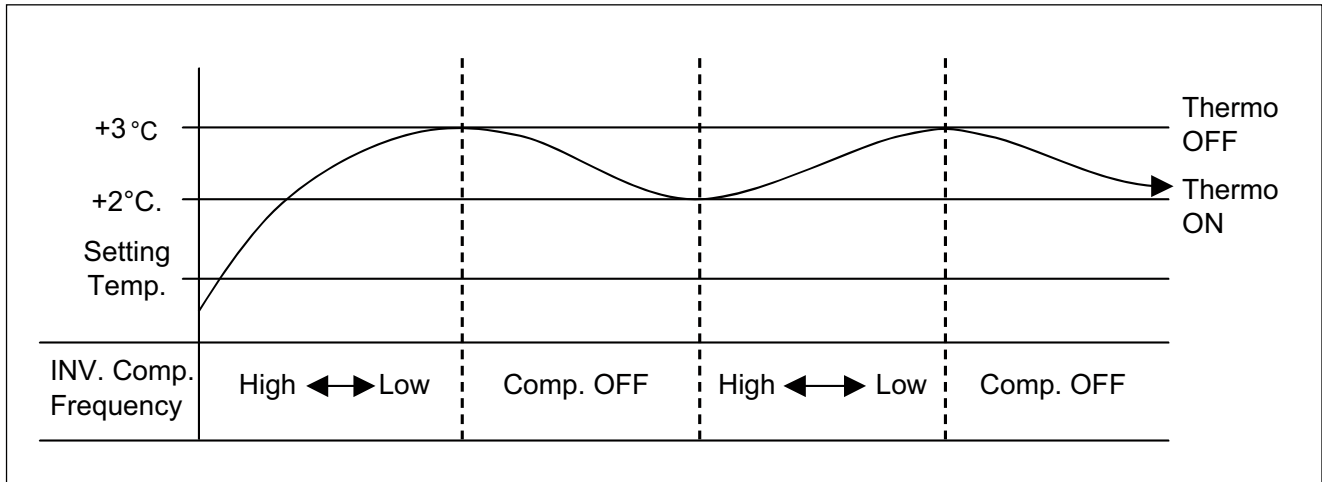
■ Cooling & heating Operation

1 Cooling Mode



- Operating frequency of compressor depends on the load condition, like the difference between the room temp. and the set temp., frequency restrictions.
- If the compressor operates at some frequency, the operating frequency of compressor cannot be changed within 30 seconds. (not emergency conditions)
- Compressor turned off when
 - intake air temperature is in between $\pm 0.5^{\circ}\text{C}$ of the setting temp. limit for three minutes continuously.
 - intake air temperature reaches below 1.0°C of the temperature of setting temp..
- Compressors 3 minutes time delay.
 - After compressor off, the compressor can restart minimum 3 minutes later.
 - Compressor operating range is form 40% to 110% of rated condition.

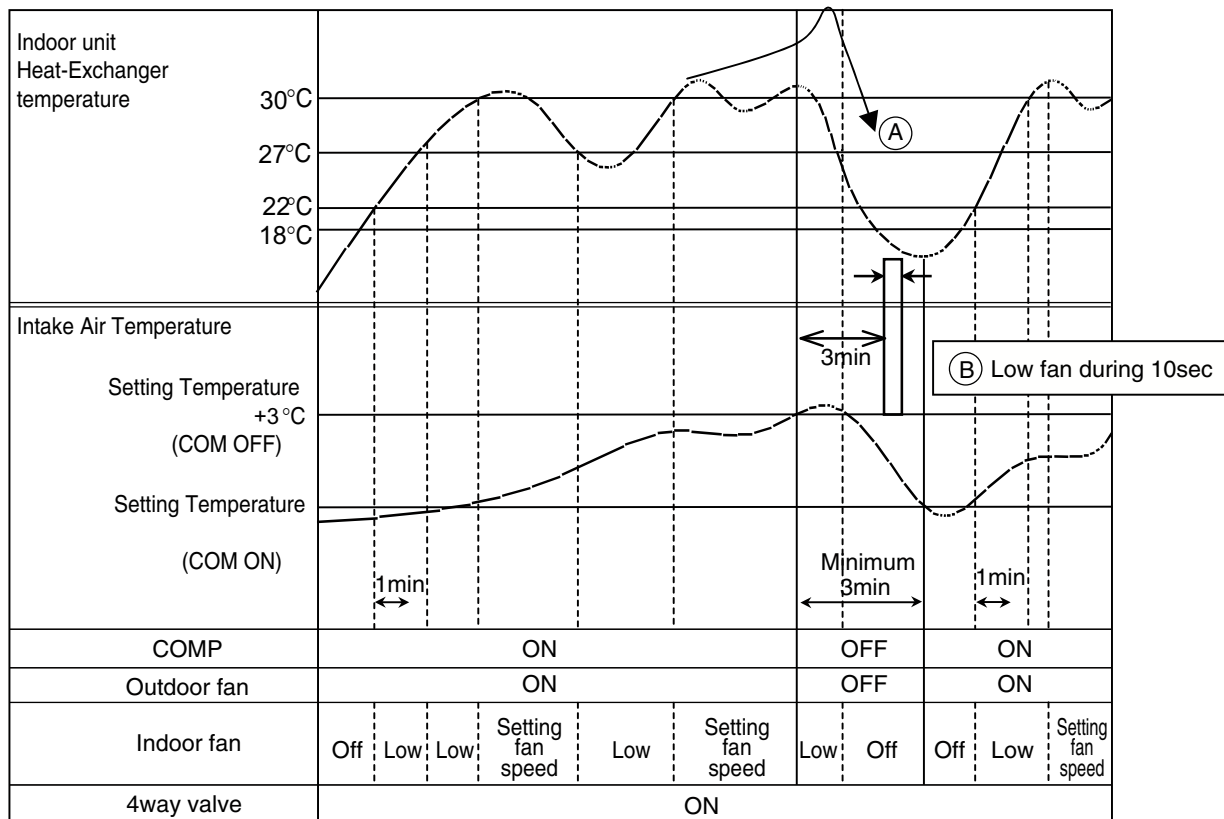
2 Heating Mode



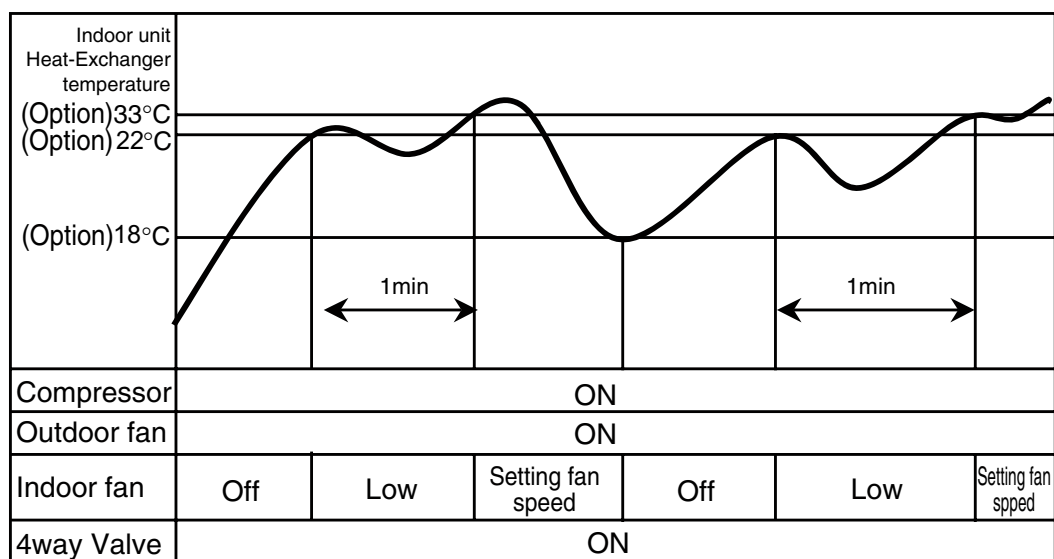
- Operating frequency of compressor depend on the load condition, The difference between the room temp. and set temp., frequency restrictions.
- If compressor operates at some frequency, the operating frequency of compressor cannot be changed within 30 seconds.
- Condition of compressor turned off
 - When intake air temperature reaches +3°C above the setting temperature.
- Condition of compressor turned on
 - When intake air temperature reaches +2°C above the setting temperature.
- * Condition of indoor fan turned off
 - While in compressor on : indoor pipe temp. < 18°C
 - While in compressor off : indoor pipe temp. < 40°C
- While in defrost control, between the indoor and outdoor fans are turned off.
- Compressor 3 minutes delay
 - After compressor off, the compressor can restart minimum 3 minutes later.

■ Heating Mode Operation Details

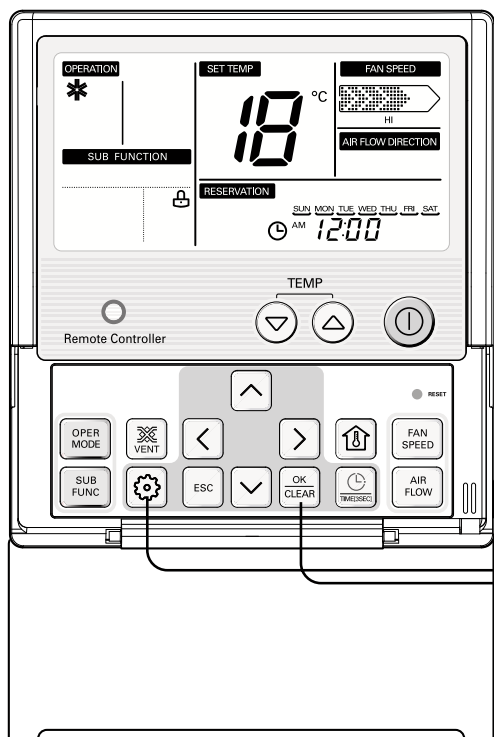
The unit will operate according to the setting by the remote controller and the operation diagram is shown as following.



- **Compressor-off interval** : - (A) While the indoor Heat-Exchanger temperature is higher than 27°C, fan operates at low speed, when it becomes lower than 26°C fan stops.
 - (B) For eliminating latent heat-loss, fan operates at low speed for 10 seconds periodically.



13.4.2 Child Lock Function



This function is to prevent children or other people from using indoor.

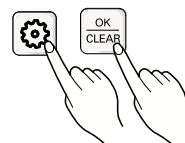
- Press  button.
- Repeat pressing  button until the  icon flash.



- When the button flashing, press  button to set this function.
- When setting this function,  icon will be displayed whenever entering any operation except for Room temperature access.

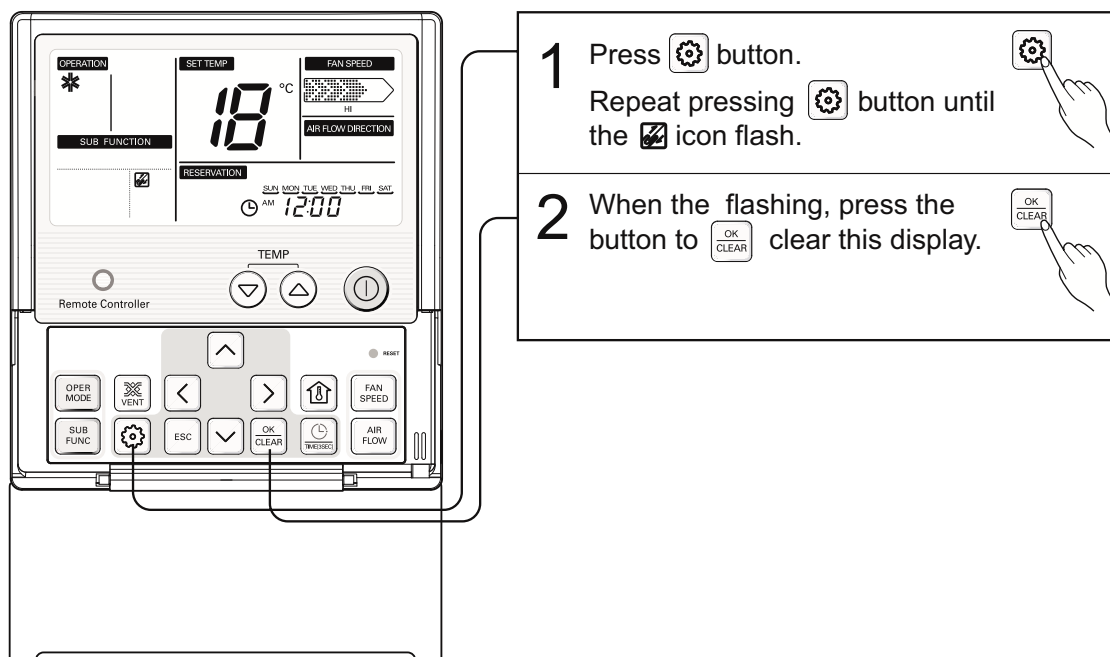


To release this setting, keep pressing  button until  icon flash and then enter  button.



13.5. Function setting : Filter Sign Clear

This function is to clear the indicator of indoor filter cleaning.



When filter is covered by dust, it will reduce cooling/ heating efficiency and accumulate more electric power. Therefore, do clean the filter whenever cleaning time is expired.

13.6 Sleep Timer Operation

- When the sleep time is reached after <1,2,3,4,5,6,7,0(cancel) hr> is input by the remote control while in appliance operation, the operation of the appliance stops.
- While the appliance is on pause, the sleep timer mode cannot be input.
- While in cooling mode operation, 30 min later since the start of the sleep timer, the setting temperature increases by 1°C. After another 30 min elapse, it increases by 1°C again.
- When the sleep timer mode is input while in cooling cycle mode, the airflow speed of the indoor fan is set to the low.
- When the sleep timer mode is input while in heating cycle mode, the airflow speed of the indoor fan is set to the medium.

13.7 Timer(On/Off)

1 On-Timer Operation

- When the set time is reached after the time is input by the remote control, the appliance starts to operate.
- The timer LED is on when the on-timer is input. It is off when the time set by the timer is reached.
- If the appliance is operating at the time set by the timer, the operation continues.
While in Fuzzy operation, the airflow speed of the indoor fan is automatically selected according to the temperature.

2 Off-Timer Operation

- When the set time is reached after the time is input by the remote control, the appliance stops operating.
- The timer LED is on when the off-timer is input. It is off when the time set by the timer is reached.
- If the appliance is on pause at the time set by the timer, the pause continues.

13.8 Weekly Program

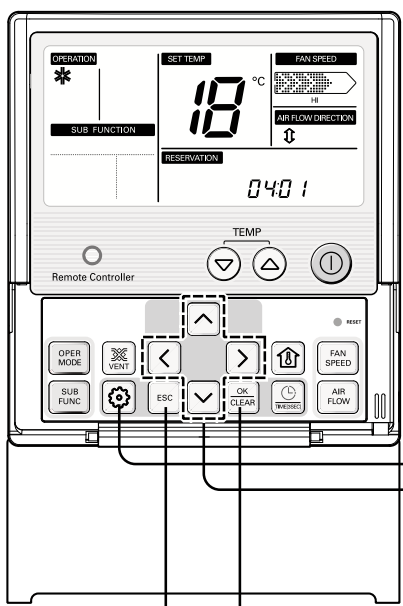
- If necessary, an operator can make an On/Off reservation of the product for a period of one week.
- On/Off schedule of operation for a period of One week.
- No need to turn the unit On/Off manually during working days.
On/Off time is scheduled in micom of the wired remote control.

Operation Time Table (Example)

Setting	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Temp.	25°C	25°C	25°C	25°C	25°C	Off	
On	09:00	08:00	09:00	08:00	09:00		
Off	12:00	17:00	12:00	12:00	12:00		

13.9 Two Thermistor Control

There may be a significant difference between temperature taken at the installed product indoor temperature. Two thermistor control provides option to control temperature by referring any of the two temperatures.



- 1 Press  button for 4 seconds to enter the installer setting mode until timer segment display "01:01".
- 2 Repeat pressing  button to select Function code 04.


 Function Code Thermistor setting

Ex) Setting Thermistor as '2TH'.
- 3 Set Thermistor mode by pressing   button. (01: Remote Controller, 02: Indoor, 03: 2TH)


- 4 Press  button to save or release.

<Thermistor Table>

- 5 Press  button to exit or system will automatically exit after 25 seconds without any input.

※ Therefore system will use value that sensed from indoor unit or remote controller

Temperature sensor location		Function
01	Indoor unit	Operation in indoor unit temperature sensor
02	Remote controller	Operation in remote controller Temperature sensor
03	2-Thermistor(2TH)	Operation in lower temperature after comparing the temperature between the indoor unit and remote controller

CAUTION

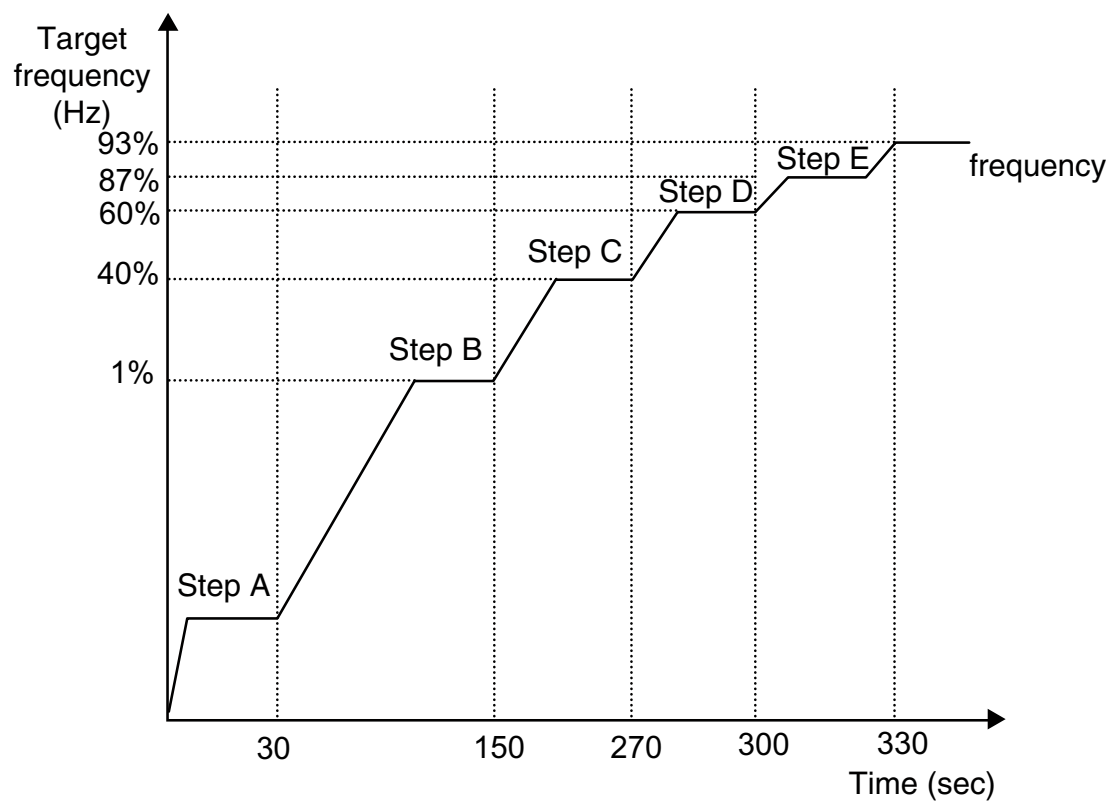
- Select the position after counselling with a customer.
- In case of cooling mode, room temperature is controlled by the main body sensor.
- To control the room temperature by a wired remote controller, install controller(room temp. sensor) to sense the temperature more accurately.
- Manufactured in the position 3.

13.10 Compressor

Inverter compressor starting

: Target Step calculation method

- Step A = Step 1 Frequency (Min. frequency : 20~25Hz)
- Step B = (Step A + Target STEP) / 2
- Step C = (Step B + Target STEP) / 2
- Step D = (Step C + Target STEP) / 2
- Step E = (Step D + Target STEP) / 2



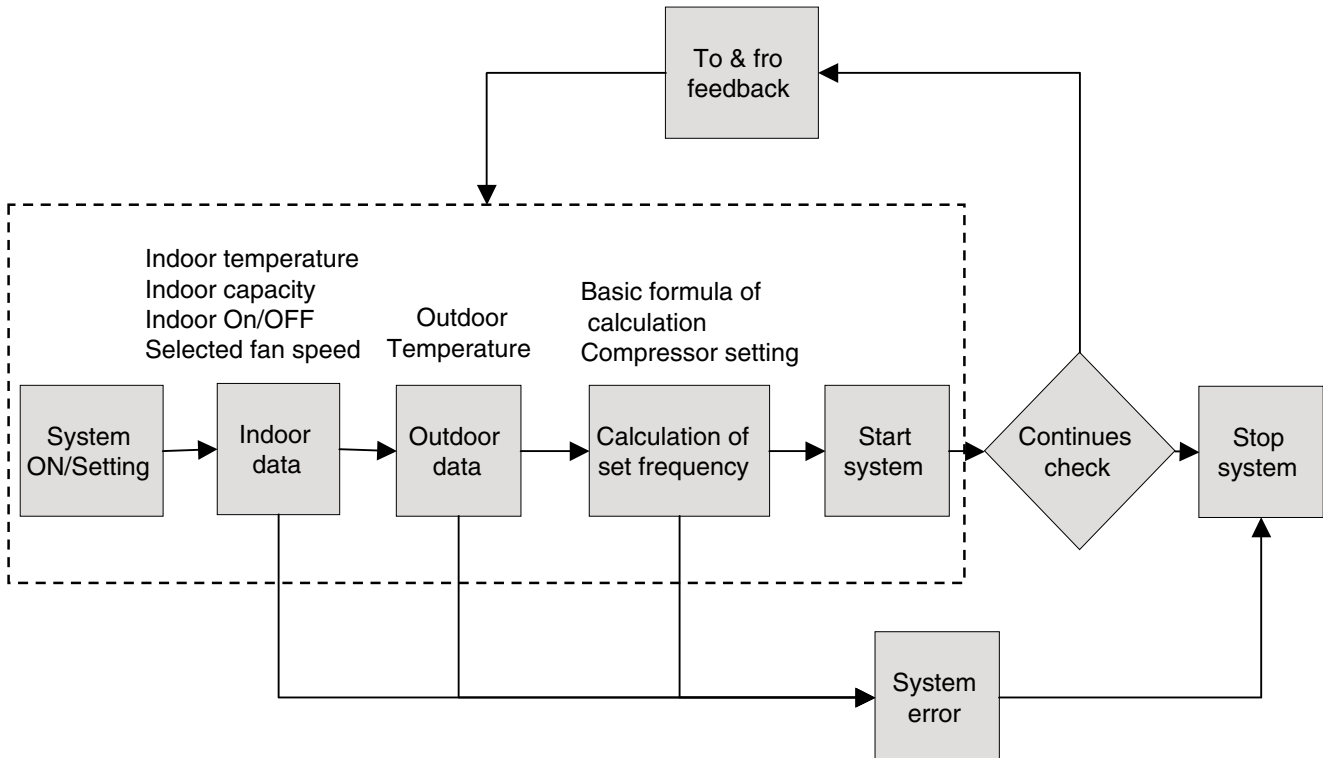
13.11 Step(frequency) control

13.11.1 Frequency control

Frequency that corresponds to each rooms capacity will be determined according to the difference in the temperature of each room and the temperature set by the remote controller.

There are various factors determining the frequency.

1. Indoor unit capacity value.
2. Temperature compensation factor
3. Initial frequency setting



13.11.2 Primary step setting

: Capacity steps of compressor are decided by ΣQ_j (Summation of capacity code), TAO(Outdoor temp.), TAI(Indoor temp.), DTAI (Step Compensation of temperature difference Indoor Temp. and Setting Temp.

$$\text{Comp Step} = (\text{Step base} + \text{Long piping compensation}) \times \Delta\text{Step TAO} \times \Delta\text{Step TAI} \times \Delta\text{Step DTAI}$$

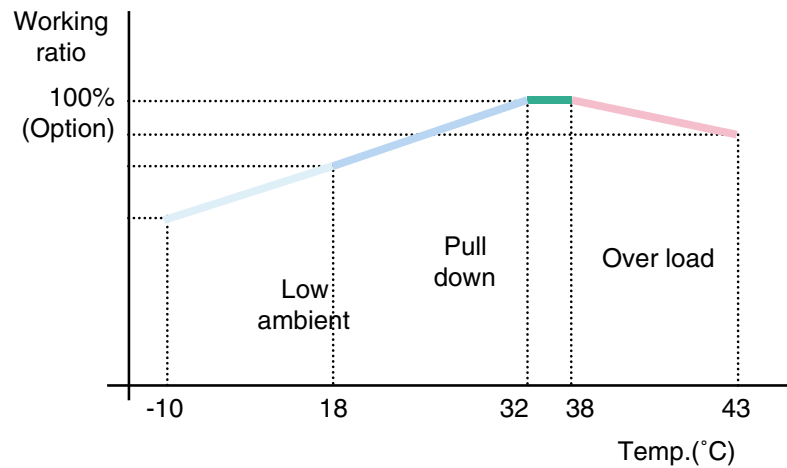
Step base	Standard frequency step by ΣQ_j (Summation of capacity code)
Long piping	Comp. Step compensation by setting long piping
$\Delta\text{Step TAO}$	step compensation by TAO (Outdoor temp.)
$\Delta\text{Step TAI}$	step compensation by TAI (Indoor temp.)
$\Delta\text{Step DTAI}$	Step Compensation of temperature difference Indoor Temp. and Setting Temp.

* Target frequency step (Step base) exceeds maximum step, the Step base value follows the maximum step value.

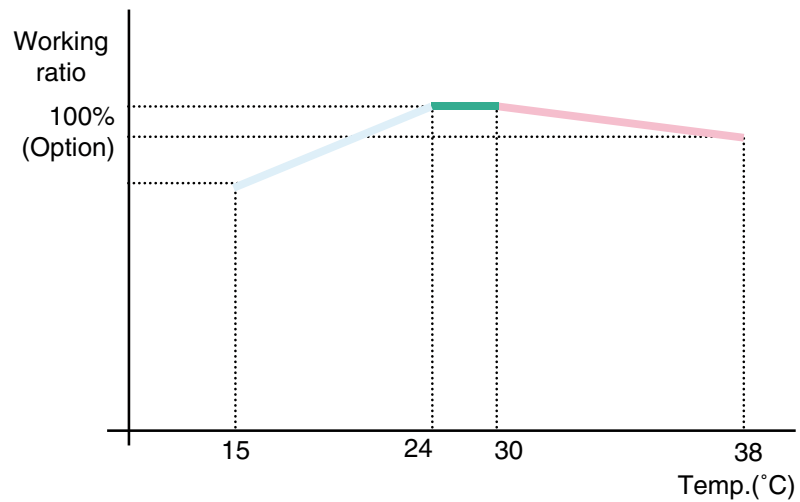
* The compressor get the minimum step in case Step base value is lower than the minimum step of operating capacity.

13.11.3 Setting value / Cooling

△Step TAO : step compensation by Outdoor temp.



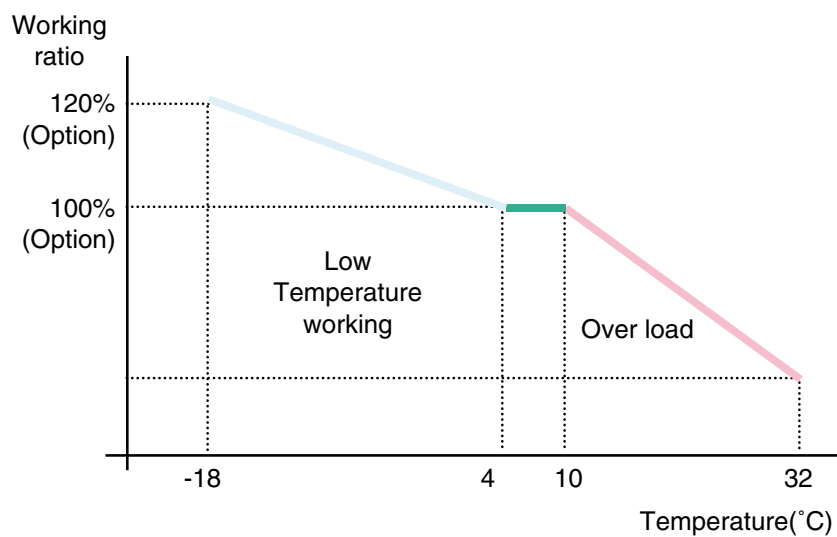
△Step TAI: step compensation by Indoor temp.



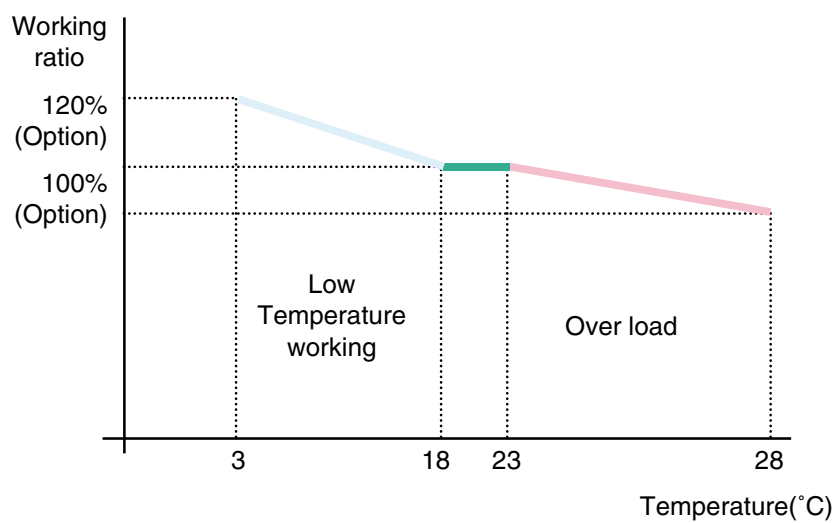
△Step DTAI : Step Compensation of temperature difference between indoor temp. and setting temp. (40-110%)

13.11.4 Setting value / Heating

△Step TAO : step compensation by Outdoor temp.



△Step TAI: step compensation by Indoor temp.



△Step DTAI : Step Compensation of temperature difference between indoor temp. and setting temp (40-110%).

13.12 Reversing valve operation

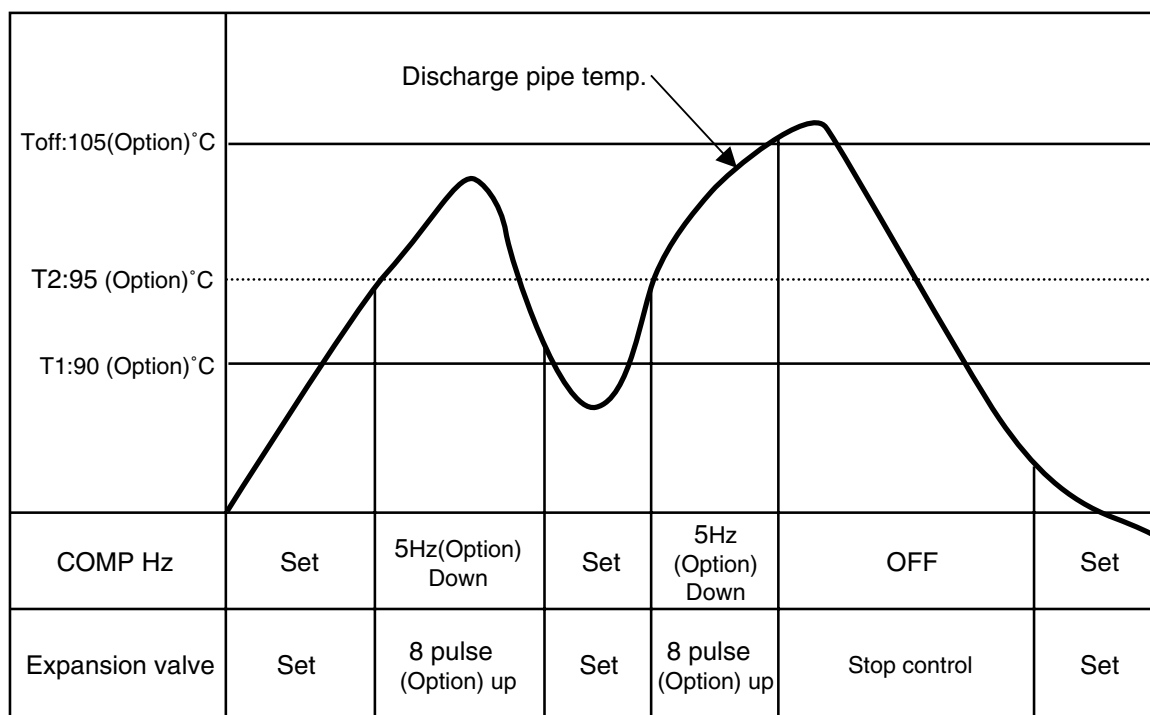
1. At the starting (outdoor is powered on, indoor is not) reversing valve continues OFF (cooling).
2. For the cooling and defrosting operation :valve OFF, for the heating operation :valve ON
3. Method of changing mode from heating to defrosting : As defrosting starts Inverter compressor Hz is lowered to 30Hz for 5 sec and the valve is OFF for the defrost mode. (refer “working process of each component in defrosting and returning to heating mode ”)
- 4 . Method of changing mode from defrosting to heating : As the defrosting is finished inverter compressor frequency is lowered to 30Hz for 10 sec. And the valve is ON for the heating mode.
5. If the operating mode is changed to heating from cooling, “3 min. restarting rule” is applied, and reversing valve position is changed within 30 sec. after compressor turns OFF.
6. If the compressor is stopped during heating mode by remote controller operation or error mode, reversing valve position is changed to OFF in 30 sec. after compressor turns OFF.
7. If the compressor is stopped during heating mode by Thermistor signal, reversing valve will remain in heating position.

13.13 Discharge pipe control

- 1) There can be two situations
 - a) Sensor is failed (error code for sensor failure will be generated)
 - b) Abnormal high temperature discharge temperature (error code for high discharge will be generated)
 Both cases unit will stop.

Compressor working

1. If discharge pipe temperature $\geq T1$ No limitation on compressor frequency
2. $T2 \leq$ discharge pipe temperature $\leq Toff$ (Hysteresis control) Compressor frequency down by 5 pulse & Expansion valve up by 10 pulse in every 1 min. If LEV is in the starting control it will follow starting control first.
3. discharge pipe temperature $\geq Toff$ then compressor will be OFF System will stop if this situation occurs 5 times in 1 hour and error code will be generated also self diagnosis will start.



13.14. Input Current Control

Function

Controlling total current to protect power semiconductor devices from burn-out by the low current (including connecting mistake) and over current.

Operating process

Detection : check the output DC voltage of Current Transformer(CT).

13.15 Defrost Control

13.15.1 Function

:These are about the control of compressor, fan of outdoor unit, reversing valve, LEV.

13.15.2 Starting to the defrosting operation

A) Defrost operation will be start when all the conditions below are matched simultaneously

Accumulation time of operation and the period after completion of defrost = 45min (Outdoor air temperature -3°C)

B) Outdoor piping temperature is below than -6 (Option) $^{\circ}\text{C}$ for starting defrosting operation.

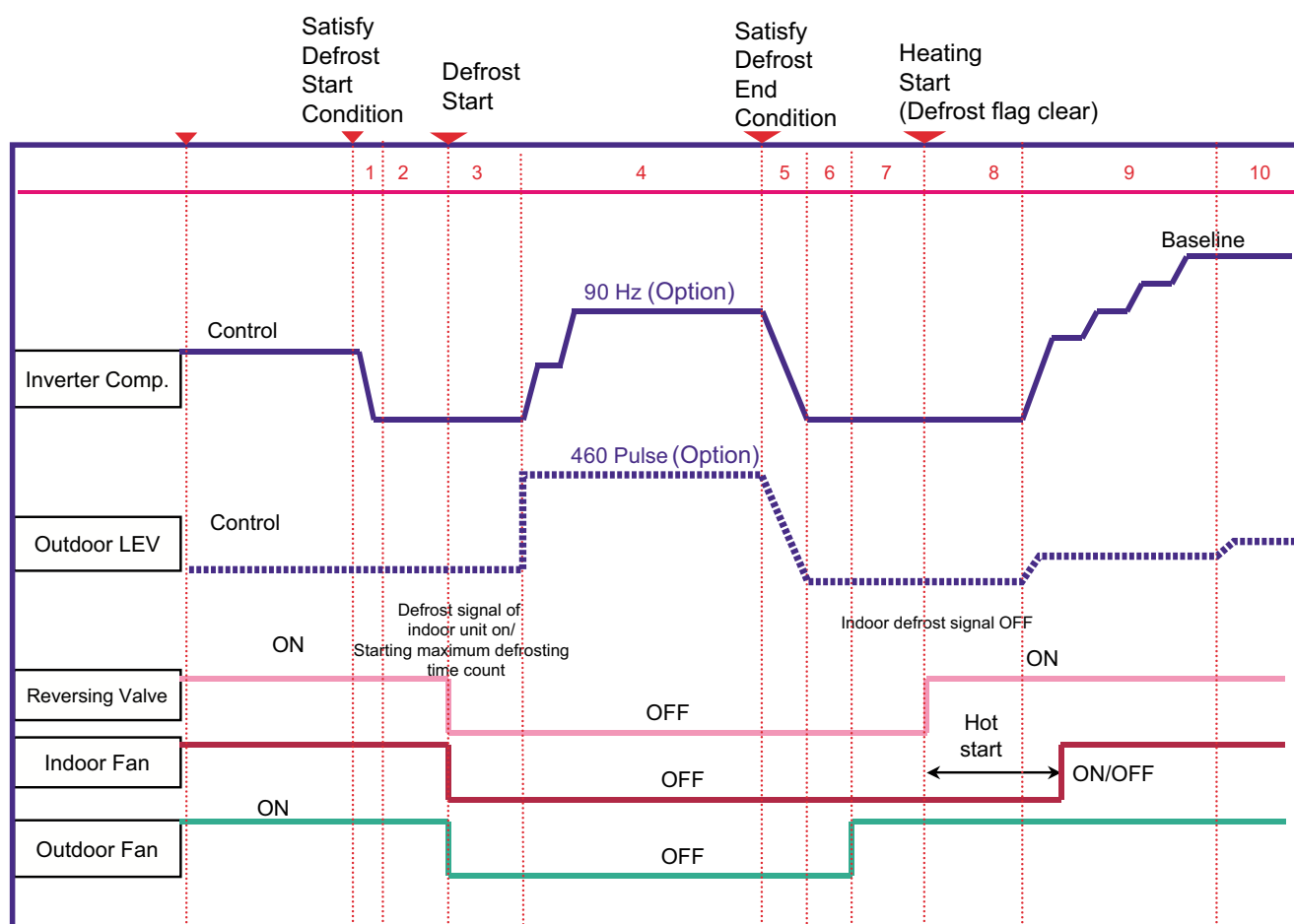
13.15.3 Completion of defrost operation

Send signal of defrost completion in case of meeting one of the condition as below.

1. Defrosting time ≈ 11 minutes

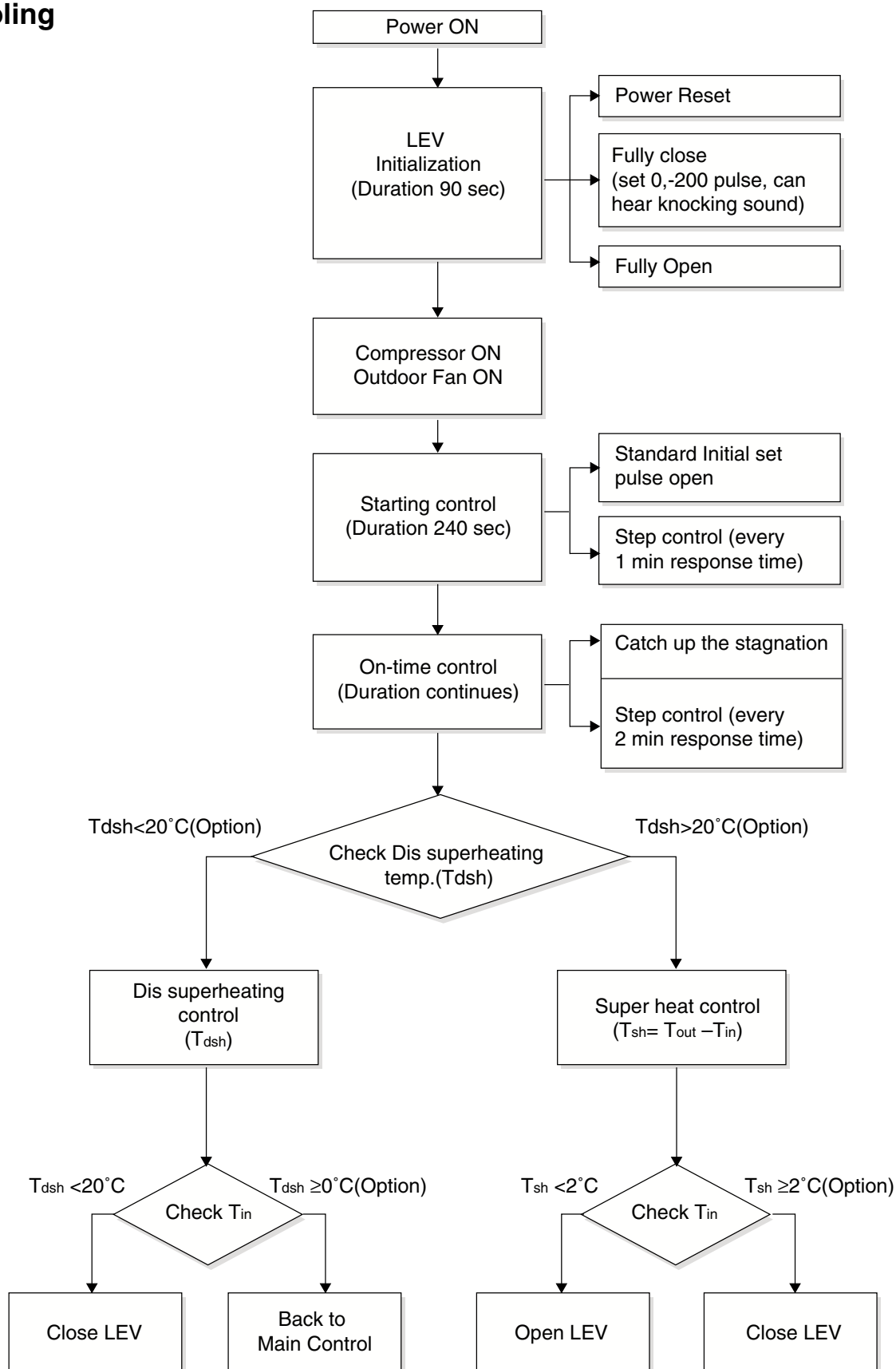
2. Piping temperature maintain 10 seconds (Option) in condition of more than 15°C (Option).

13.15.4 Defrosting Control Algorithm

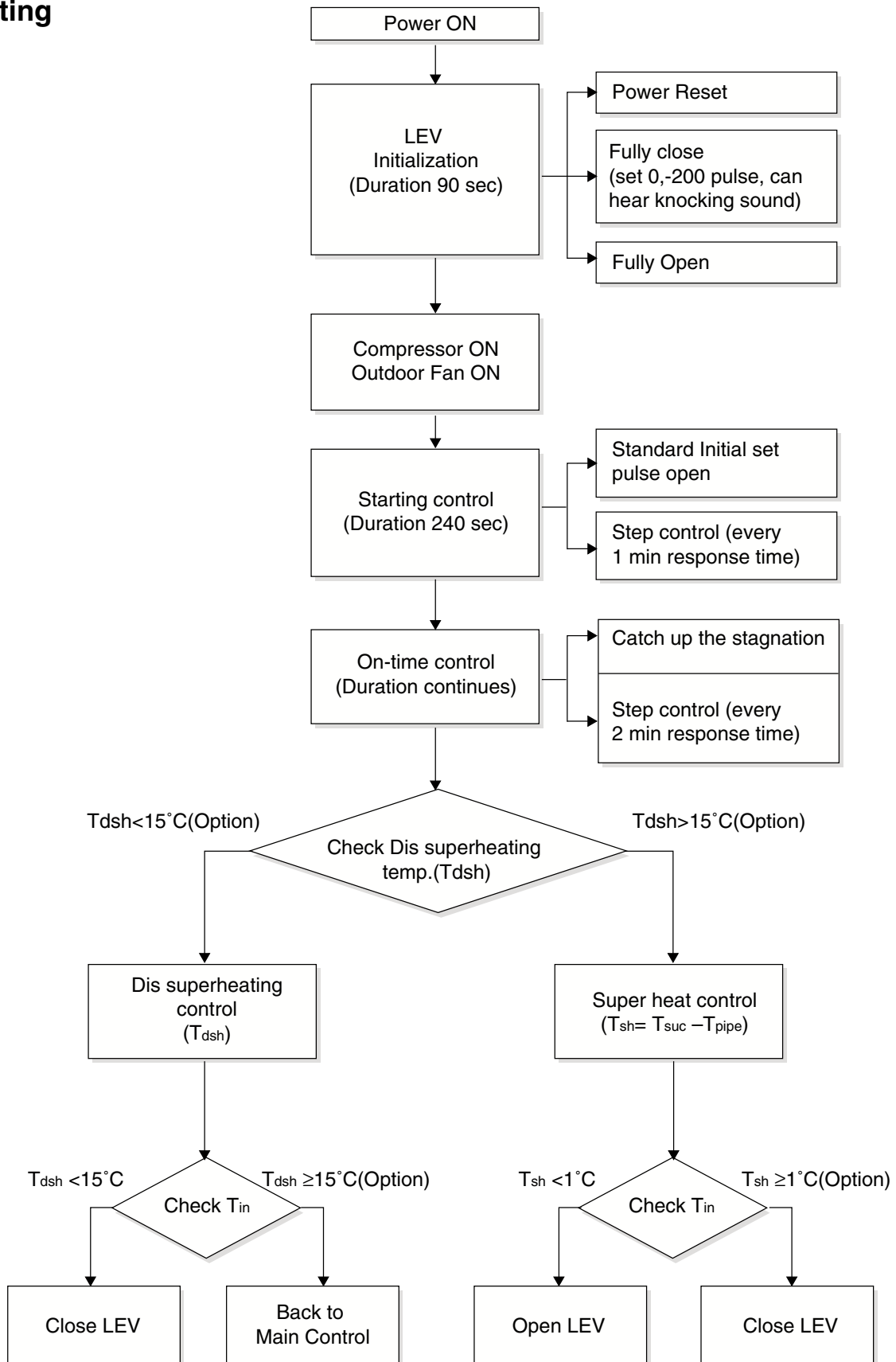


13.16 Flow Chart

• Cooling



• Heating

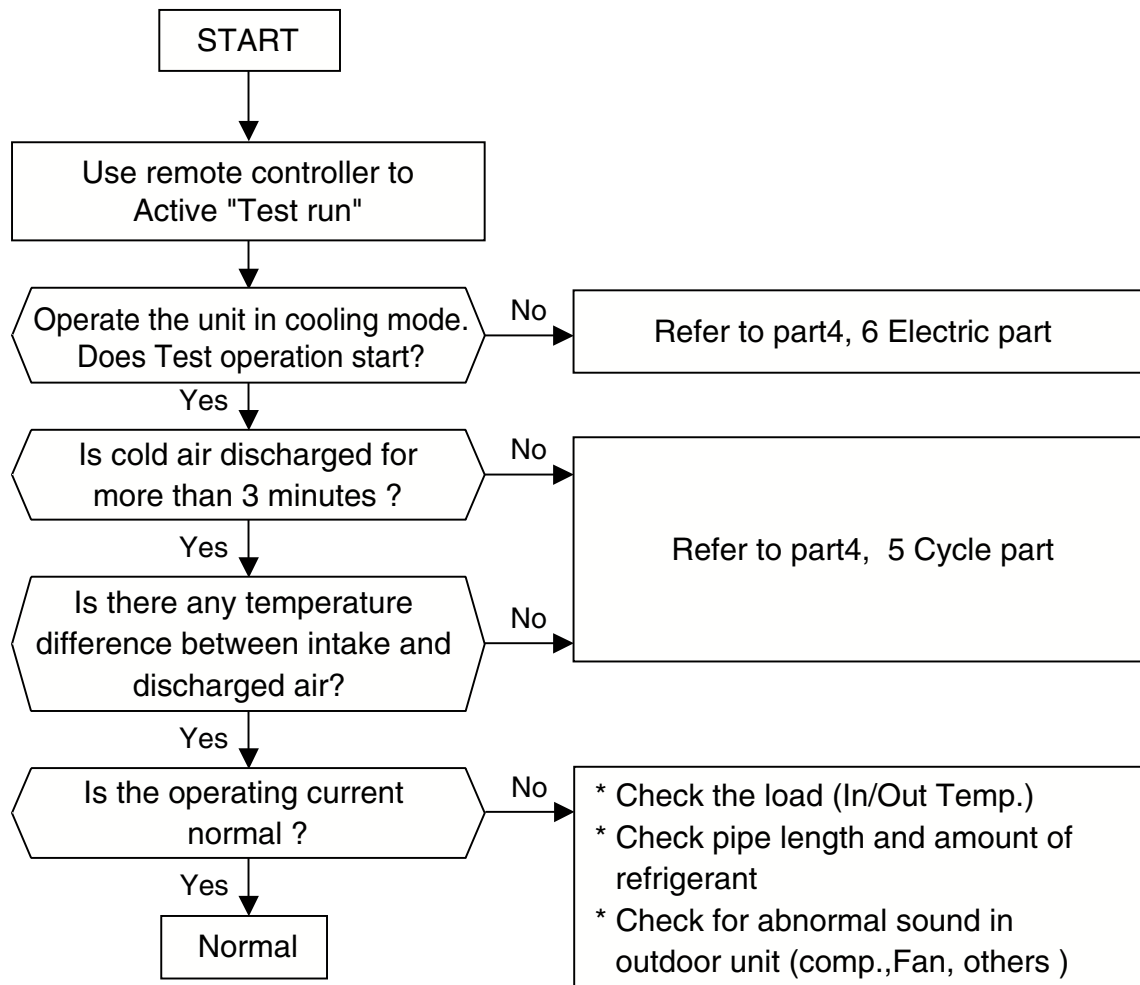


14. Test Run

14.1. Check before Test Run

1	Check to see whether there is any refrigerant leakage, and check whether the power or transmission cable is connected properly.
2	Check liquid pipe and gas pipe valves are fully opened. NOTE: Be sure to tighten caps.
3	Confirm that 500 V megger shows 2.0 MΩ or more between power supply terminal block and ground. Do not operate in the case of 2.0 MΩ or less. NOTE: Never carry out mega ohm check over terminal control board. Otherwise the control board may break. Immediately after mounting the unit or after leaving it turned off for an extended length of time, the resistance of the insulation between the power supply terminal board and the ground may decrease to approx. 2.0 MΩ as a result of refrigerant accumulation in the internal compressor. If the insulation resistance is less than 2.0 MΩ, turn on the main power supply.

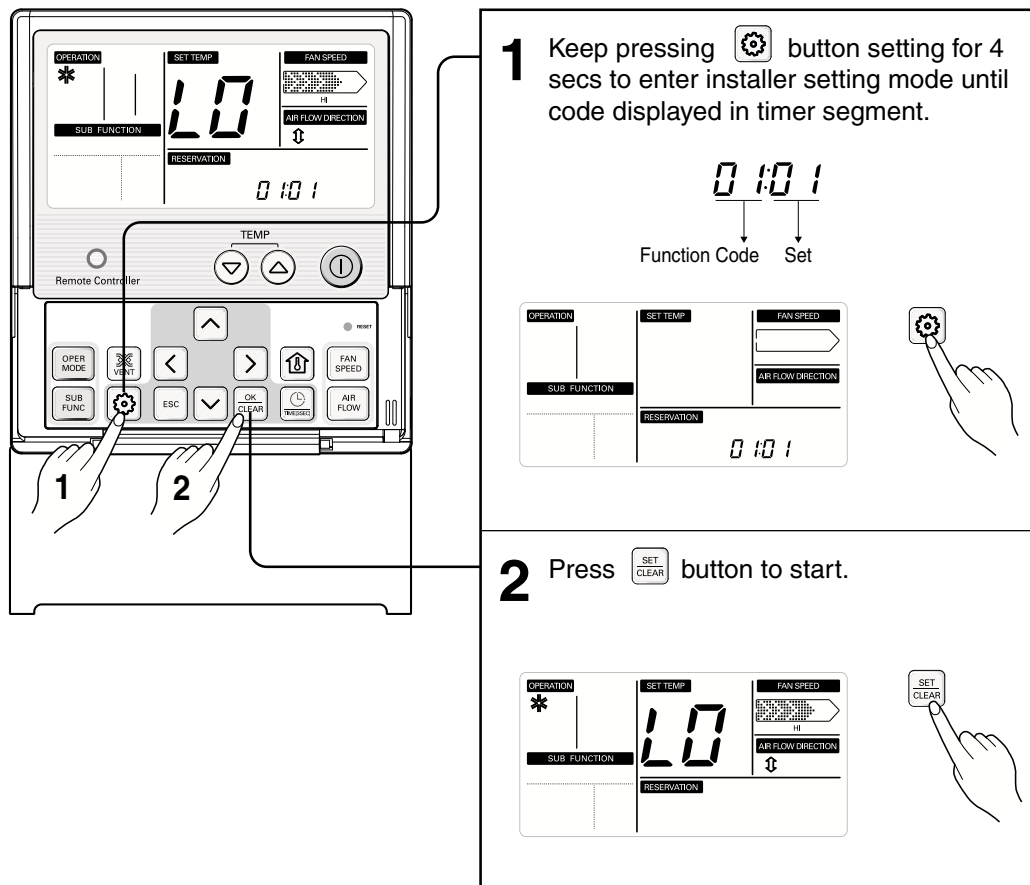
14.2 Test Run Flow chart



- If the unit has accessory, it should be tested.

1. PRECAUTIONS IN TEST RUN

- The initial power supply must provide at least 90% of the rated voltage. Otherwise, the air conditioner should not be operated.
- The test run is started by pressing the button on the remote controller.



- * 18°C cooling, High Fan Speed, Airflow direction mode will be operated during 18 minutes with ignoring room temperature.
 - * After running 18 minutes under test run mode, system will automatically turn OFF.
 - * In case of duct type, the Airflow UP/DOWN function is not displayed.
 - * During test run mode, receiving signal from wireless remote controller will release this operation.
- If you press any kind of button, Test Run mode will be released.

2. CHECK THE FOLLOWING ITEMS WHEN INSTALLATION IS COMPLETE

- After completing work, be sure to measure and record trial run properties, and store measured data, etc.
- Measuring data are room temperature, outside temperature, suction temperature, blow out temperature, air velocity, air volume, voltage, current, presence of abnormal vibration and noise, operating pressure, piping temperature.
- As to the structure and appearance, check following items.

- ☐ Is the circulation of air adequate?
- ☐ Is the drainage OK?
- ☐ Is the heat insulation complete (refrigerant and drain piping)?
- ☐ Is there any leakage of refrigerant?

- ☐ Does the remote controller work properly?
- ☐ Is there any error on wiring?
- ☐ Aren't terminal screws loosened?

M4.....118N.cm{12kgf.cm} M5.....196N.cm{20kgf.cm}
M6.....245N.cm{25kgf.cm} M8.....588N.cm{60kgf.cm}

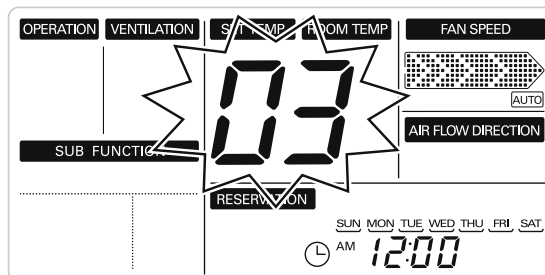
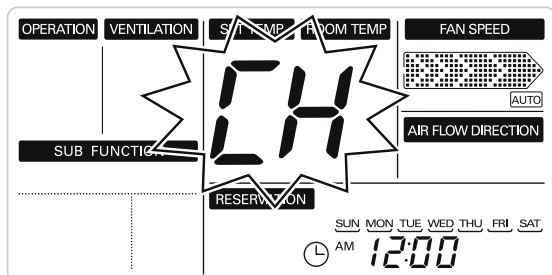
15.Troubleshooting Guide

15. 1. Error code display function

1 Error code displayed on remote controller

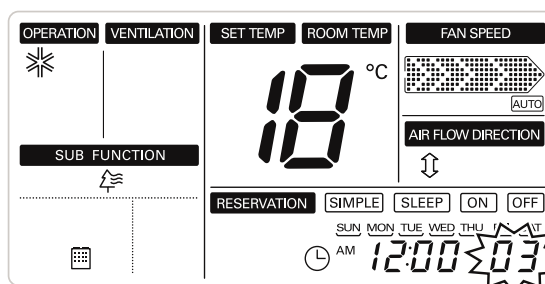
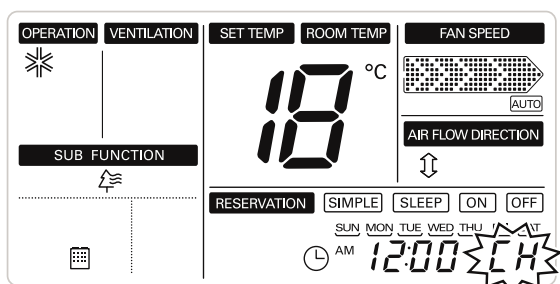
When the error occurred in the indoor unit, then the error code will be displayed on remote controller

- "CH" and error code number is displayed at regular intervals.



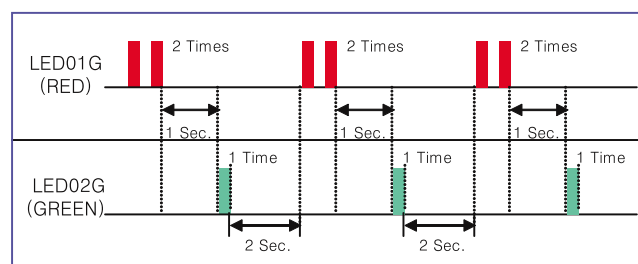
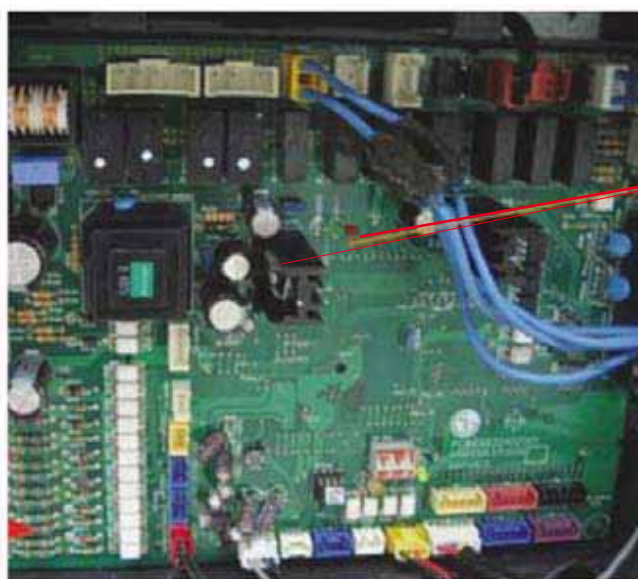
Ex) In case of "CH03"

When the remote controller is used in Group Control, the error code will be displayed as shown below(This is only apply for"CH03")



- "CH" and error code number is displayed at regular intervals beside the Clock (as shown in figure).

2 Error code displayed on outdoor PCB

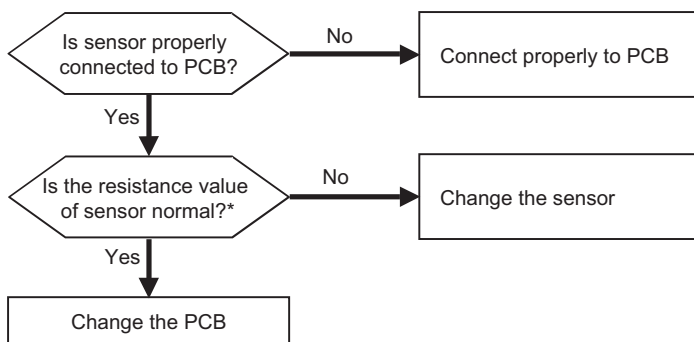


Ex) In case of "CH21"

15.2 Error code

Error No.	Error Type	Error Point	Main Reasons
01	Indoor unit air sensor error	Indoor unit sensor is open/short	1. Indoor unit PCB wrong connection 2. Indoor unit PCB failure 3. Sensor problem (main reason)
02	Indoor unit pipe inlet sensor error		
06	Indoor unit pipe outlet sensor error		

■ Error diagnosis and countermeasure flow chart

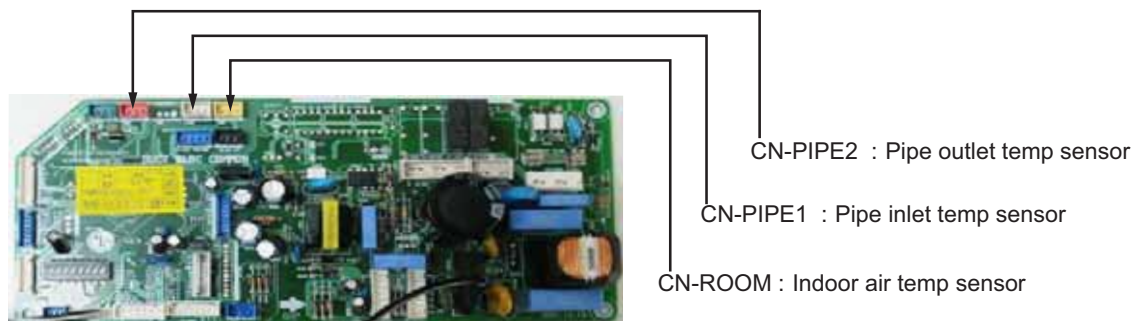


** In case the value is more than 100kΩ (open) or less than 100Ω (short), Error occurs

Refer: Resistance value maybe change according to temperature of temp sensor,
It shows according to criteria of current temperature(±5% margin) → Normal

Air temp sensor: 10°C = 20.7kΩ : 25°C = 10kΩ : 50°C = 3.4kΩ

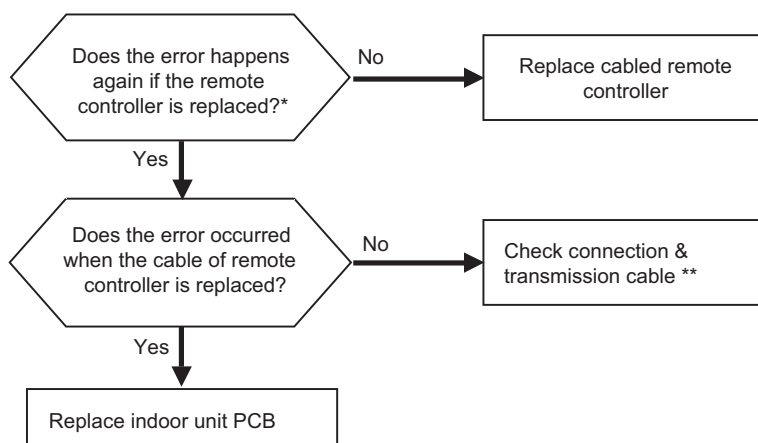
Pipe temp sensor: 10°C = 10kΩ : 25°C = 5kΩ : 50°C = 1.8kΩ



Measure the resistance of outlet pipe temp sensor.

Error No.	Error Type	Error Point	Main Reasons
03	No transmission between cabled remote controller & indoor unit	The remote controller did not receive the signal from indoor unit during specific time	1. Remote controller fault 2. Indoor unit PCB fault 3. Connector fault, Wrong connection 4. transmission cable problem

■ Error diagnosis and countermeasure flow chart



* If there is no remote controller to replace : Use another unit's remote controller doing well

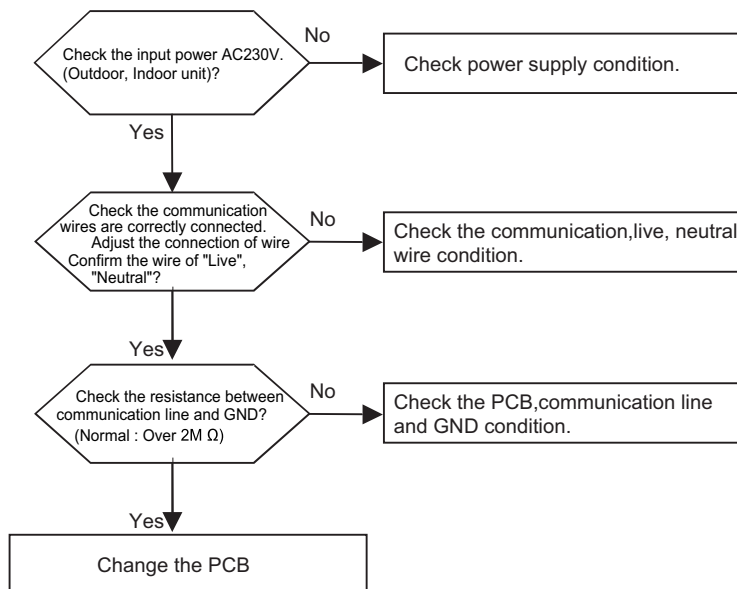
** Check cable : Contact failure of connected portion or extension of cable are main cause
 Check any surrounded noise (check the distance with main power cable)
 → make safe distance from the devices generate electromagnetic wave



CN-REMO : Remote controller connection

Error No.	Error Type	Error Point	Main Reasons
05 *	Title Communication (Indoor → Outdoor)	• Communication poorly	1.Power input AC 230V. (Outdoor, Indoor) 2.The connector for transmission is disconnected. 3.The connecting wires are misconnected. 4.The communication line is shorted at GND. 5.Transmission circuit of outdoor PCB is abnormal. 6.Transmission circuit of indoor PCB is abnormal.

■ Error diagnosis and countermeasure flow chart



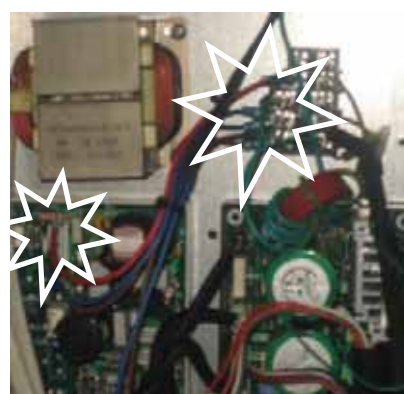
* CH05 is displayed at indoor unit, CH53 is displayed at outdoor unit.

* CH53 : Check the CN_COM and CN_POWER is correctly connected.

Outdoor Unit CN_COM



Indoor Unit CN_COM and CN_POWER



Error No.	Error Type	Error Point	Main Reasons
06	Indoor unit outlet pipe temperature sensor error	Indoor unit outlet pipe temperature sensor open or short	Refer to CH02

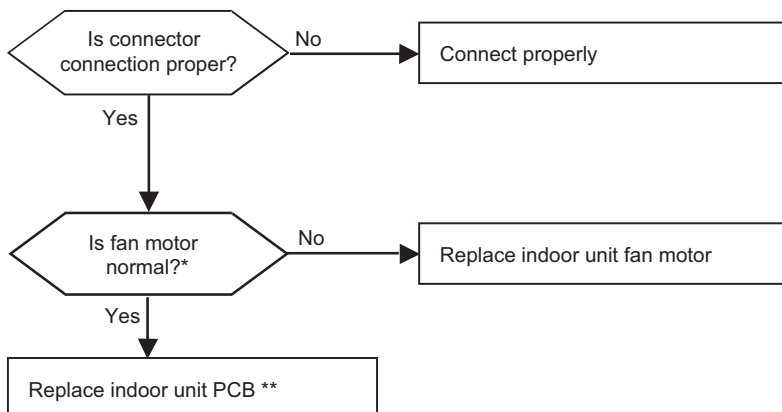
Error No.	Error Type	Error Point	Main Reasons
09	Indoor unit EEPROM error		1. Error developed in transmission between the micro- processor and the EEPROM on the surface of the PCB. 2. ERROR due to the EEPROM damage

■ Error diagnosis and countermeasure flow chart

- Replace the indoor unit PCB.

Error No.	Error Type	Error Point	Main Reasons
10	Indoor unit BLDC fan motor failure	Indoor BLDC fan motor feedback signal is absent (for 50 sec.)	1. Motor connector connection fault 2. Indoor PCB fault 3. Motor fault

■ Error diagnosis and countermeasure flow chart



* It is normal when check hall sensor of indoor fan motor as shown below



① ④⑤⑥⑦

Each terminal with the tester

Tester		Normal resistance($\pm 10\%$)
+	-	DUCT chassis
①	④	∞
⑤	④	hundreds k Ω
⑥	④	∞
⑦	④	hundreds k Ω

<Checking connection state of fan motor connector>

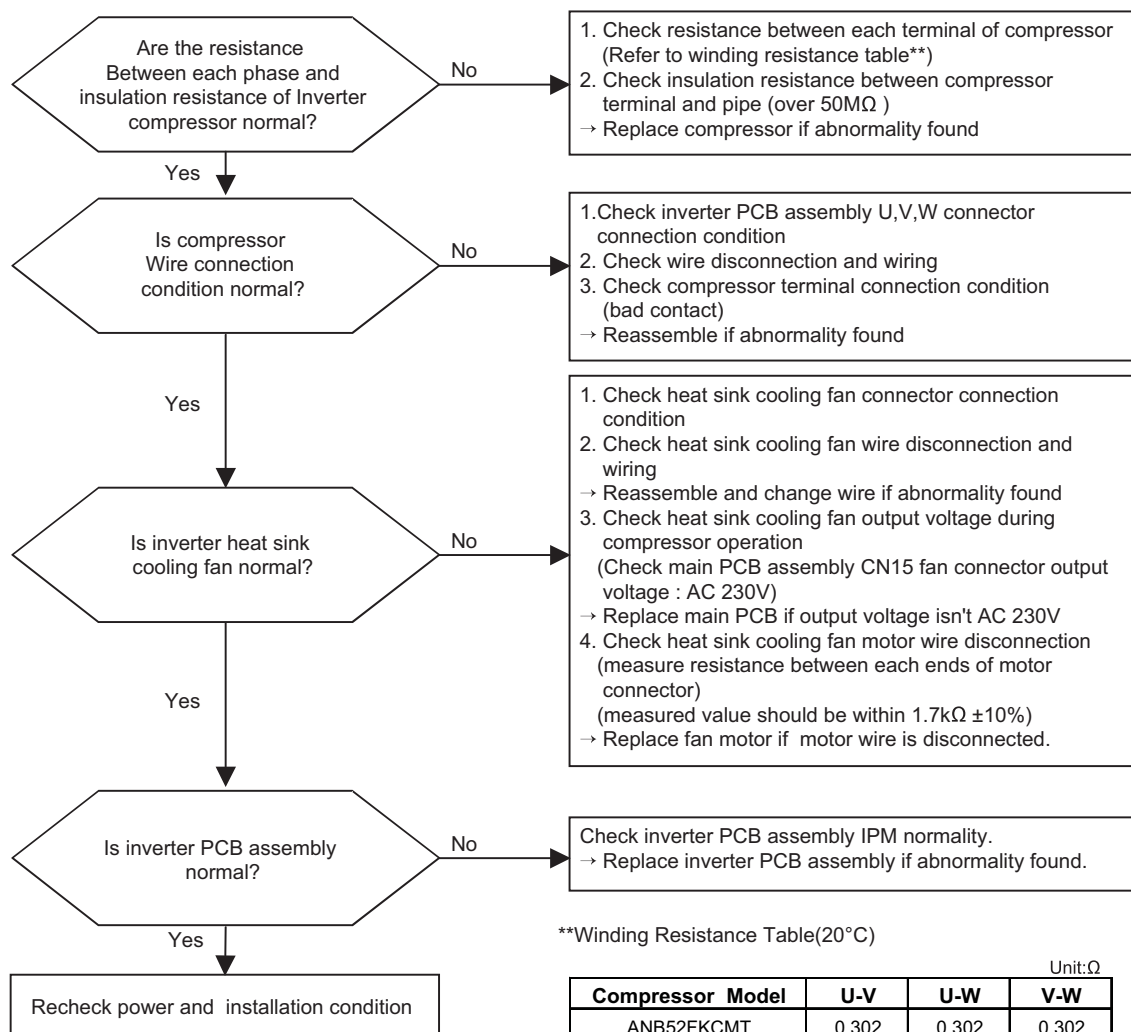


** Replace the indoor unit PCB.

(Notice: The connection of motor connector to PCB should be done under no power supplying to PCB)

Error No.	Error Type	Error Point	Main Reasons
21*	Inverter PCB Assy IPM Fault occur	IPM self protection circuit activation (Overcurrent/IPM overheating/Vcc low voltage)	1. Over current detection at Inverter compressor(U,V,W) 2. Compressor damaged (insulation damaged/Motor damaged) 3. IPM overheating (Heat sink fan damaged/Heat sink fan connector disconnected/Heat sink disassembled) 4. Inverter compressor terminal disconnected or loose 5. Inverter PCB assembly damaged 6. Outdoor Unit input current low

■ Error diagnosis and countermeasure flow chart



* Measuring resistance between each terminal of compressor



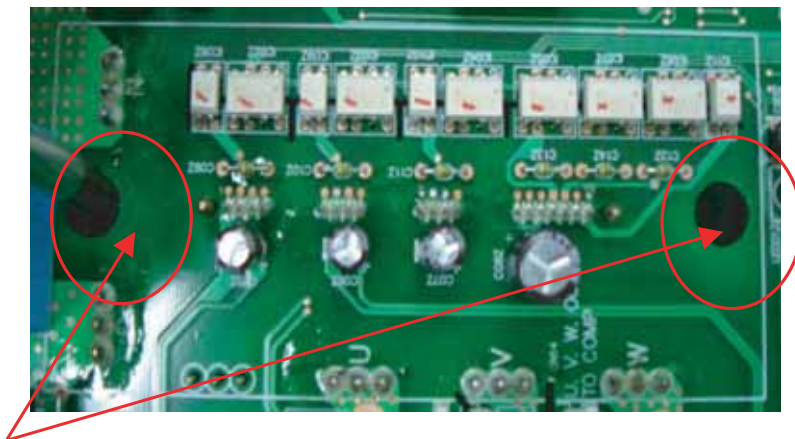
* Heat sink cooling fan connector



* Heat sink cooling fan



* IPM joining point

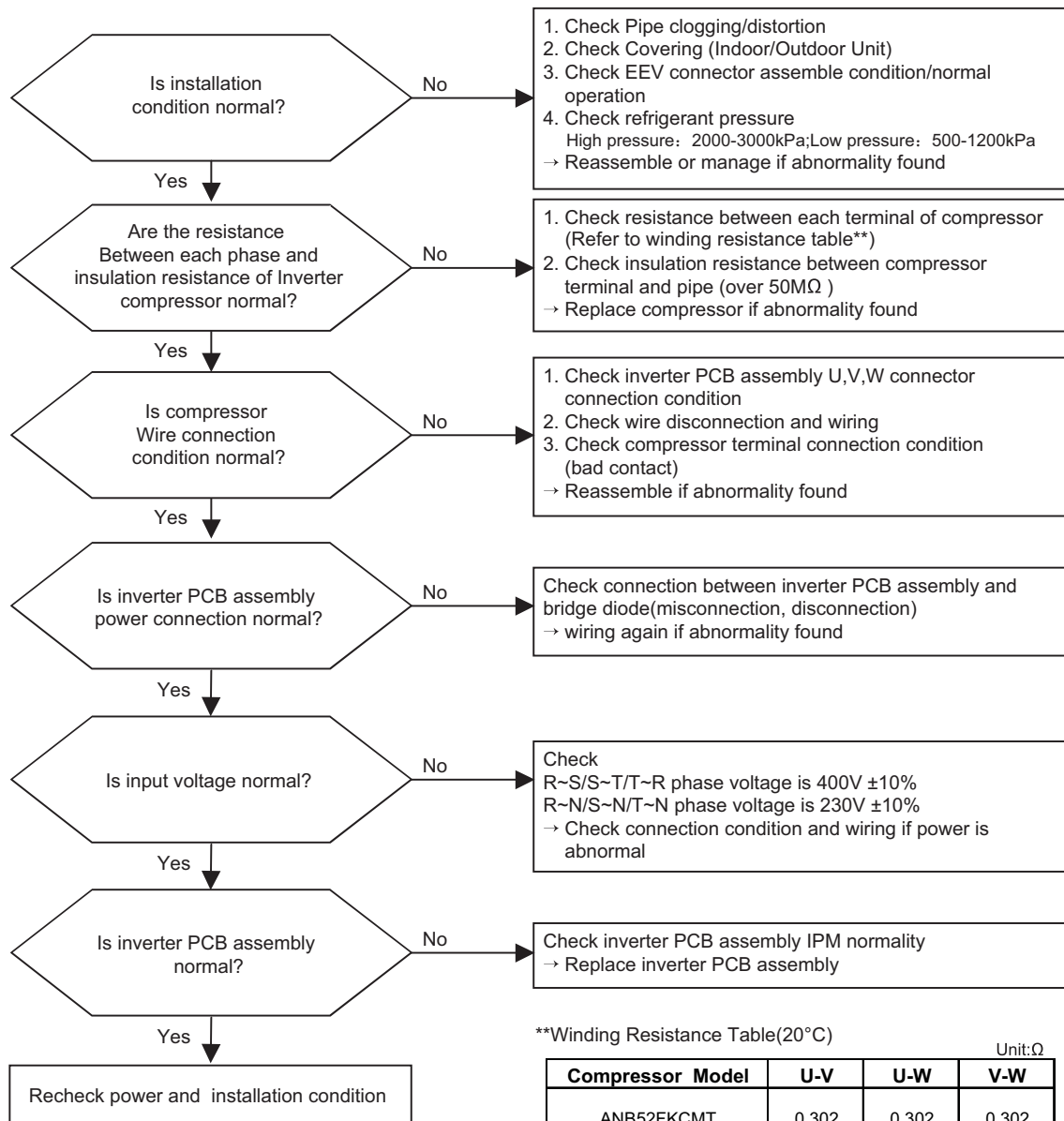


Check joining condition

There is no gap between IPM and heat sink.

Error No.	Error Type	Error Point	Main Reasons
22*	AC Input Current Over Error	Inverter PCB Assembly input 3 phase power current is over limited value(20A)	1. Overload operation (Pipe clogging/Covering/EEV defect/Ref. overcharge) 2. Compressor damage(Insulation damage/Motor damage) 3. Input voltage low 4. Power Line Misconnection 5. Inverter PCB Assembly damage (Input current sensing part)

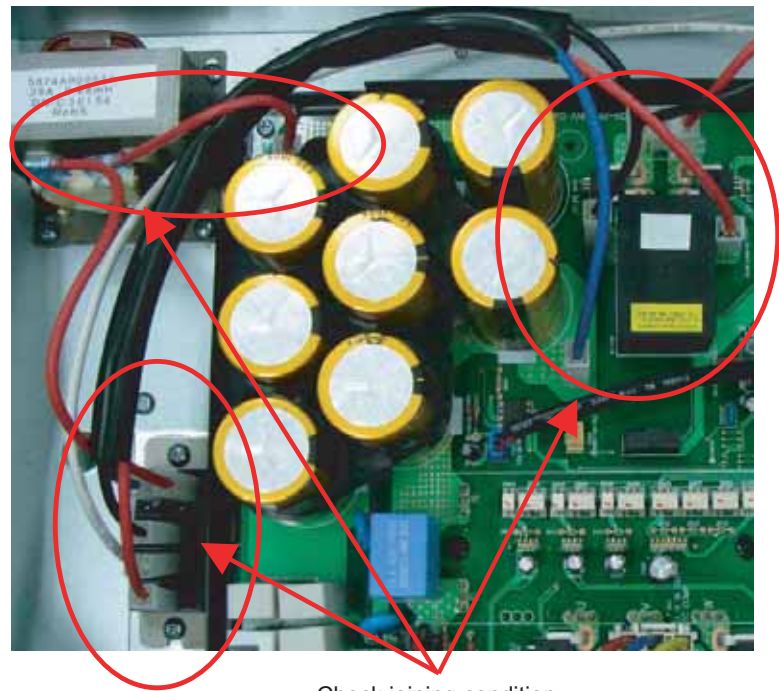
■ Error Diagnosis and Countermeasure Flow Chart



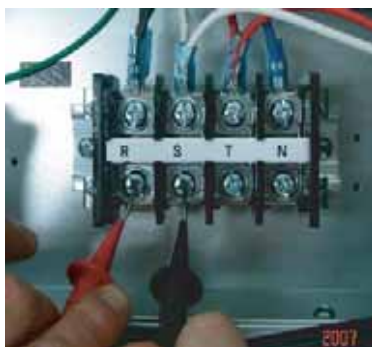
* Measuring resistance between each terminal of compressor



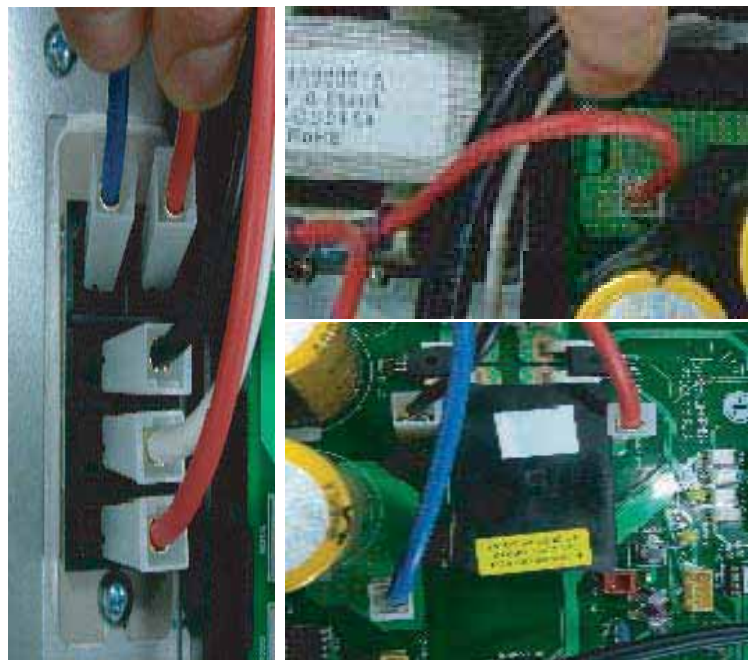
* Inverter PCB & bridge diode wiring



* Measuring input voltage

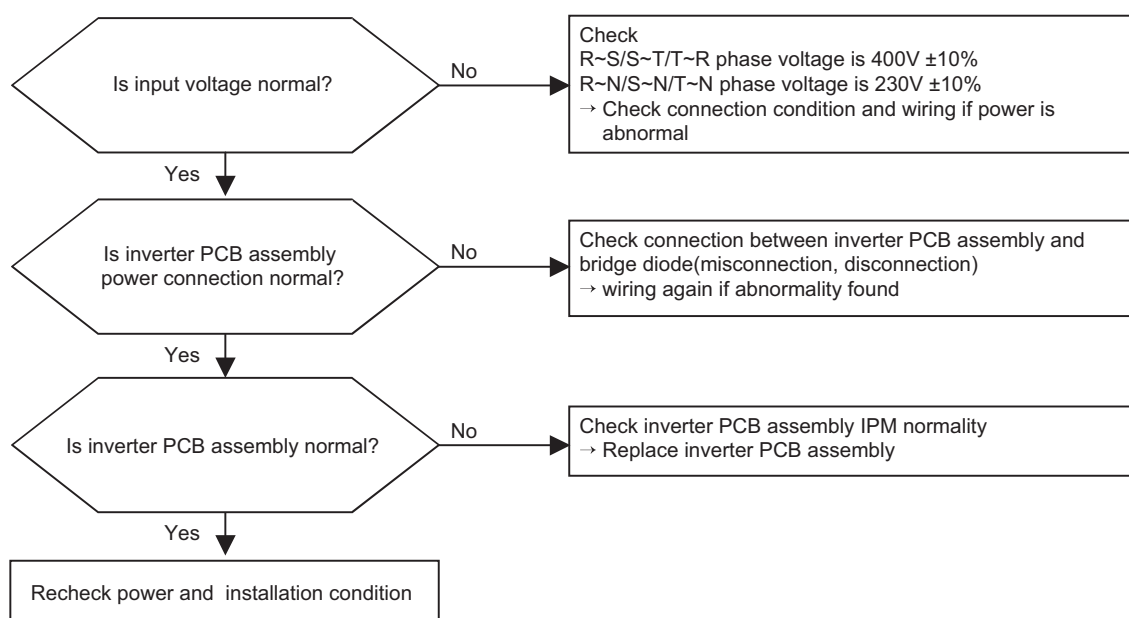


Check joining condition

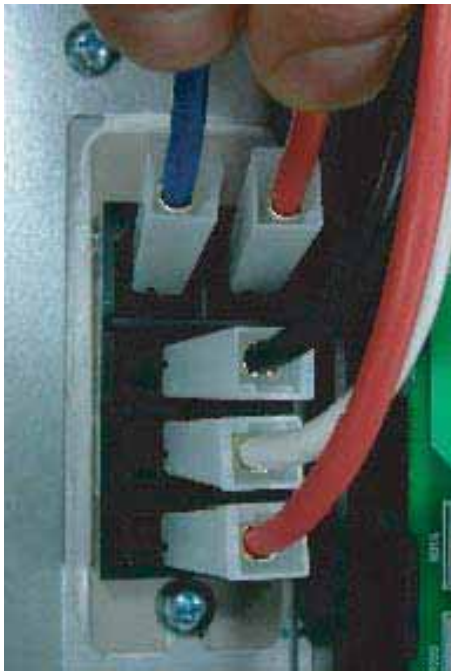


Error No.	Error Type	Error Point	Main Reasons
23*	Inverter Compressor DC Link Low Voltage	DC Voltage isn't charged after starting relay on	1. DC Link terminal misconnection/terminal contact fault 2. Starting relay damage 3. Condenser damage 4. Inverter PCB assembly damage (DC Link voltage sensing part) 5. Input voltage low

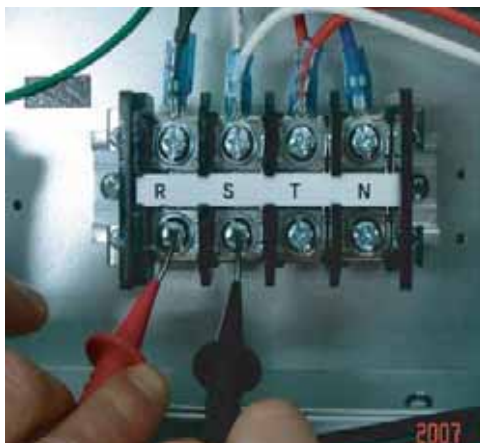
■ Error Diagnosis and Countermeasure Flow Chart



* Inverter PCB & bridge diode wiring

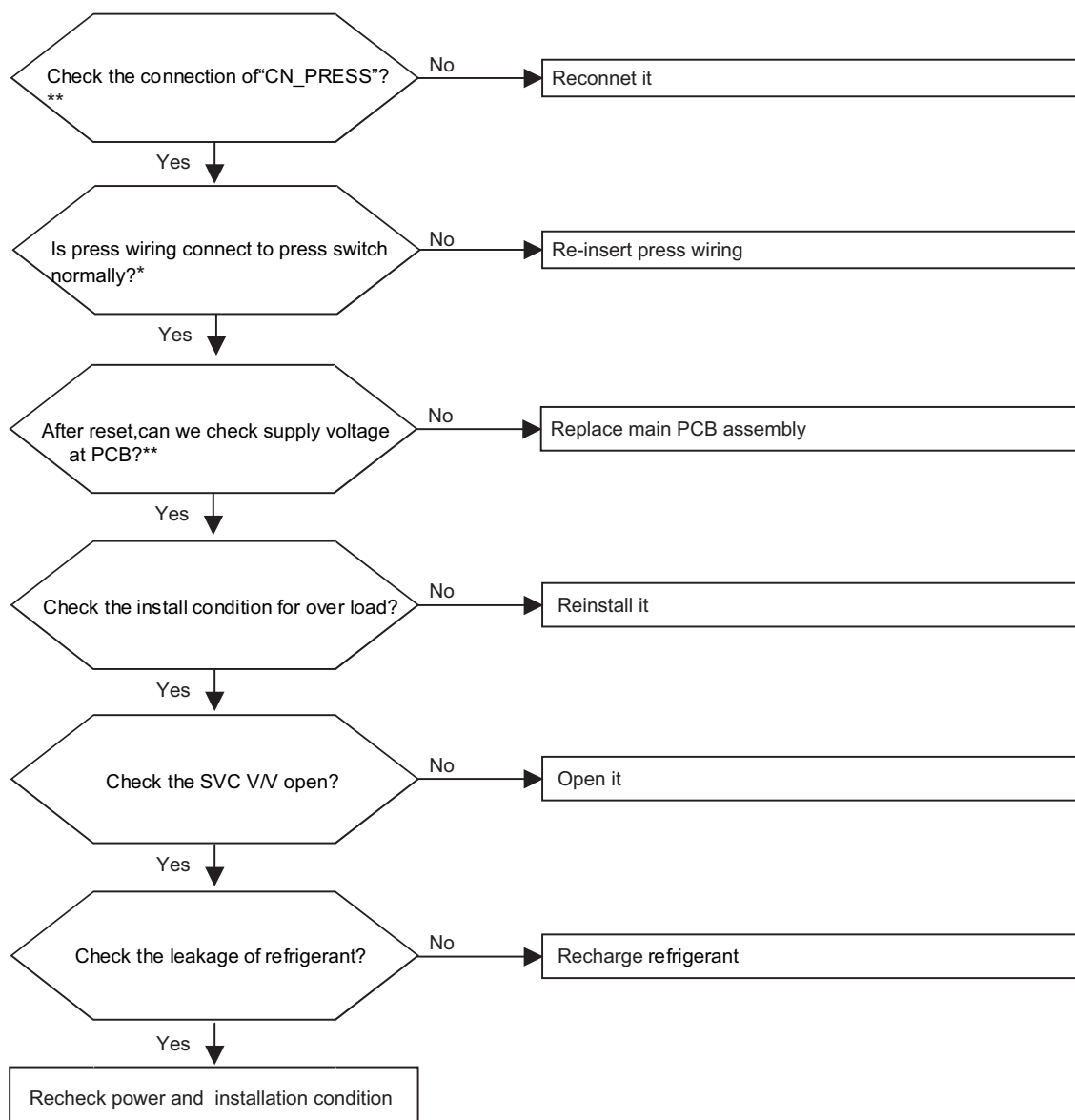


* Measuring input voltage



Error No.	Error Type	Error Point	Main Reasons
24*	Press SW Open	Low / High press S/W open.	1.Check the connection of "CN_Press". 2.Check the components.

■ Error Diagnosis and Countermeasure Flow Chart



* Low PRESS wiring &
LowPRESS switch

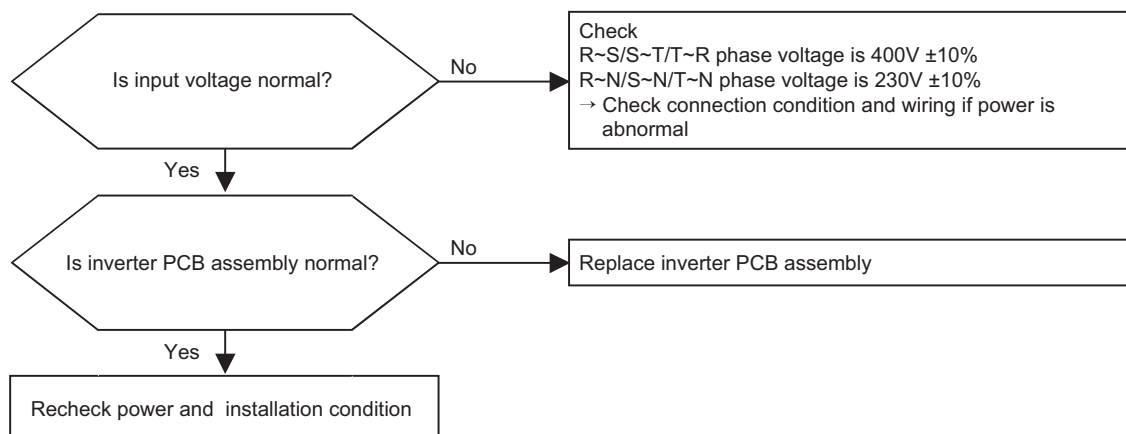


**Main PCB CN_PRESS
Check the output voltage of terminal socket



Error No.	Error Type	Error Point	Main Reasons
25*	Input Voltage high/low	Input voltage is over limited value of the product (Refer to table**)	1. Input voltage abnormal (T-N) 2. Outdoor unit inverter PCB assembly damage (input voltage sensing part)

■ Error Diagnosis and Countermeasure Flow Chart



* Measuring input voltage



* Inverter PCB assembly power wiring



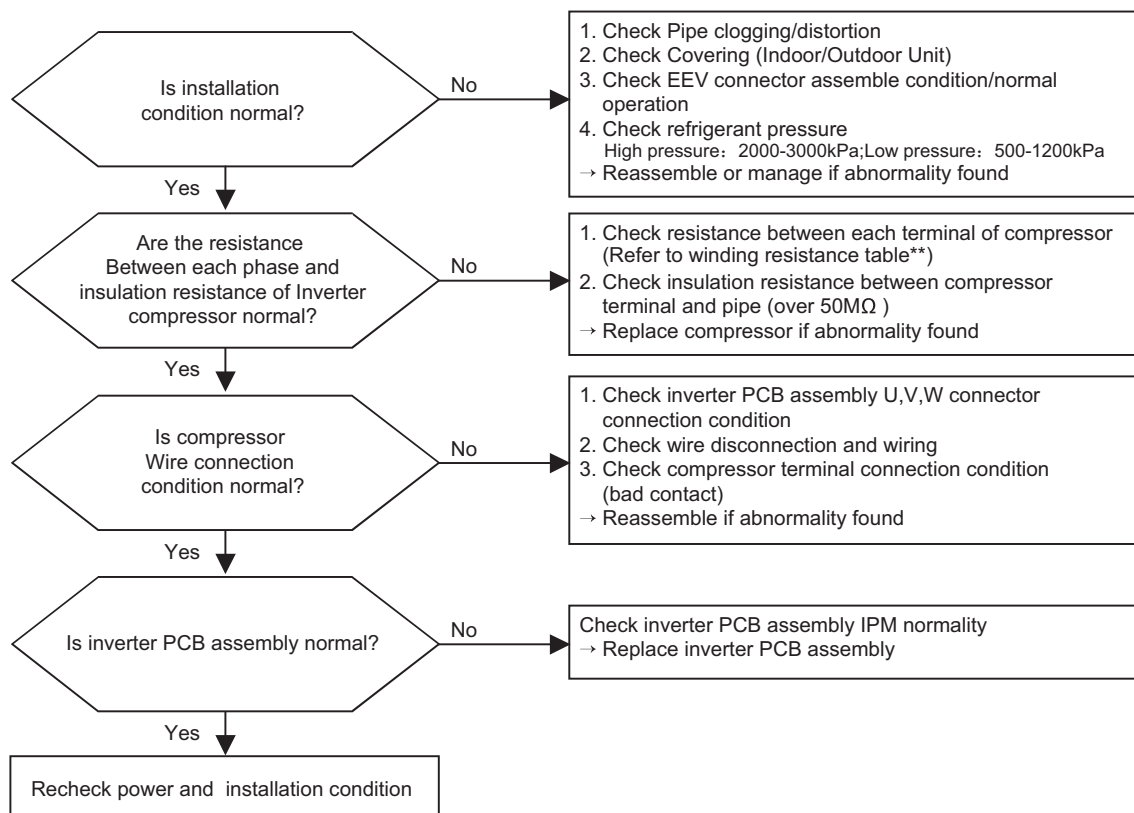
**Table

Unit : V

Phase Voltage	Input Voltage (low)	Input Voltage (high)
R-N	173 or less	289 or more
S-N	173 or less	289 or more
T-N	173 or less	289 or more

Error No.	Error Type	Error Point	Main Reasons
26*	Inverter compressor starting failure Error	Starting failure because of compressor abnormality	1. Overload operation (Pipe clogging/Covering/EEV defect/Ref. overcharge) 2. Compressor damage (Insulation damage/Motor damage) 3. Compressor wiring fault 4.Outdoor Unit inverter PCB damage (CT)

■ Error Diagnosis and Countermeasure Flow Chart



**Winding Resistance Table(20°C)

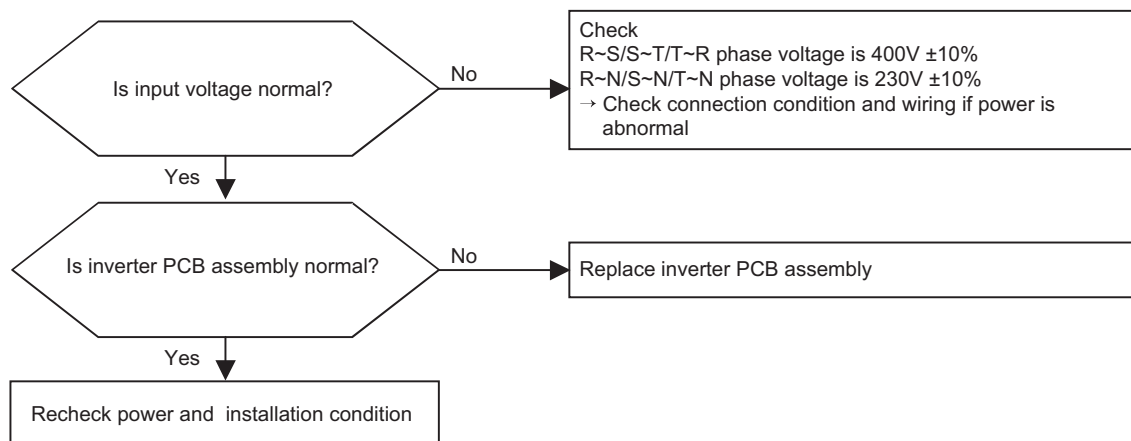
Compressor Model	Unit:Ω		
	U-V	U-W	V-W
ANB52FKCMT	0.302	0.302	0.302

* Measuring resistance between each terminal of compressor



Error No.	Error Type	Error Point	Main Reasons
28*	Inverter DC link high voltage error	Inv PCB DC link voltage supplied over 780V	1. Input voltage abnormal (R,S,T,N) 2. Outdoor Unit inverter PCB damage (DC Link voltage sensing part)

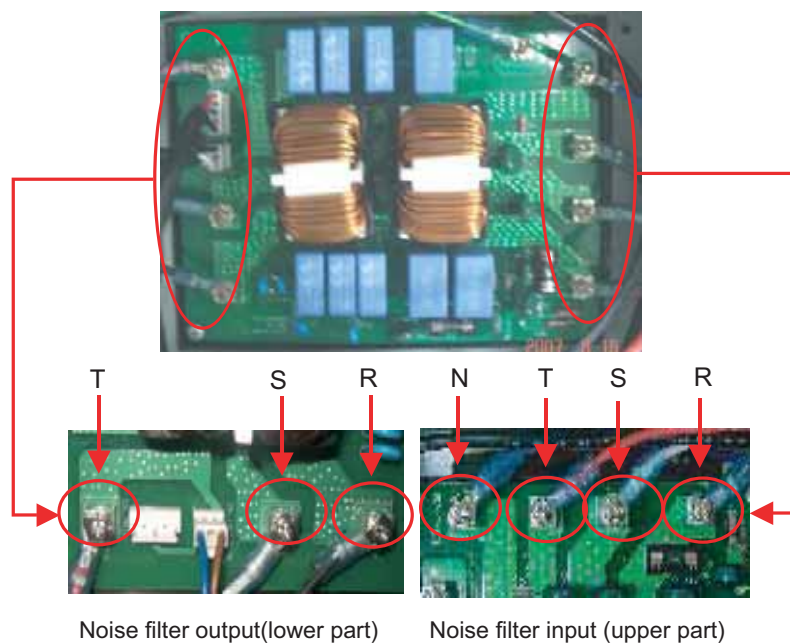
■ Error Diagnosis and Countermeasure Flow Chart



* Measuring input voltage

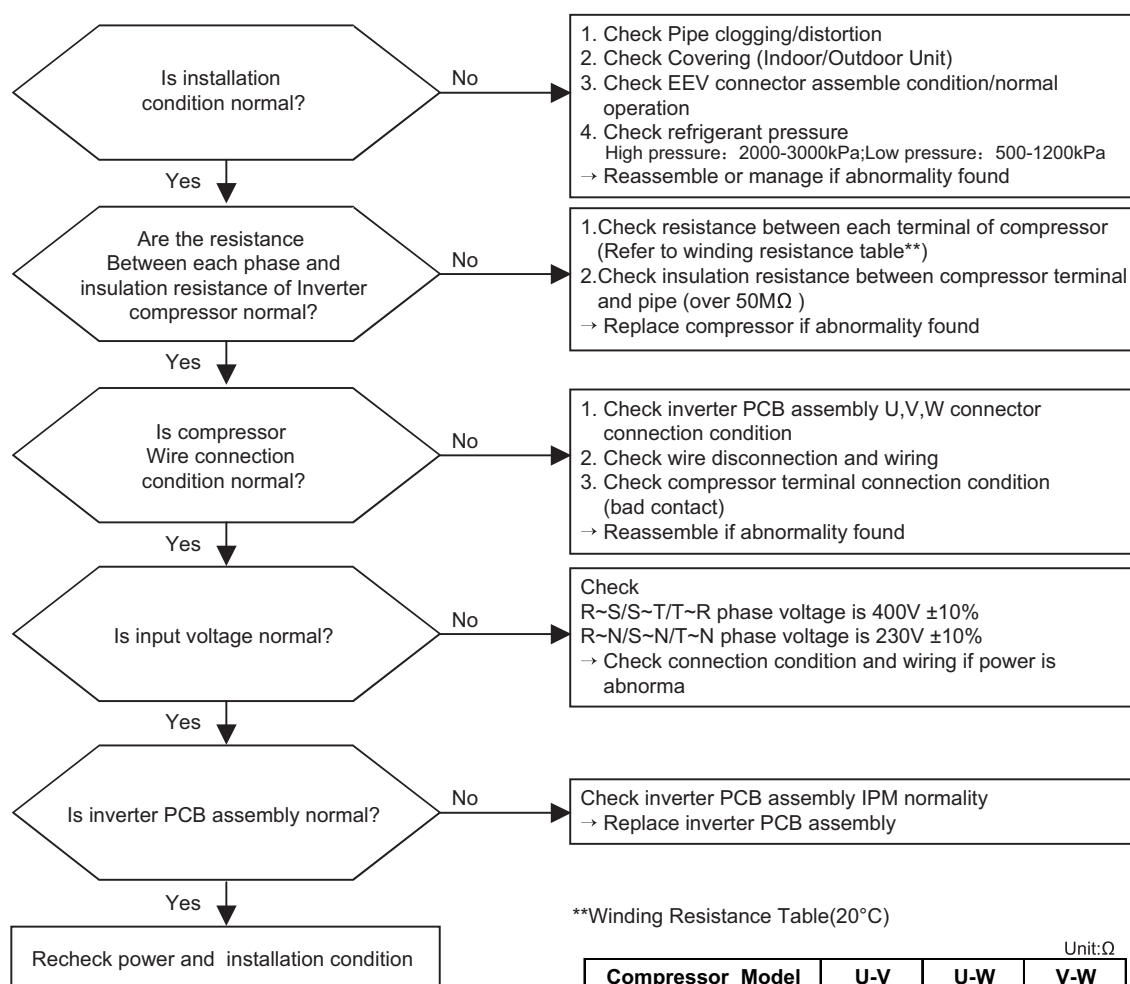


* Noise filter wiring



Error No.	Error Type	Error Point	Main Reasons
29*	Inverter compressor over current	Inverter compressor input current is over 30A	1. Overload operation (Pipe clogging/Covering/EEV defect/Ref. overcharge) 2. Compressor damage(Insulation damage/Motor damage) 3. Input voltage low 4. Outdoor Unit inverter PCB assembly damage

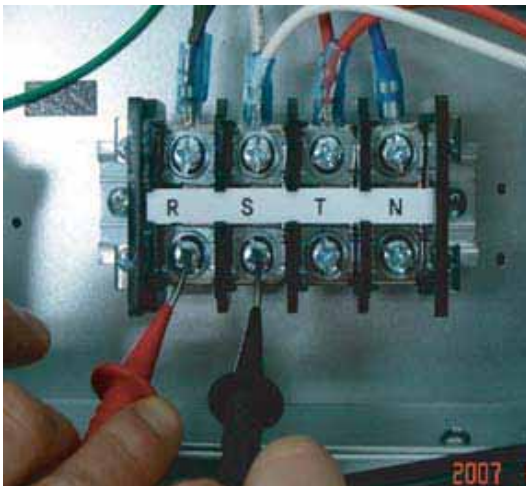
■ Error Diagnosis and Countermeasure Flow Chart



* Measuring resistance between each terminal of compressor

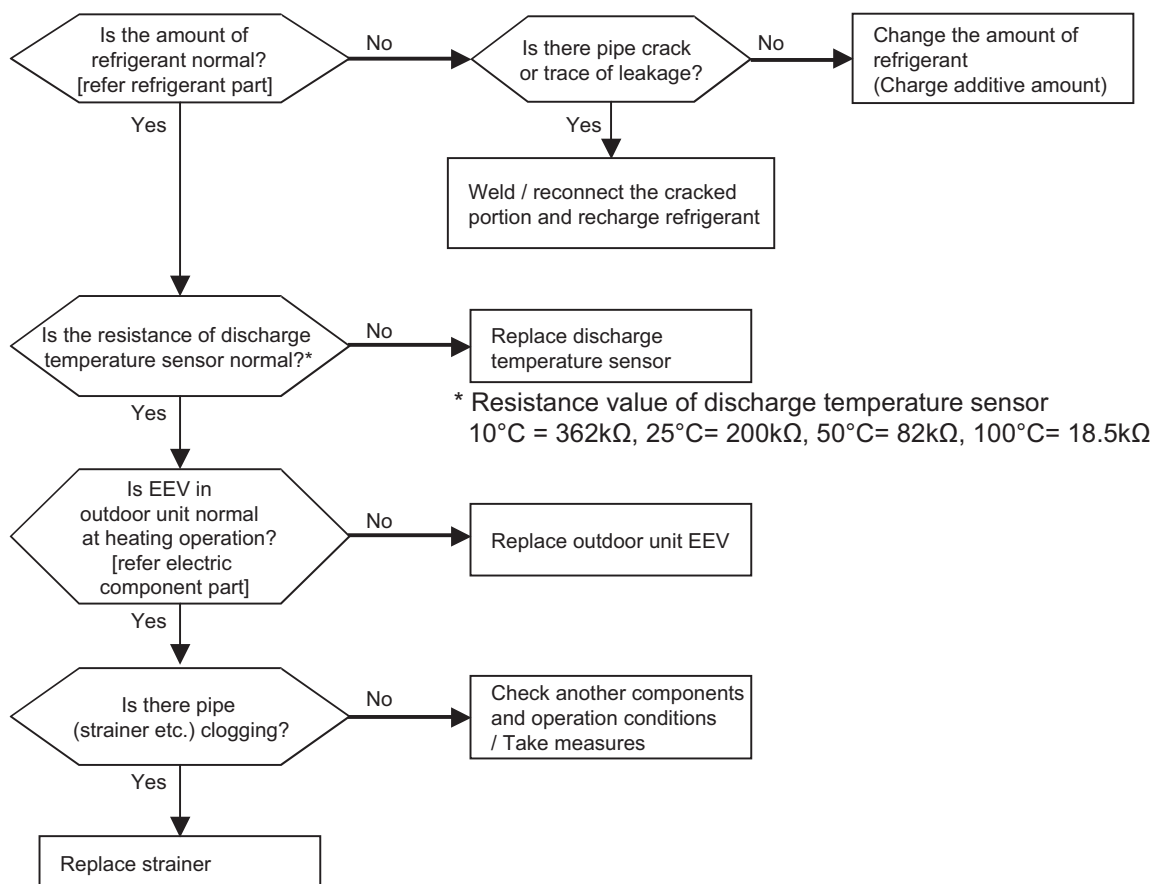


* Measuring input voltage



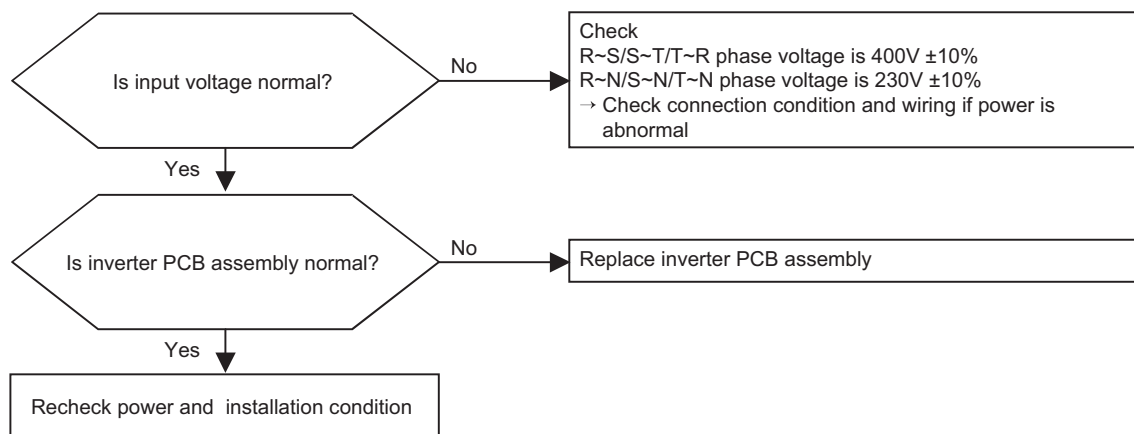
Error No.	Error Type	Error Point	Main Reasons
32*	Over-increase discharge temperature of inverter compressor at main outdoor unit	Compressor is off because of over-increase discharge temperature of inverter compressor	1. Temperature sensor defect of inverter compressor discharge pipe 2. Refrigerant shortage / leak 3. EEV defect 4. Liquid injection valve defect

■ Error diagnosis and countermeasure flow chart

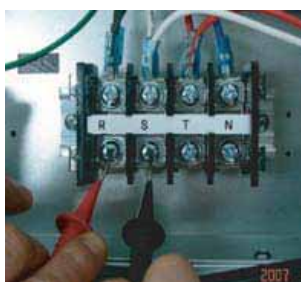


Error No.	Error Type	Error Point	Main Reasons
40*	Inverter compressor CT sensor error	Micom input voltage isn't within $2.5V \pm 0.3V$ at initial state of power supply	1. Input voltage abnormal (T-N) 2. Outdoor Unit inverter PCB damage (CT sensing part)

■ Error Diagnosis and Countermeasure Flow Chart

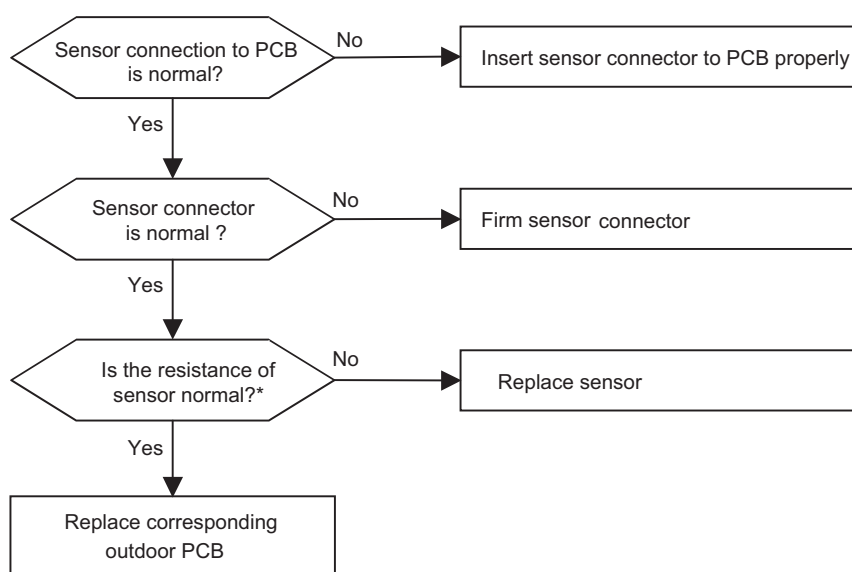


* Measuring input voltage



Error No.	Error Type	Error Point	Main Reasons
41*	Compressor discharge pipe temperature sensor error	Sensor measurement valve is abnormal (Open/Short)	1. Defective connection of the compressor discharge pipe temperature sensor 2. Defective discharge pipe compressor sensor of the compressor (open/short) 3. Defective outdoor PCB

■ Error diagnosis and countermeasure flow chart



* Error is generated if the resistance is more than 5MΩ(open) and less than 2kΩ (short)

Note: Standard values of resistance of sensors at different temperatures (±5% variation)
 10°C = 362kΩ : 25°C= 200kΩ : 50°C= 82kΩ : 100°C= 18.5kΩ



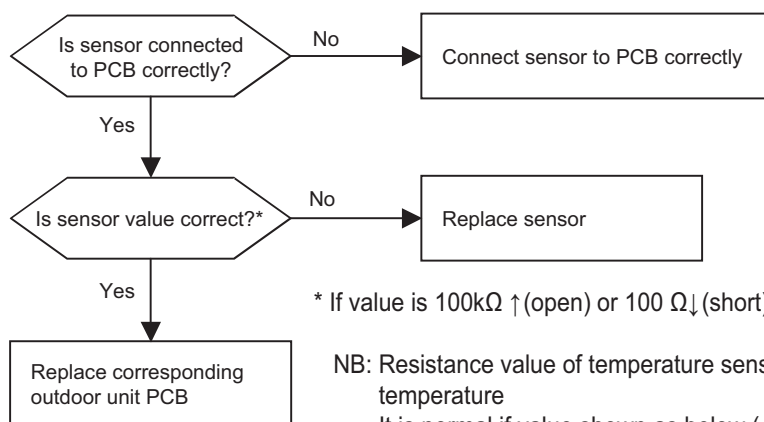
Check the resistance inverter compressor discharge temperature sensor



Check the resistance of constant compressor discharge temperature sensor.
 (Apply for two compressors model)

Error No.	Error Type	Error Point	Main Reasons
44*	Sensor error of outdoor air temperature	Abnormal value of sensor (Open/Short)	1. Bad connection of air temperature connector 2. Defect of air temperature connector(Open/Short) 3. Defect of outdoor PCB
45*	Condenser pipe sensor	Abnormal value of sensor (Open/Short)	1. Bad connection of Condenser pipe sensor 2. Defect of Condenser pipe sensor (Open/Short) 3. Defect of outdoor PCB
46*	Compressor suction temperature sensor error	Abnormal value of sensor (Open/Short)	1. Bad connection of suction temperature connector 2. Defect of suction sensor (Open/Short) 3. Defect of outdoor PCB

■ Error diagnosis and countermeasure flow chart



* If value is 100kΩ ↑ (open) or 100 Ω ↓ (short), error occurs

NB: Resistance value of temperature sensor change according to temperature

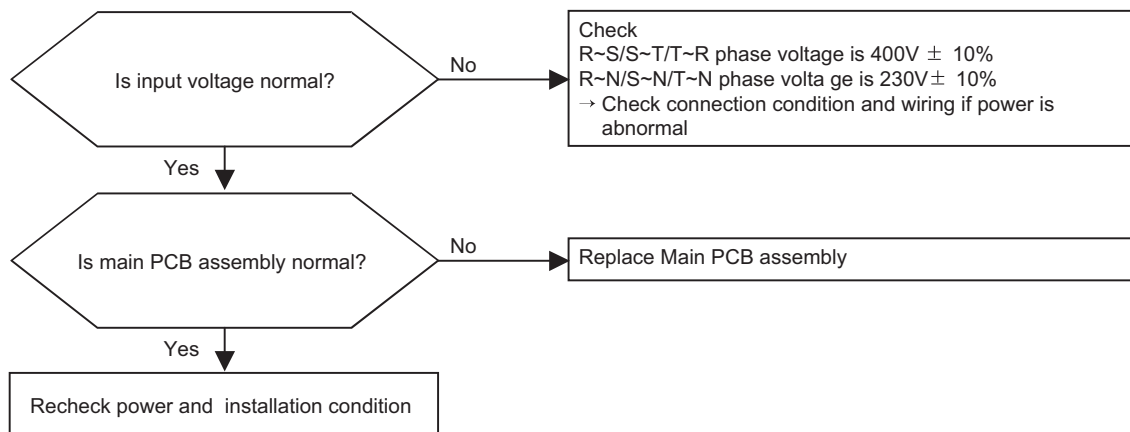
It is normal if value shown as below (±5% error)

Sensor of air temperature: 10°C = 20.7kΩ : 25°C= 10kΩ : 50°C= 3.4kΩ

Sensor of piping temperature: 10°C = 10kΩ : 25°C= 5kΩ : 50°C= 1.8kΩ

Error No.	Error Type	Error Point	Main Reasons
50*	Outdoor Unit 3phase power omission error	Omitting one or more of R,S,T input power	1. Input Voltage abnormal (R,S,T,N) 2. Check power Line connection condition 3. Main PCB damage(**)

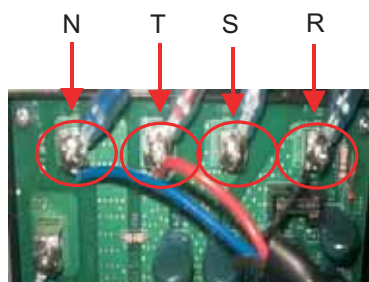
■ Error Diagnosis and Countermeasure Flow Chart



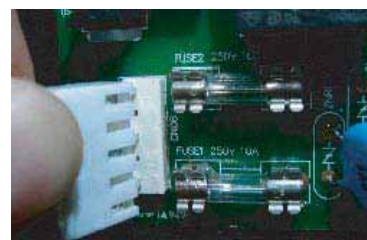
* Measuring input voltage



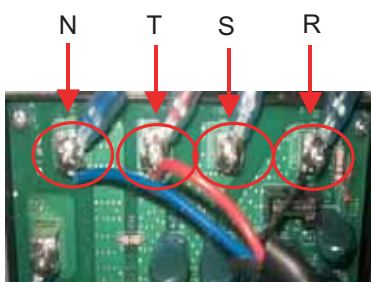
* Noise filter wiring



* Main PCB power connection



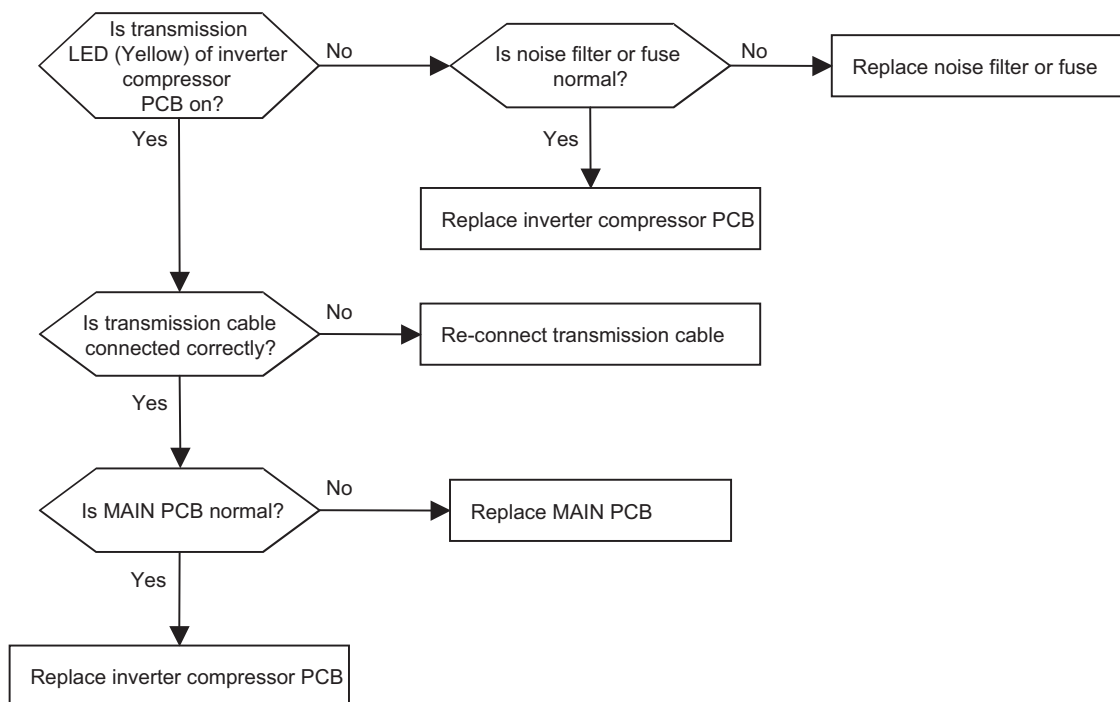
* Main PCB power supply



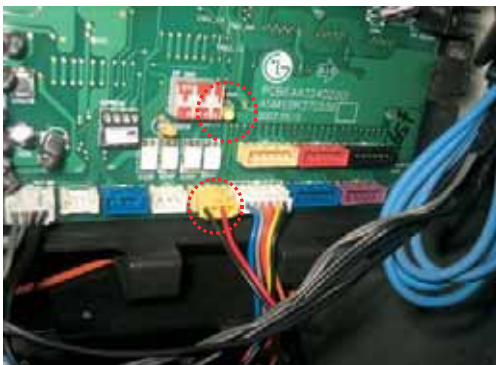
** Note: Main PCB power supply by noise filter PCB low part.

Error No.	Error Type	Error Point	Main Reasons
52*	Transmission error between (Inverter PCB → Main PCB)	Main controller can't receive signal from inverter controller	1. Power cable or transmission cable is not connected 2. Defect of outdoor Main fuse/Noise Filter 3. Defect of outdoor Main / inverter PCB

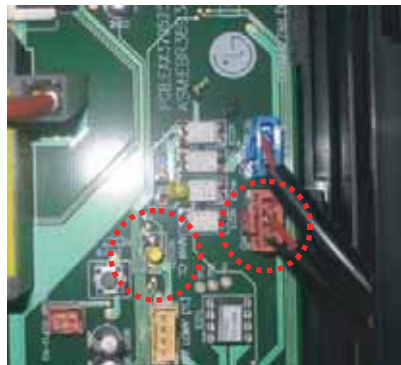
■ Error diagnosis and countermeasure flow chart



* The method of checking MAIN PCB and inverter compressor PCB (If normal, transmission LED blinks)



Transmission connector & LED in MAIN PCB

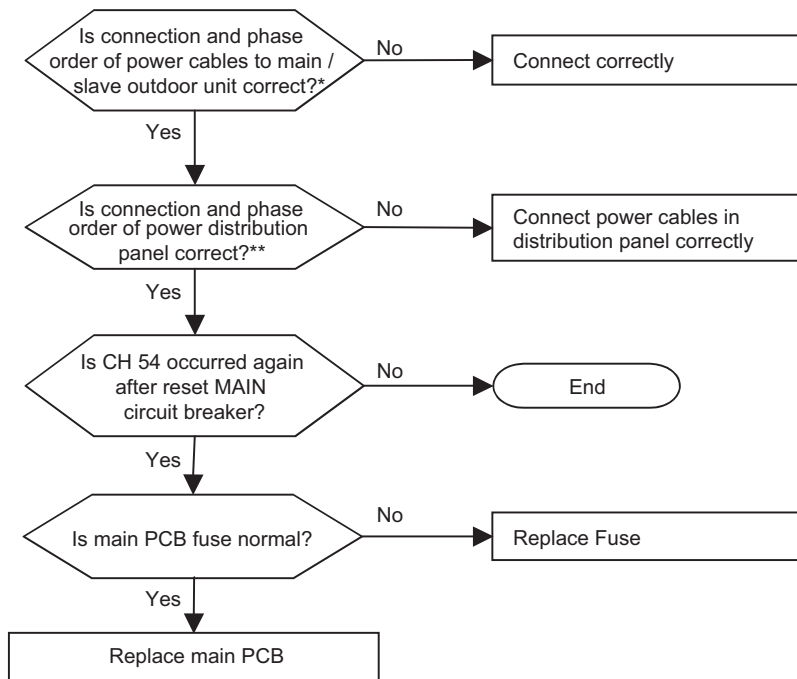


Transmission connector & LED in inverter compressor PCB

Error No.	Error Type	Error Point	Main Reasons
53	Transmission error (Indoor unit → Main PCB)	In case Main PCB can't receive signal from indoor unit	1. Transmission cables are not connected 2. Transmission cables are short / open 3. Defect of outdoor Main / indoor PCB

Error No.	Error Type	Error Point	Main Reasons
54*	Wrong connection of 3Ø power supply cable (Reverse direction / missing a phase)	Wrong connection of 3Ø power supply cable (Reverse direction / missing a phase)	1. Main PCB defect 2. No power of R,S,T supplied 3. Wrong connection of R,S,T cables 4. Main PCB Fuse failure

■ Error diagnosis and countermeasure flow chart



* Check power cable connection state, phase (R-S-T) order, power supply state in control box of product

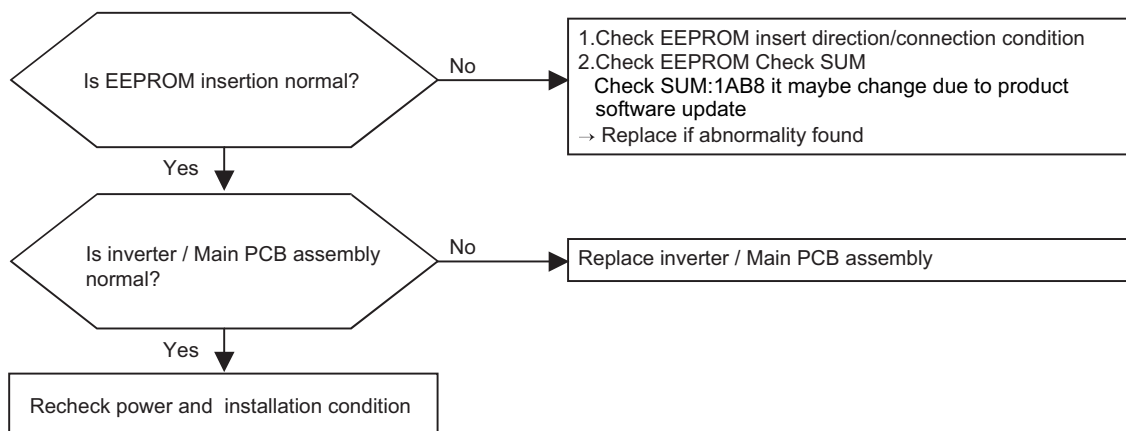


** Check power cable connection state, phase order, power supply state in distribution panel

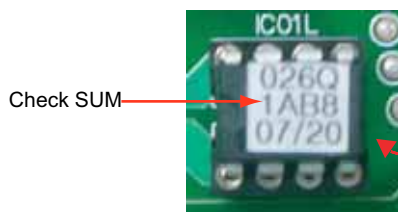


Error No.	Error Type	Error Point	Main Reasons
60*	Inverter / Main PCB EEPROM EEOR	EEPROM Access error and Check SUM error	1. EEPROM contact defect/wrong insertion 2. Different EEPROM Version 3. Outdoor Unit inverter PCB assembly damage

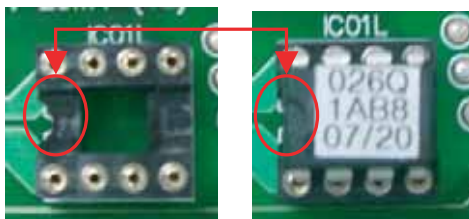
■ Error Diagnosis and Countermeasure Flow Chart



* Inverter EEPROM inserting point



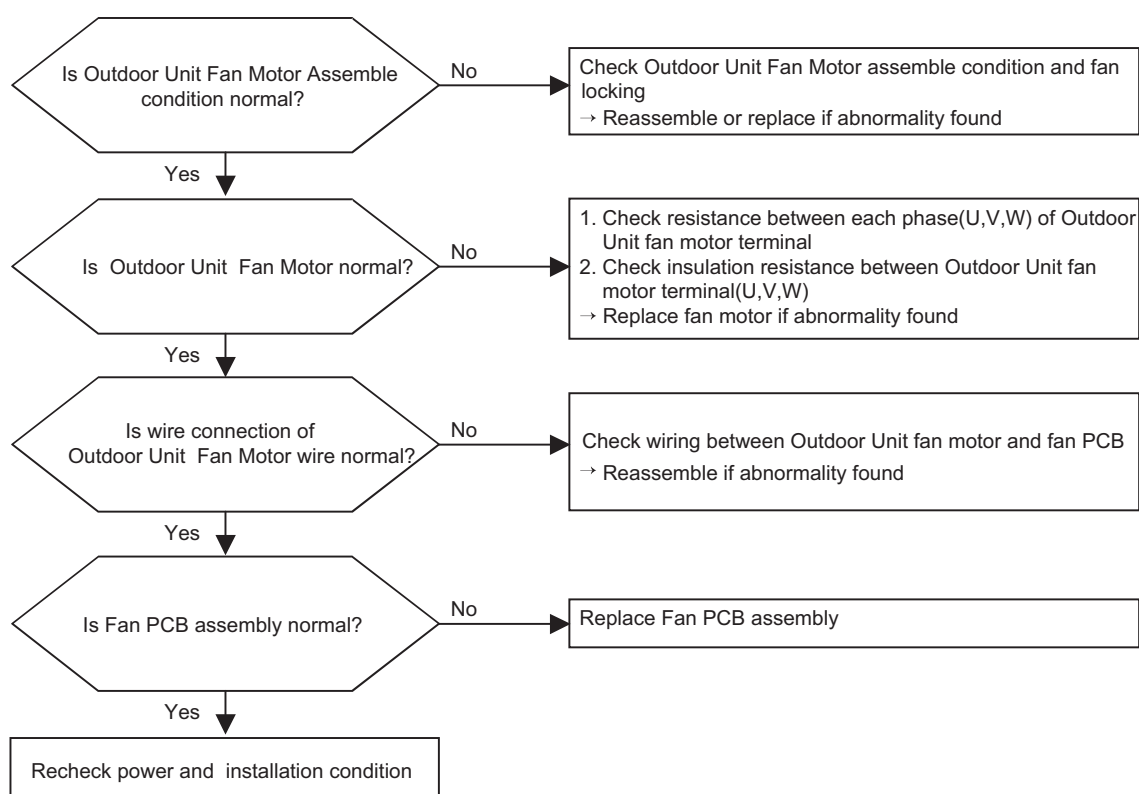
* Right inserting direction of inverter / Main EEPROM



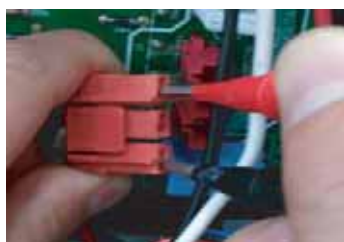
※ Note : Replace after power off

Error No.	Error Type	Error Point	Main Reasons
67*	Fan Lock Error	Fan RPM is 10RPM or less for 5 sec. when Outdoor Unit fan starts or 40 RPM or less after fan starting.	1. Fan motor defect / assembly condition abnormal 2. Wrong connection of fan motor connector (Hall sensor, U,V,W output) 3. Reversing rotation after RPM target apply 4. Fan PCB assembly defect

■ Error Diagnosis and Countermeasure Flow Chart



* Fan Motor resistance measuring between each phase

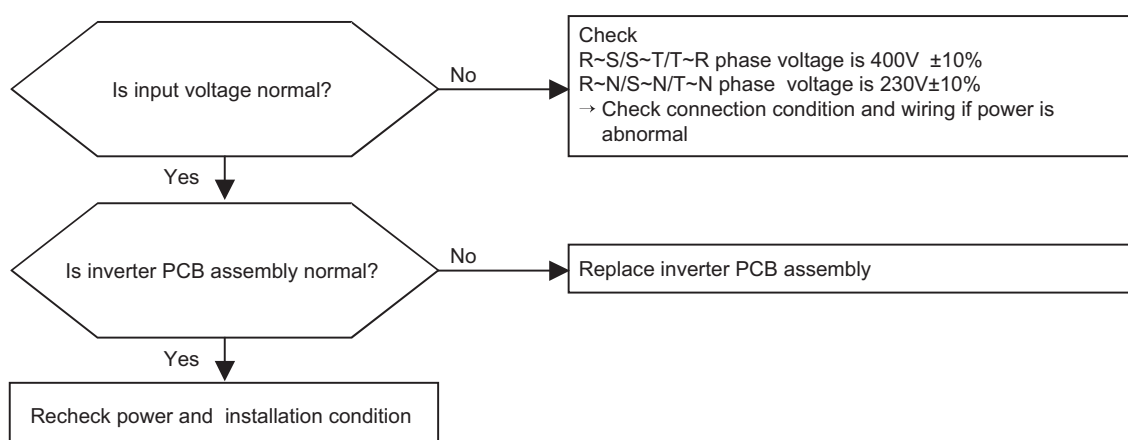


* Fan Motor Wire connection

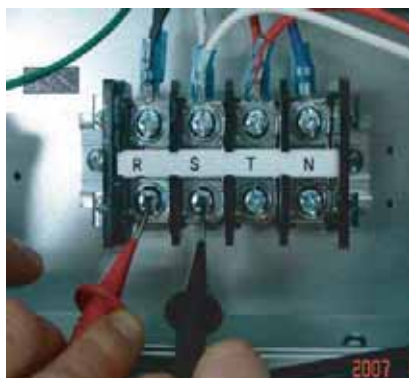


Error No.	Error Type	Error Point	Main Reasons
71*	Inverter input current CT sensor error	Micom input voltage isn't within $2.5V \pm 0.3V$ at initial state of power supply	1. Input voltage abnormal (T-N) 2. Outdoor Unit inverter PCB damage (CT sensing part)

■ Error Diagnosis and Countermeasure Flow Chart

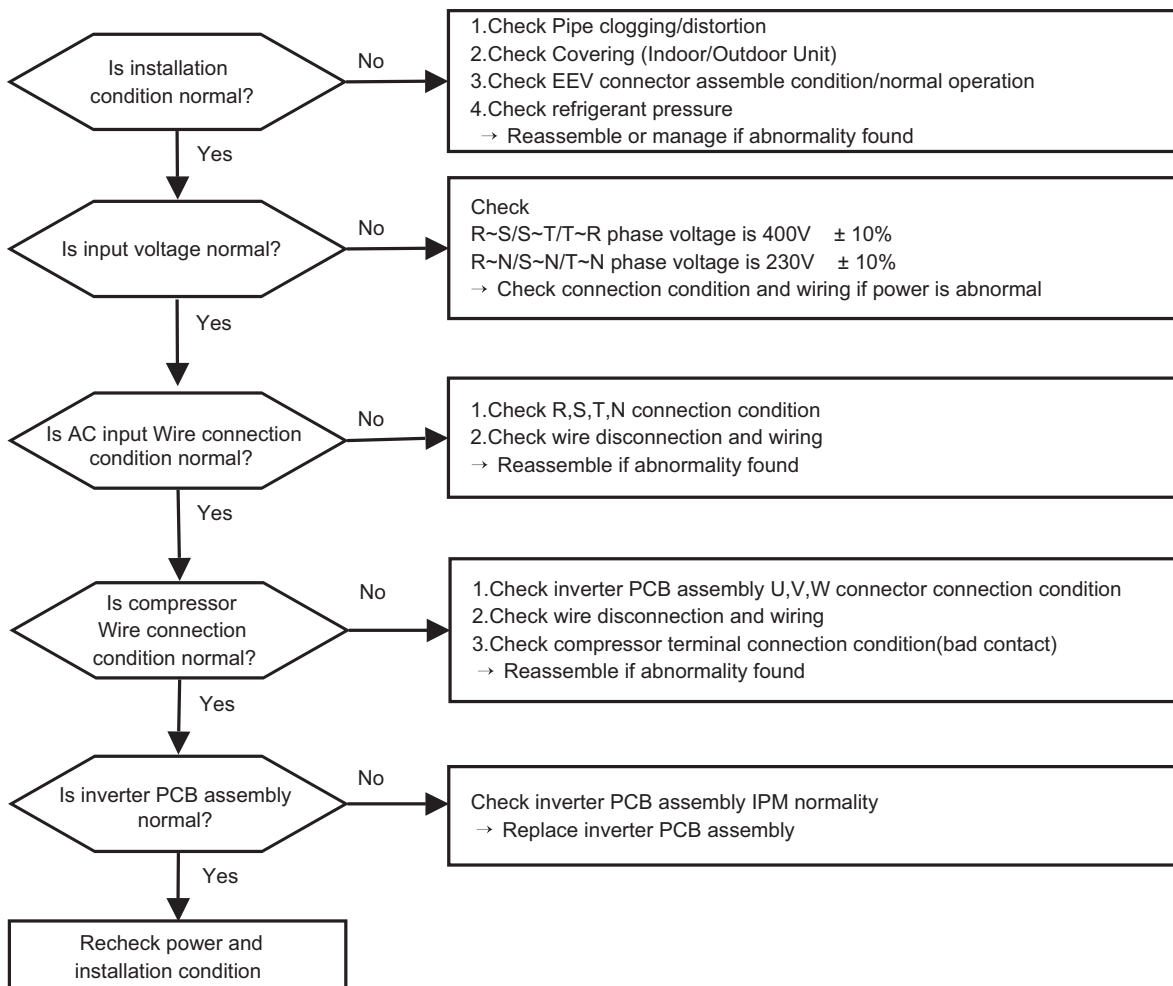


* Measuring input voltage

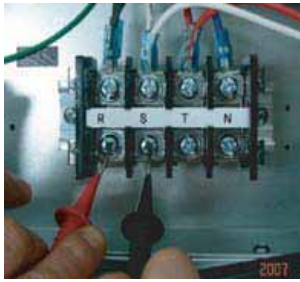


Error No.	Error Type	Error Point	Main Reasons
73*	AC input instant over current error (Matter of software)	Inverter PCB input 3 phase power current is over 50A(peak) for 2ms	1. Overload operation (Pipe clogging/Covering/EEV defect/Ref. overcharge) 2. Compressor damage (Insulation damage/Motor damage) 3. Input voltage abnormal (R,S,T,N) 4. Power line assemble condition abnormal 5. Inverter PCB assembly damage (input current sensing part)

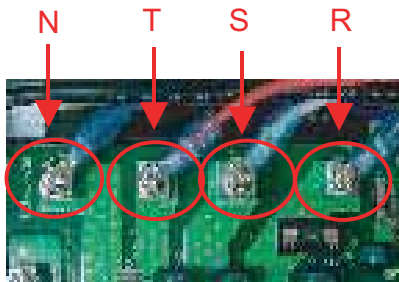
■ Error Diagnosis and Countermeasure Flow Chart



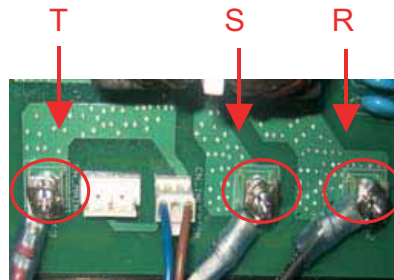
Measuring input voltage



Noise filter wiring



Noise filter input (upper part)



Noise filter output(lower part)

Inverter PCB assembly/Wiring power to inverter PCB on Noise filter



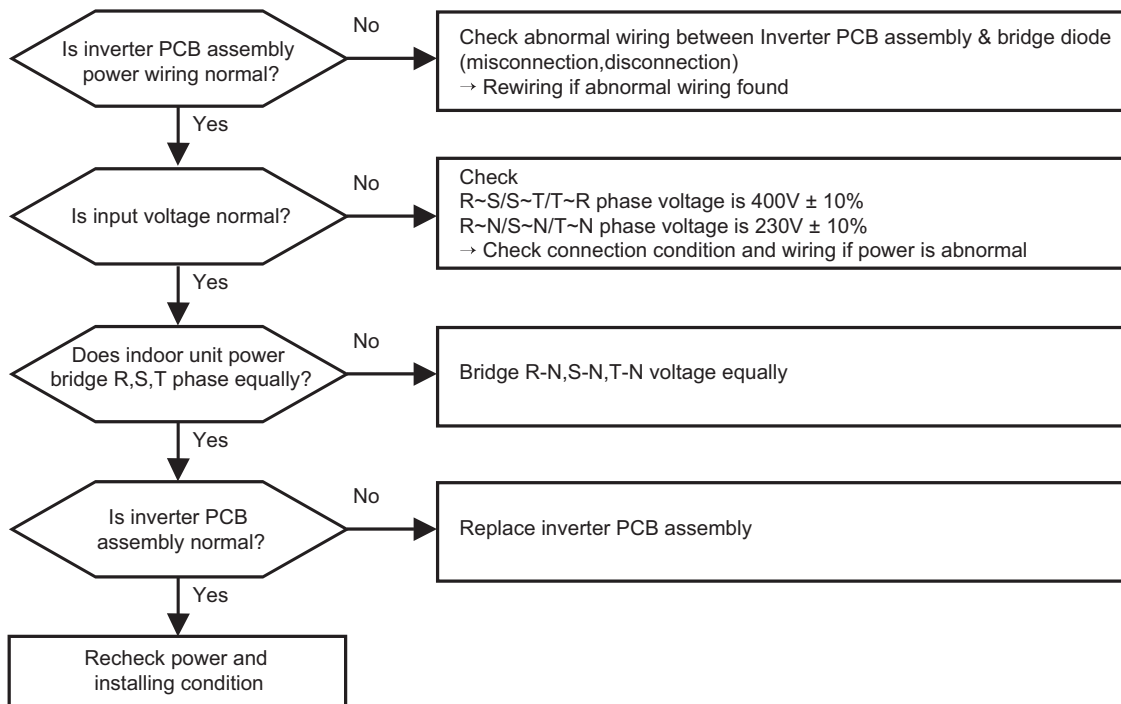
Inverter PCB assembly power connection



Noise filter power connection

Error No.	Error Type	Error Point	Main Reasons
74*	3 Phase Power Unbalance	During operation(compressor frequency is over 50Hz), difference between R & T phase is 5A for 10 seconds.	1. CT sensor defect 2. Capacity over of AVR 3. Bridge R-N or S-N or T-N phase voltage unequally at indoor unit etc.

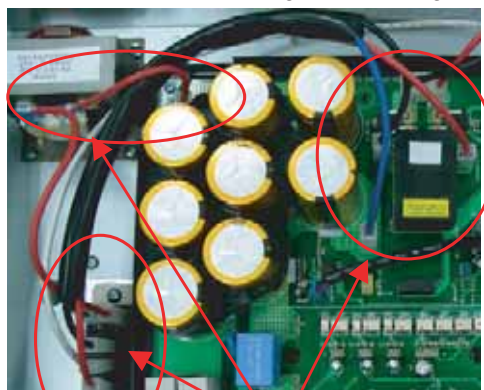
■ Error Diagnosis and Countermeasure Flow Chart



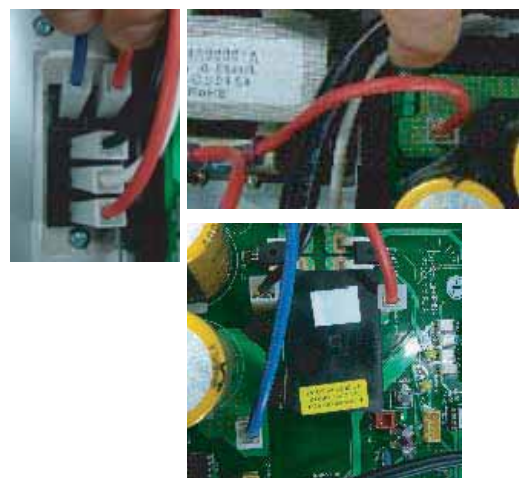
Measuring input voltage



Inverter PCB & Bridge Diode wiring

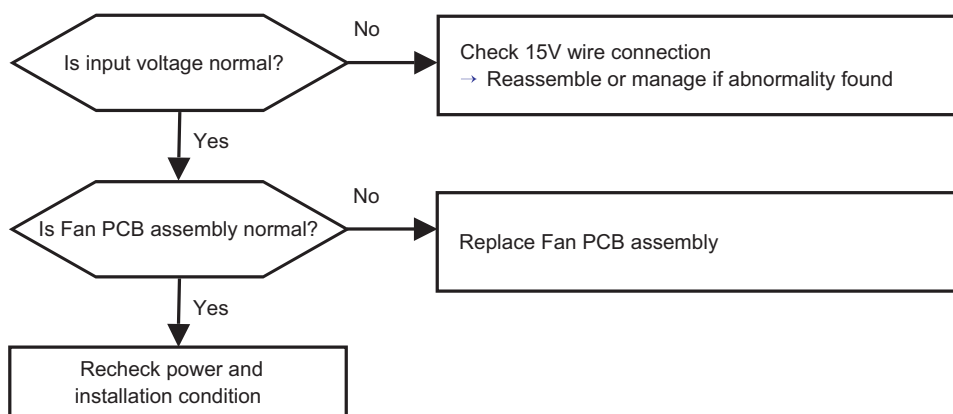


Checking joining condition



Error No.	Error Type	Error Point	Main Reasons
75*	Fan CT sensor error	Offset of micom which senses the fan motor phase current is not 2.5V	1.Input voltage is abnormal(not 15V) 2.Fan PCB assembly defect

■ Error Diagnosis and Countermeasure Flow Chart

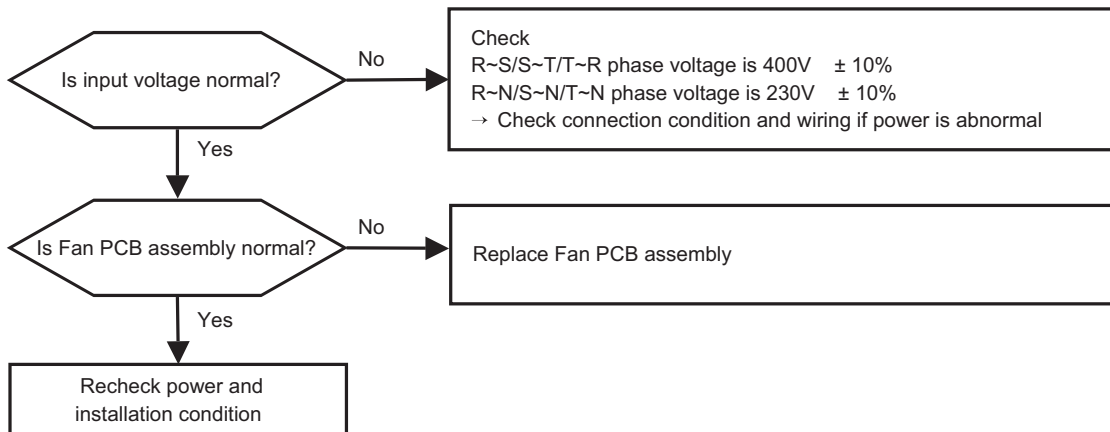


Checking 15V



Error No.	Error Type	Error Point	Main Reasons
76*	Fan DC Link High Voltage Error	Fan PCB DC link voltage supplied over 780V	1.Input power abnormal 2.Fan PCB assembly defect

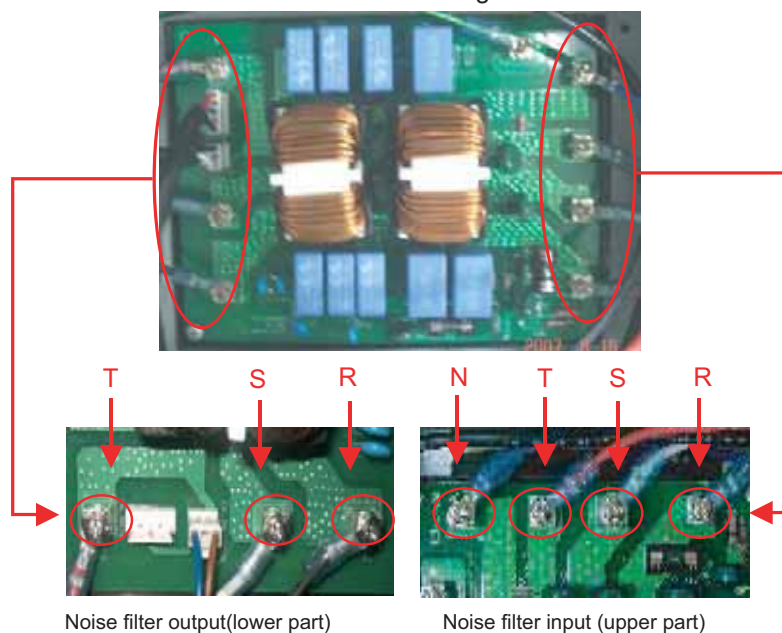
■ Error Diagnosis and Countermeasure Flow Chart



Measuring input voltage

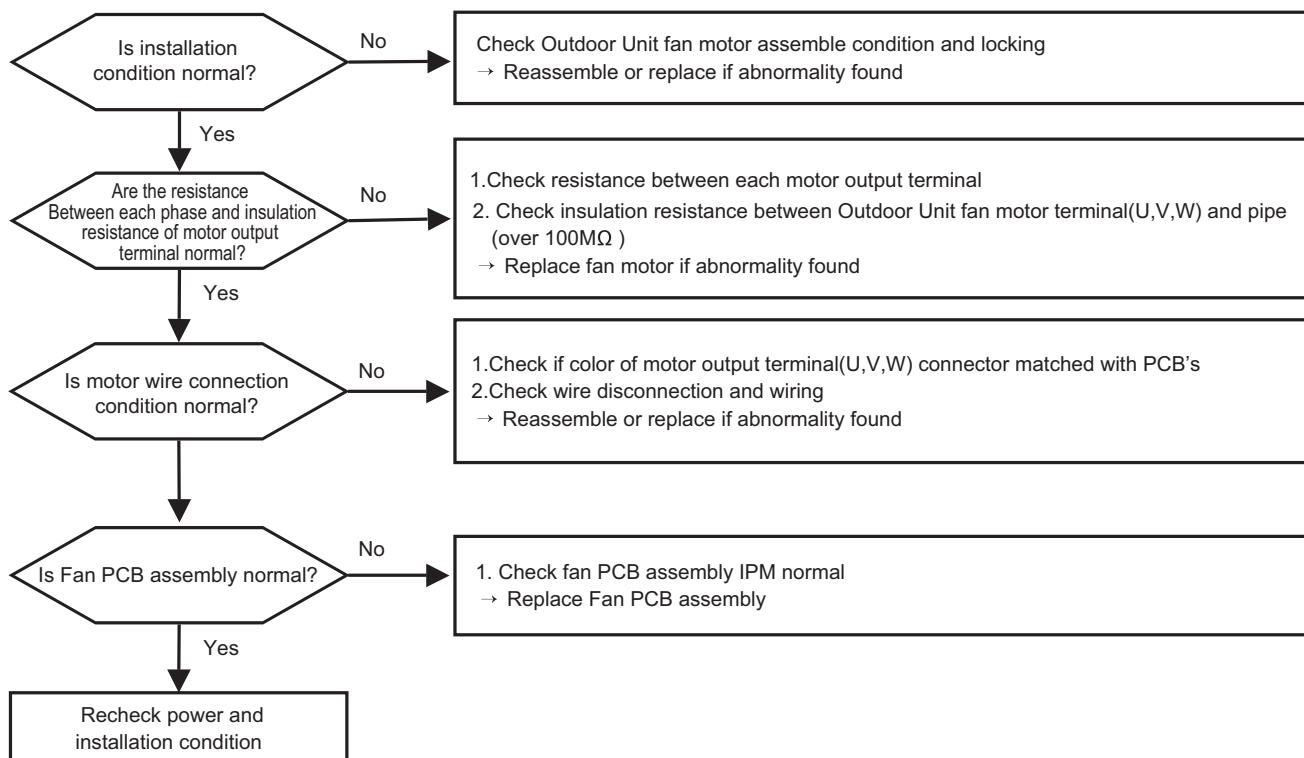


Noise filter wiring

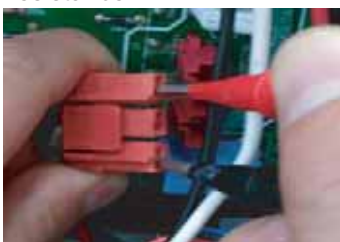


Error No.	Error Type	Error Point	Main Reasons
77*	Fan Over Current Error	Output current is over 5A for 40ms	1. Overload operation 2. Fan Motor defect 3. Fan PCB assembly defect

■ Error Diagnosis and Countermeasure Flow Chart



Measuring fan motor phase resistance

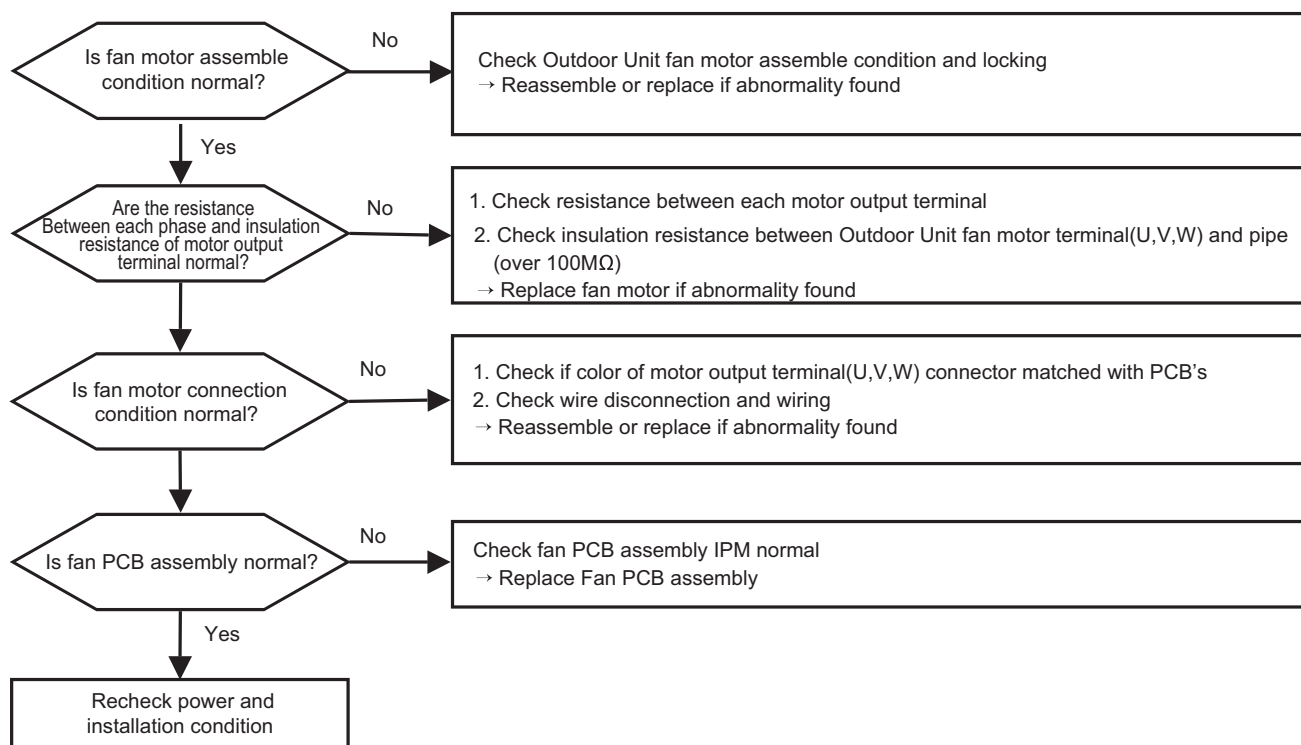


Fan motor wire connection

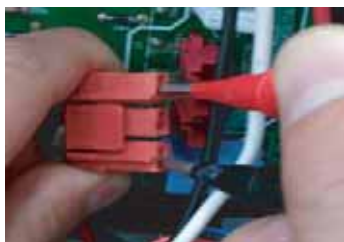


Error No.	Error Type	Error Point	Main Reasons
79*	Fan Starting Failure Error	Fan Motor initial starting failure	1.Fan motor defect/ assemble condition abnormal 2.Fan PCB defect

■ Error Diagnosis and Countermeasure Flow Chart



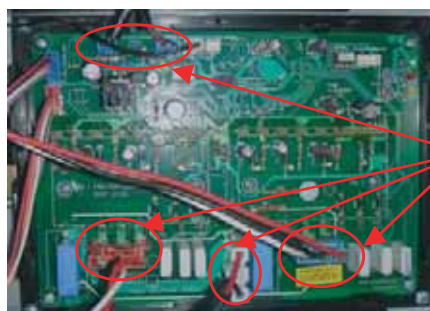
Measuring fan motor phase resistance



Measuring insulation resistance between fan terminal & chassis

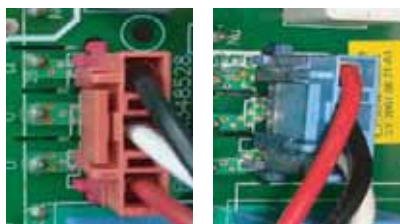


Fan wire wiring



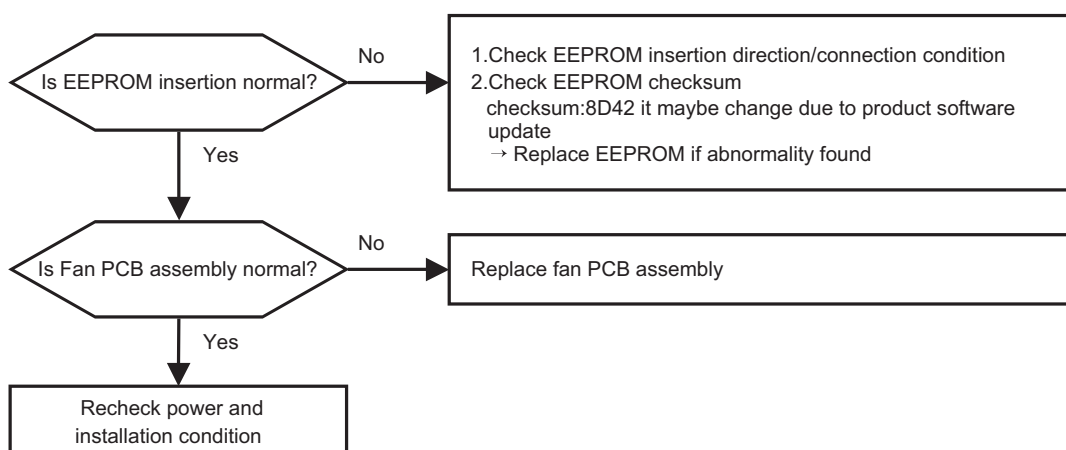
Refer to this photograph to install

Fan motor wire connection

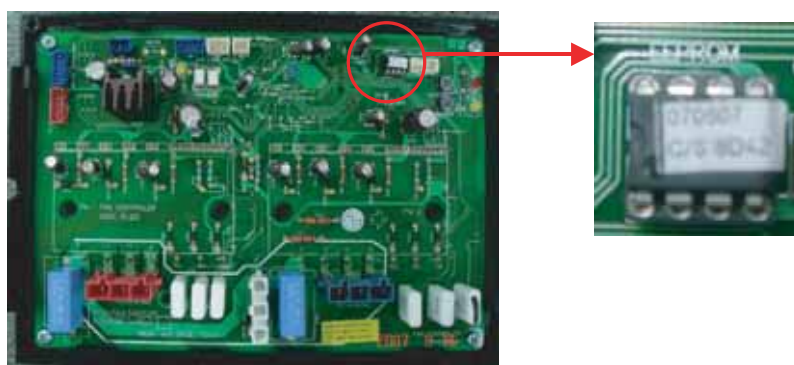


Error No.	Error Type	Error Point	Main Reasons
87*	Fan PCB EEPROM Error	Error occurs when checking the EEPROM checksum as initializing after power is supplied	1.EEPROM bad contact/wrong insertion 2.EEPROM Version is different 3.Outdoor Unit fan PCB assembly damage

■ Error Diagnosis and Countermeasure Flow Chart



Fan EEPROM insertion



Inverter EEPROM insertion direction

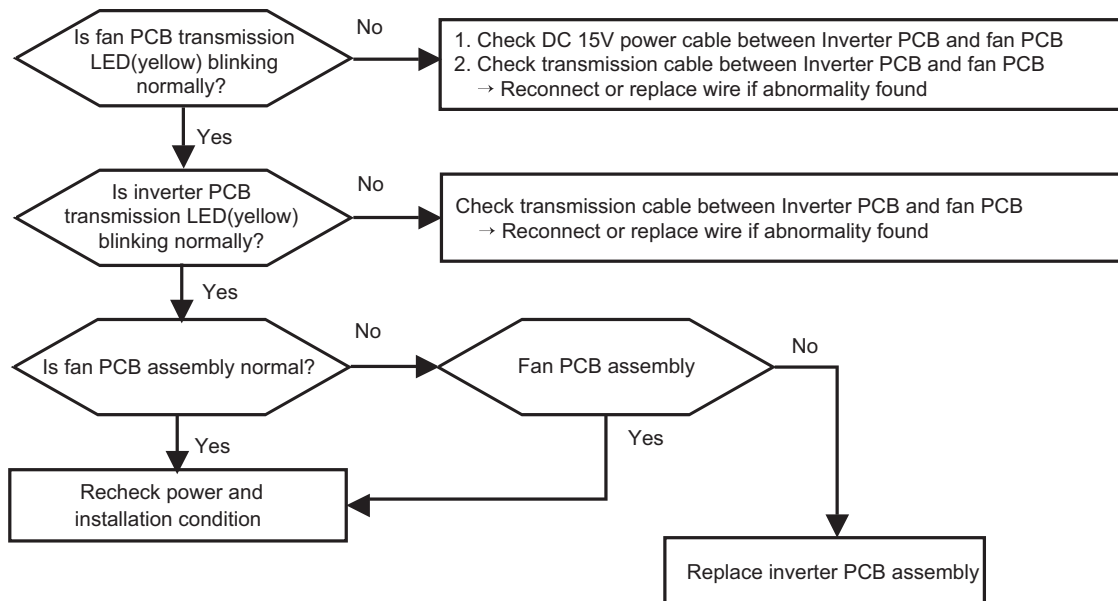


Same direction both socket hole and EEPROM hole

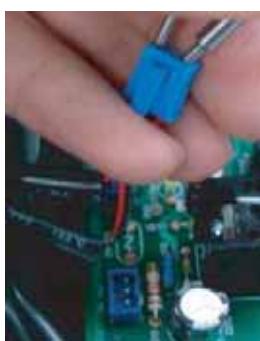
* Note : Replace after power off

Error No.	Error Type	Error Point	Main Reasons
105*	Transmission error (Fan PCB ↔ Inverter PCB)	Fan controller didn't receive signal from inverter controller	1. Wrong connection between Inverter and Fan PCB 2. Fan PCB power not supplied 3. Outdoor Unit Inv/Fan PCB defect

■ Error Diagnosis and Countermeasure Flow Chart



Checking 15V input



Fan transmission connection



Transmission connector

Transmission LED

Fan Error LED

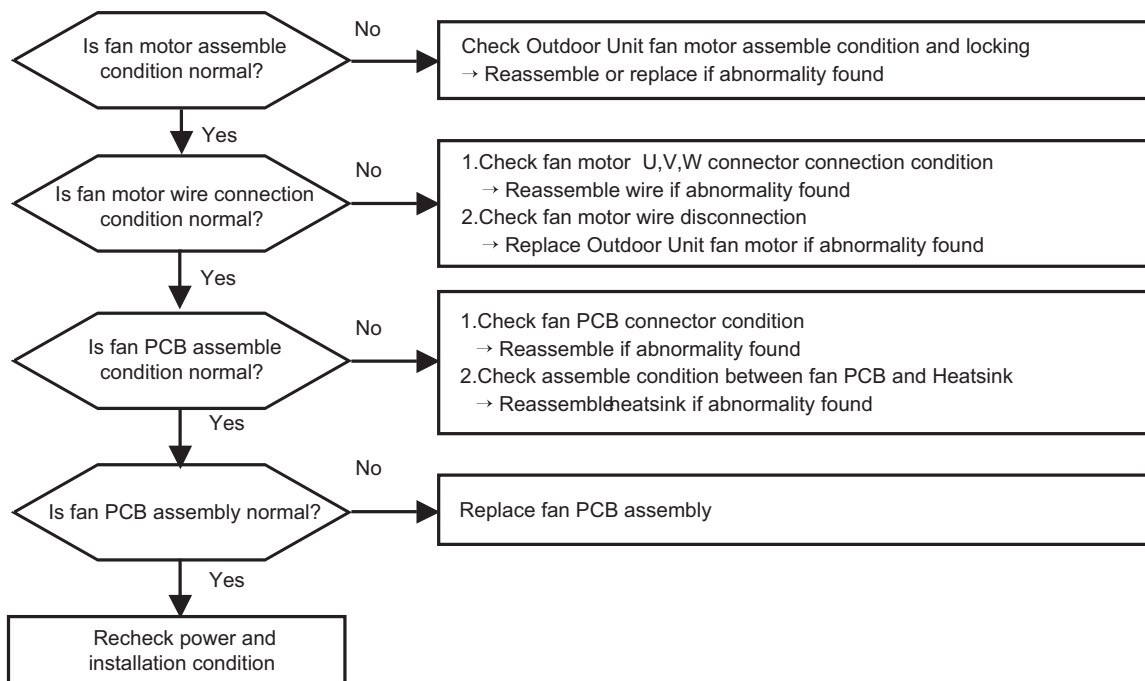


Error LED

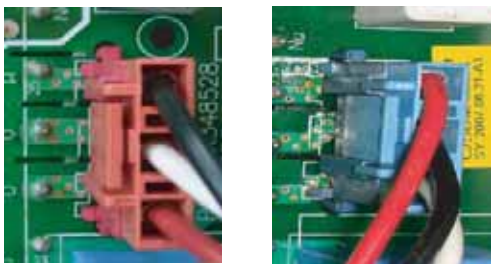
* Note : Check fan PCB assembly Error LED blinking (Check 108 Error)

Error No.	Error Type	Error Point	Main Reasons
106*	Outdoor Unit Fan PCB IPM Fault	IPM protection circuit activation (over current / overheating)	1. Overload operation (Pipe clogging/Covering/EEV defect/Ref. overcharge) 2. Outdoor Unit fan motor assemble condition abnormal (Coil disconnection/Short/Insulation damage) 3. Fan PCB heatsink assemble condition abnormal 4. Fan PCB assembly defect

■ Error Diagnosis and Countermeasure Flow Chart



Fan Motor Wire connection



Fan IPM assemble position



Check assemble condition

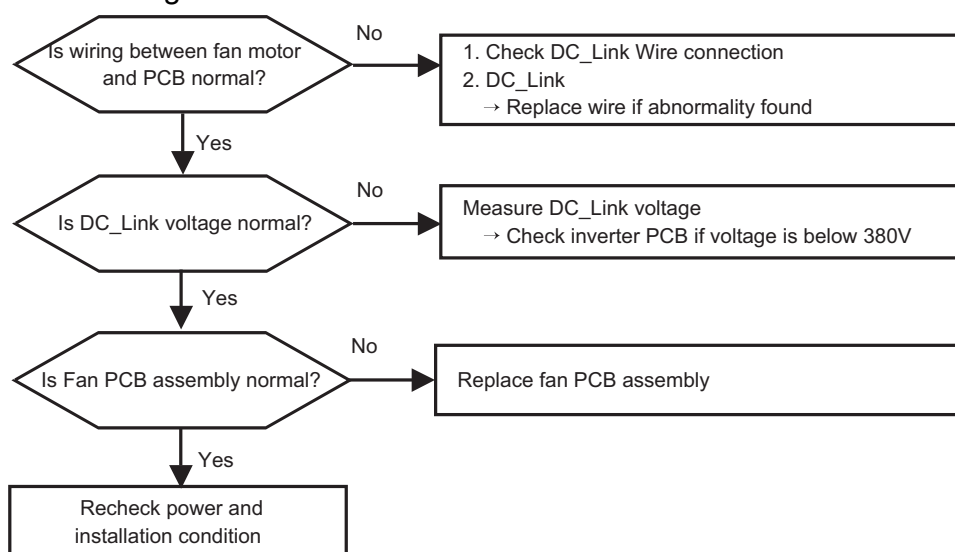
Fan Heatsink assemble position



Check assemble condition

Error No.	Error Type	Error Point	Main Reasons
107*	Fan DC Link Low Voltage Error	Fan PCB DC link voltage supplied below 380V	1. Wrong wiring between inverter PCB and Fan PCB 2. Fan PCB assembly defect

■ Error Diagnosis and Countermeasure Flow Chart



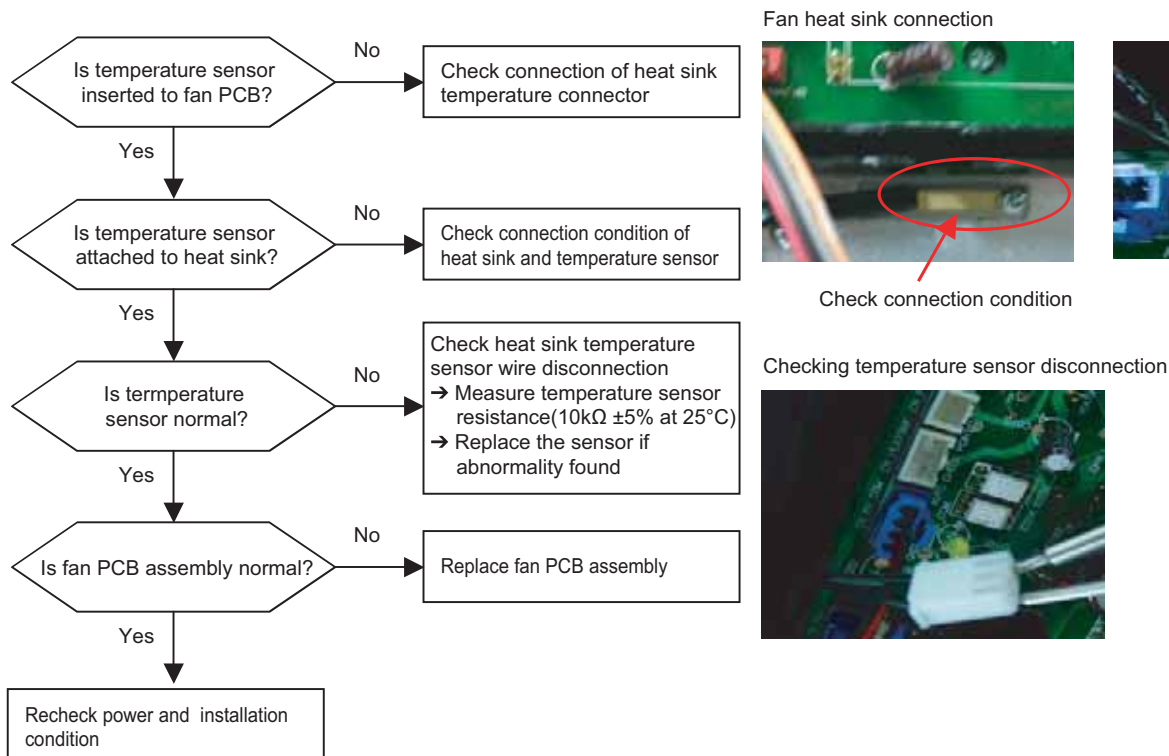
DC voltage connection



DC Volt connected

Error No.	Error Type	Error Point	Main Reasons
193*	Fan PCB heatsink temperature high	Heat sink temperature is over 95°C	1. Heatsink temperature sensor defect 2. Fan PCB assembly defect
194*	Fan PCB heatsink temperature sensor error	Heatsink temperature sensor abnormal	1. Heatsink temperature sensor defect(Open/Short) 2. Wrong connection of temperature sensor connector 3. Fan PCB assembly defect

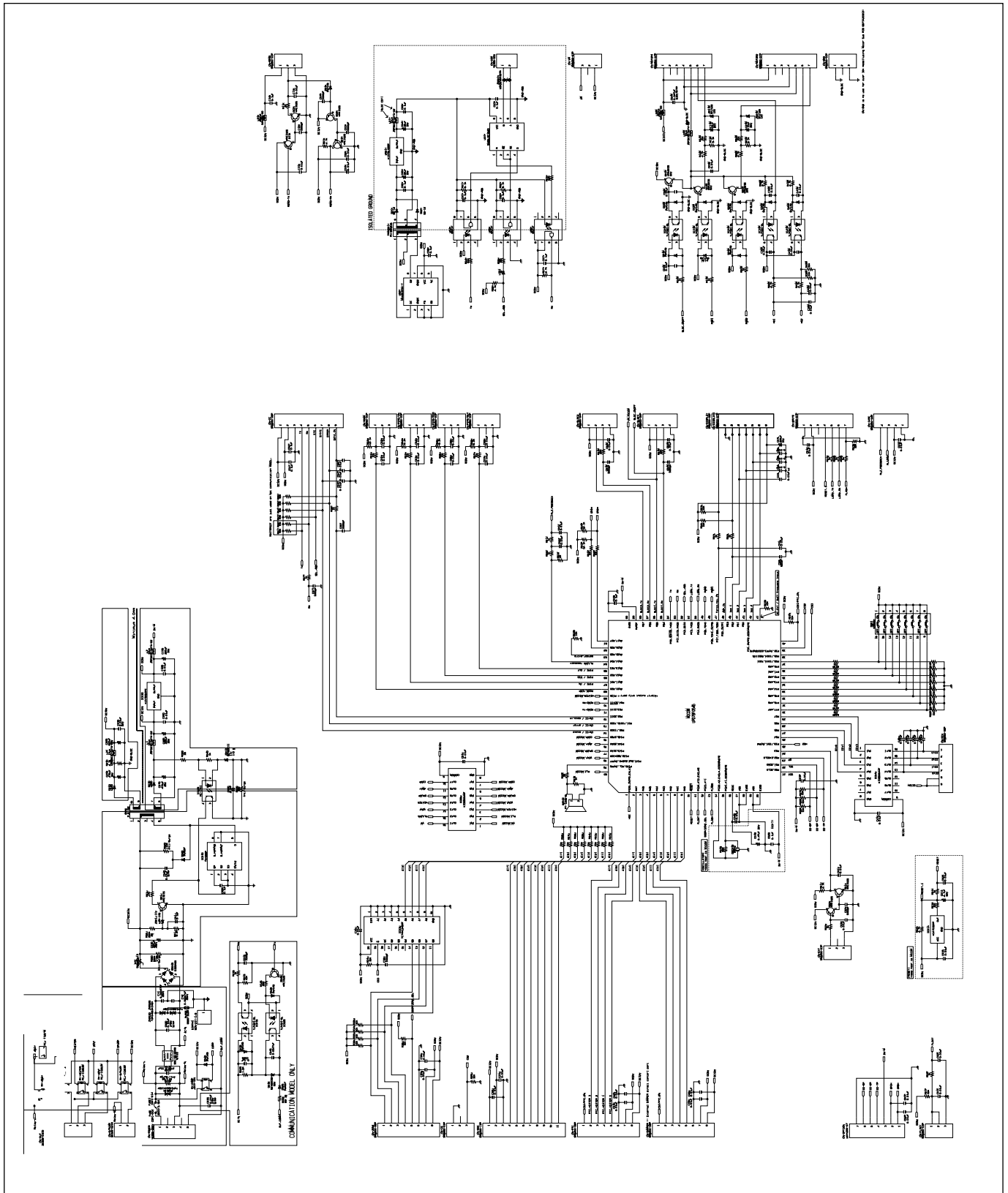
■ Error diagnosis and countermeasure flow chart



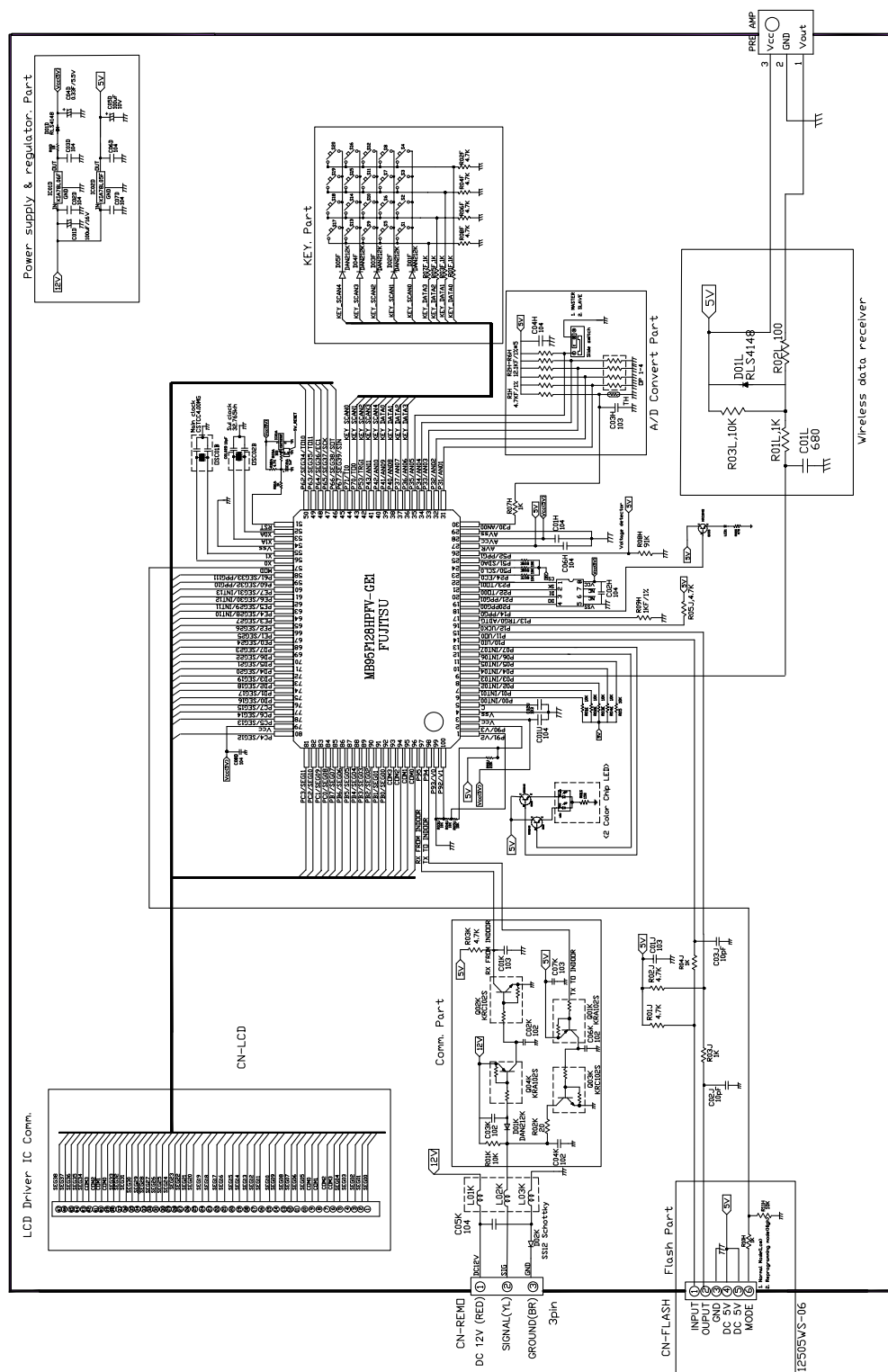
16.Electronic control device

16.1 Indoor Unit

Main PCB Circuit Diagram

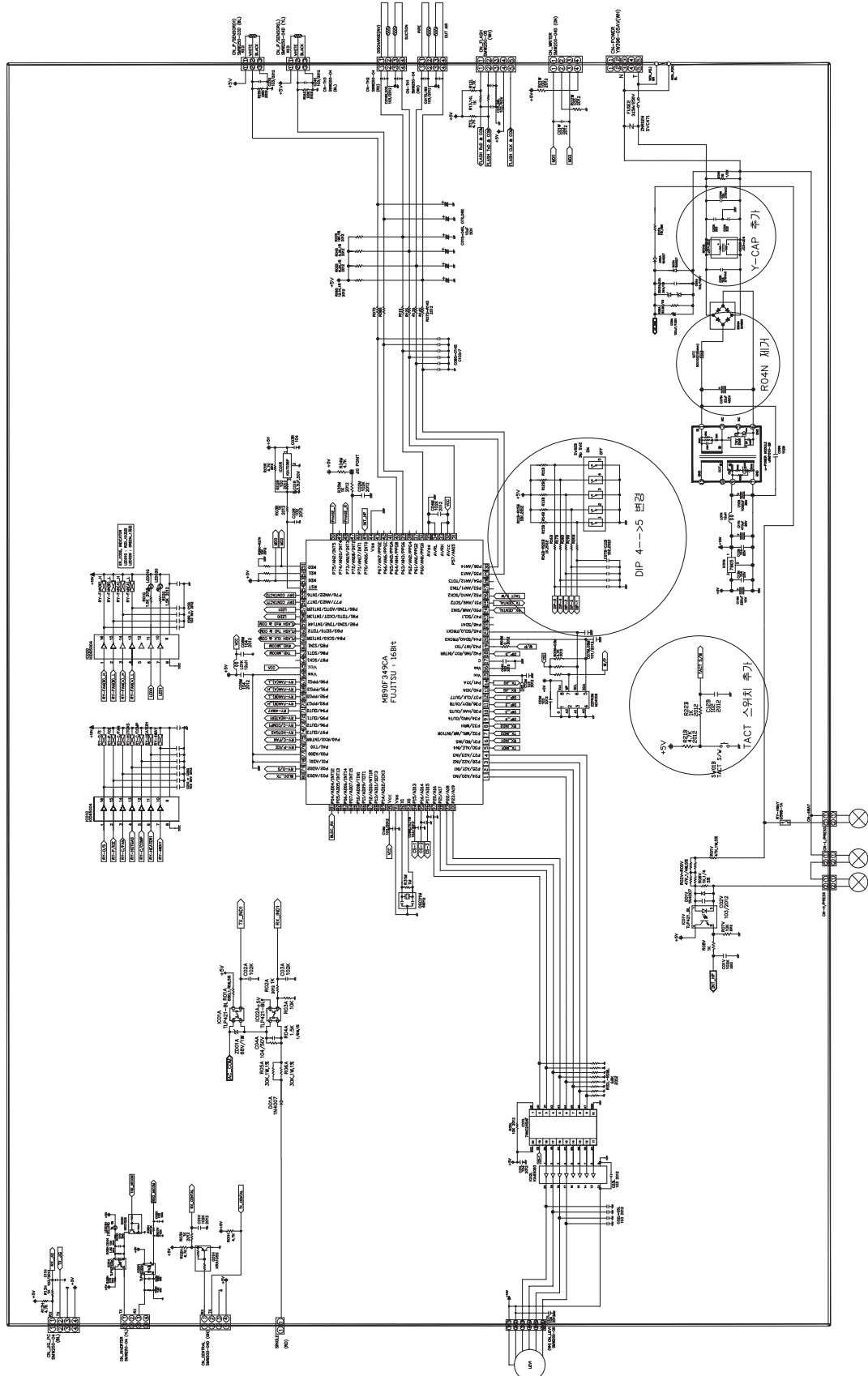


Feature of Remote-Controller PCB ASS'Y

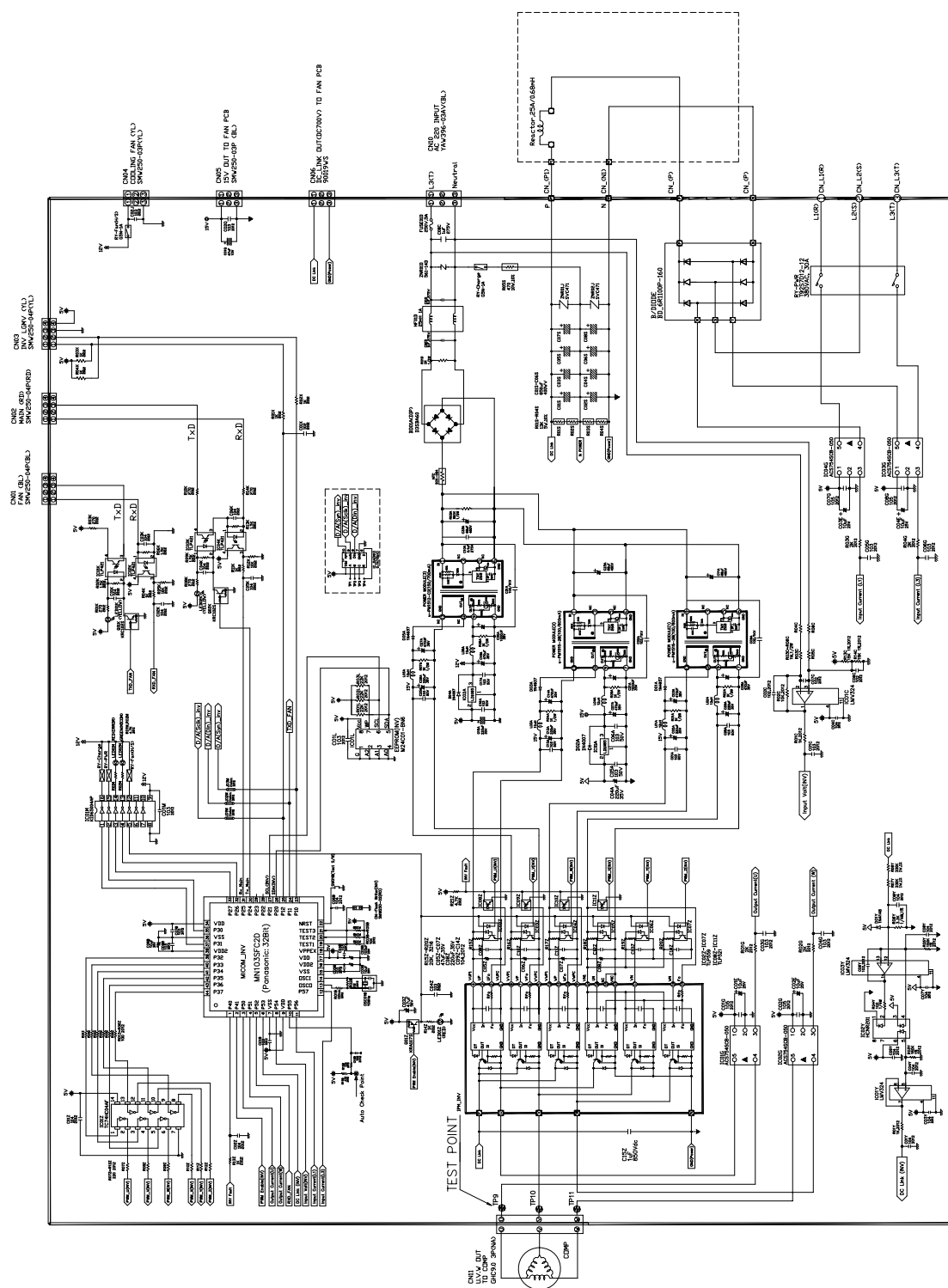


16.2 Outdoor Unit

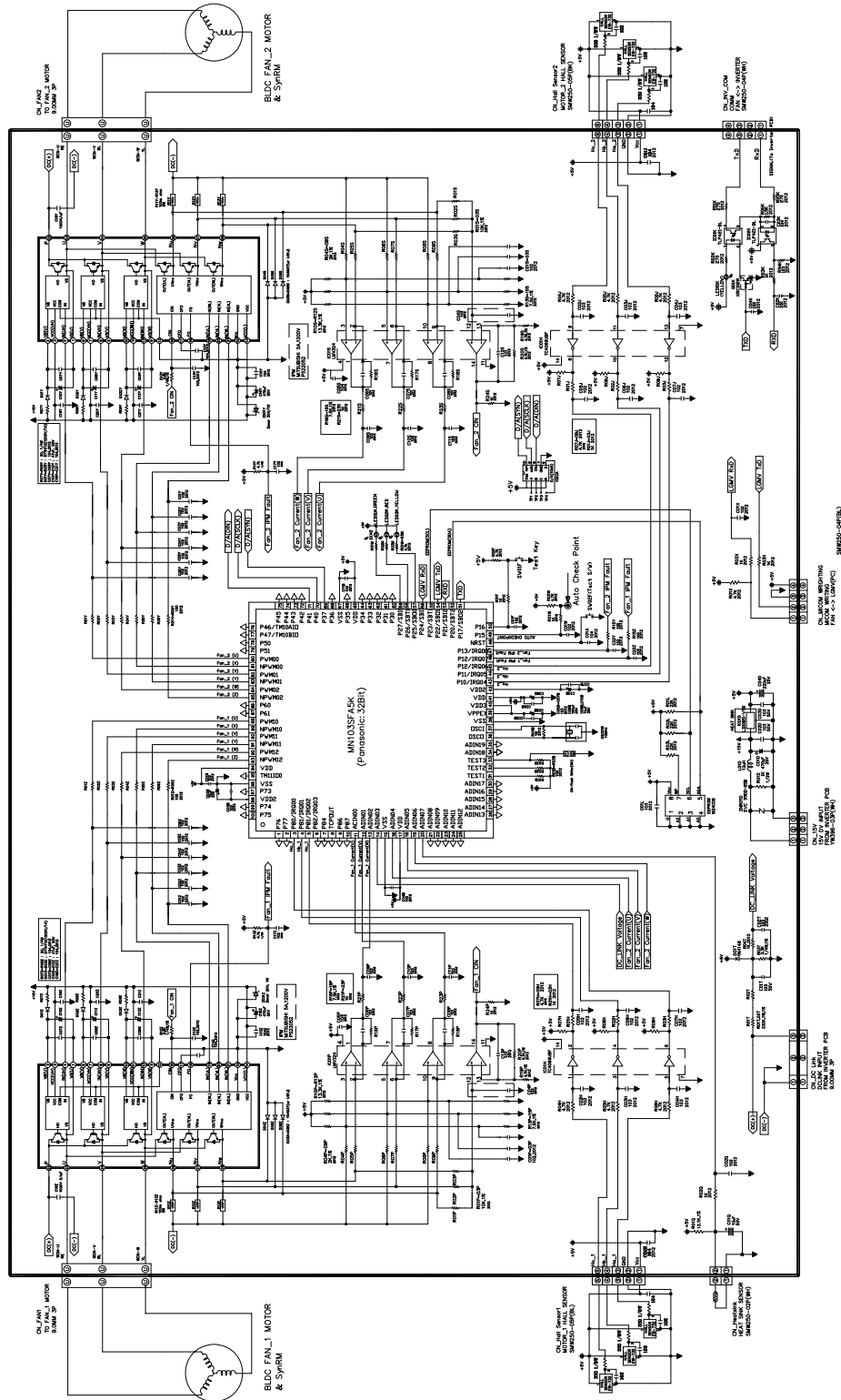
Main PCB Circuit Diagram



3 Phase -Inverter PCB Circuit Diagram



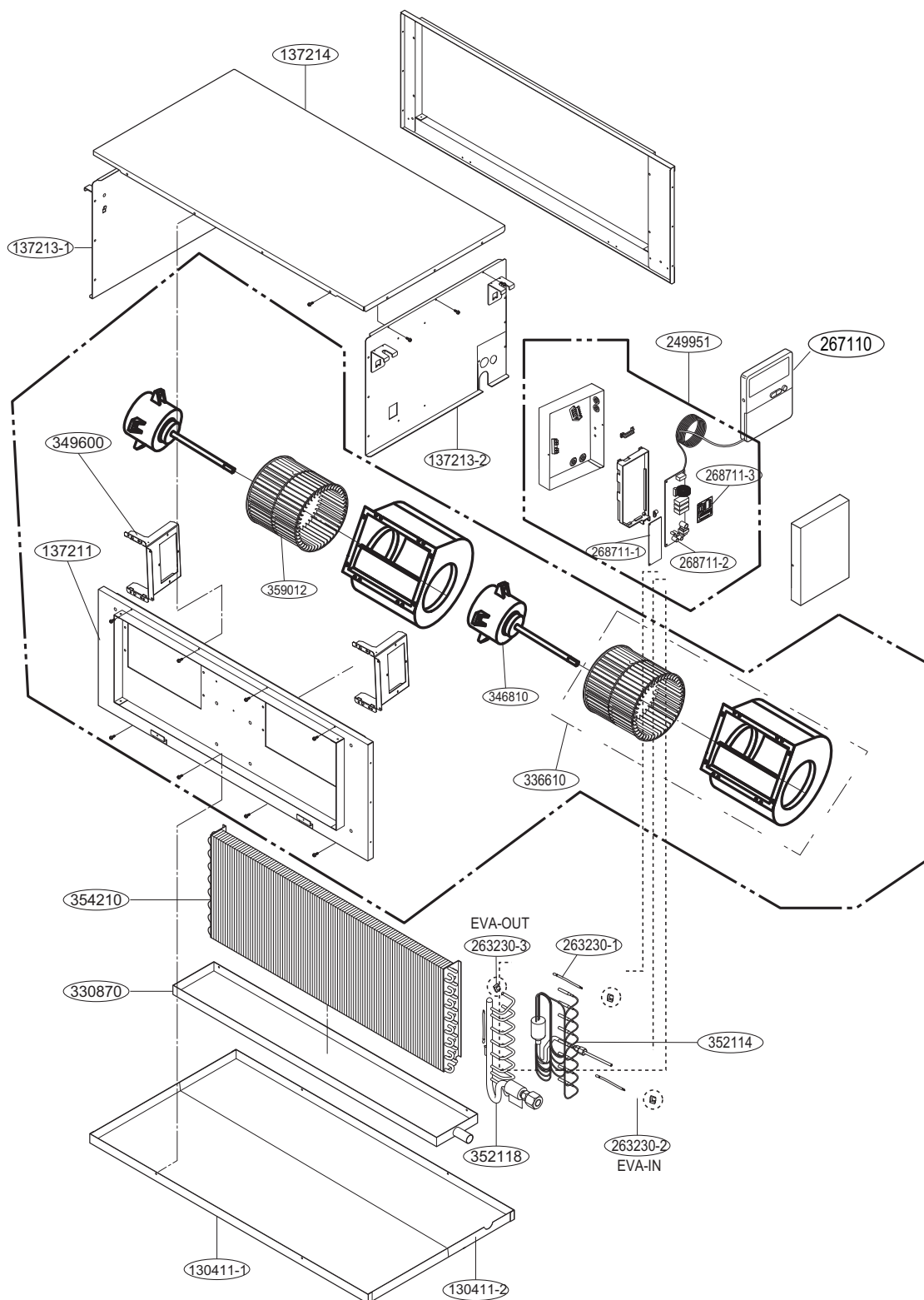
Fan PCB Circuit Diagram



17. Exploded View

17.1 Indoor Unit

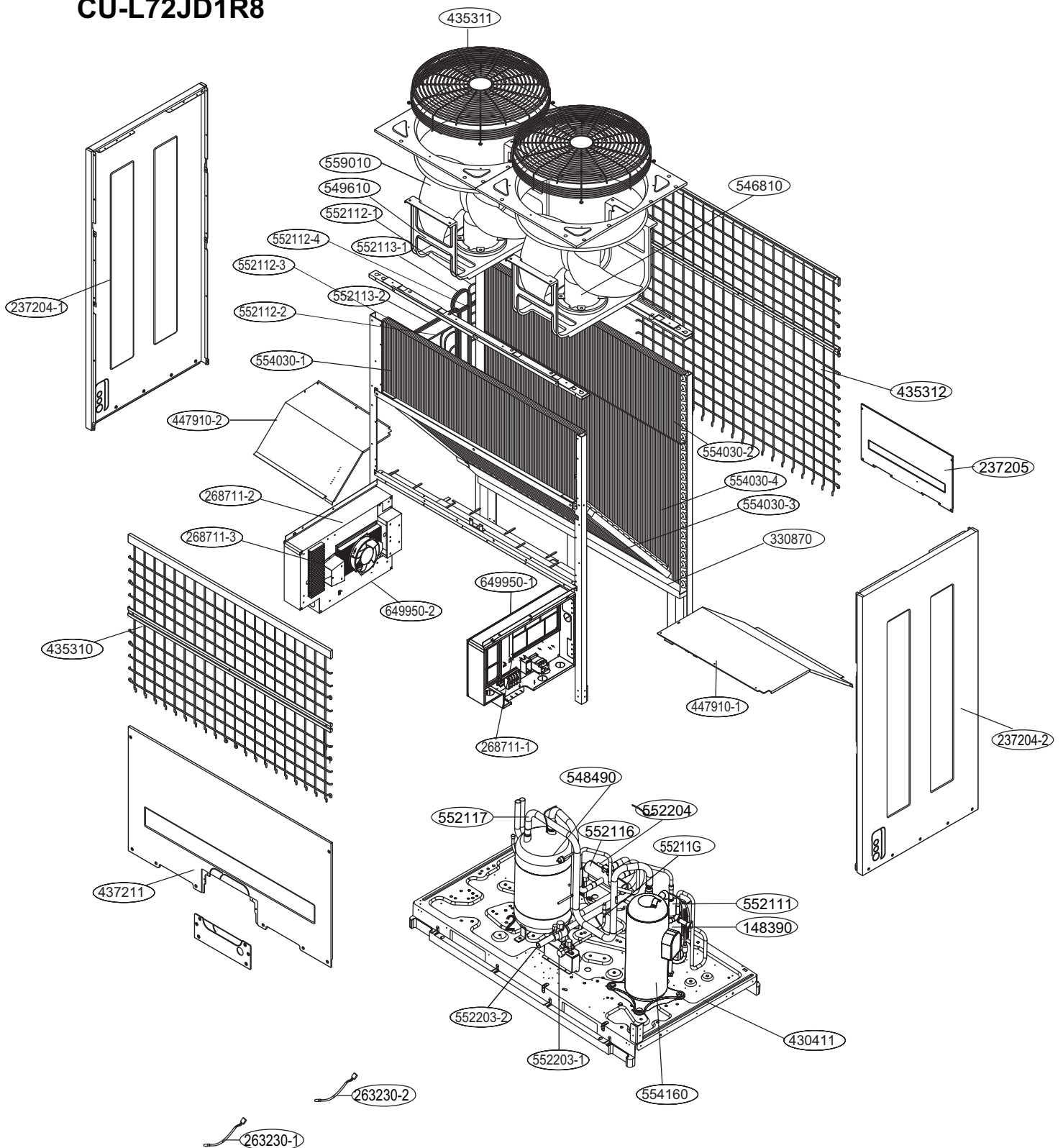
CS-L72JD1R5



Service Part List

LOC	DESCRIPTION	CS-L72JD1R5		SVC
		Factory	Customer	
137211	Panel Assembly,Front(Indoor)	AGL35764801	CWAGL3J04801	R
349600	Bracket,Motor	MAZ39910701	CWMAZ3YZ0701	R
346810	Motor Assembly,DC,Indoor	EAU39857401	CWEAU3YT7401	R
336610	Housing Assembly	3661A20025H	CW366120025H	R
359012	Fan,Blower	5900A20052D	CW590020052D	R
137214	Panel Assembly,Upper	AGL35765201	CWAGL3J05201	R
137213-1	Panel Assembly,Side	AGL35764401	CWAGL3J04401	R
137213-2	Panel Assembly,Side	AGL35764701	CWAGL3J04701	R
354210	Evaporator Assembly,First	5421A20217E	CW542120217E	R
352114	Tube Assembly,Distributor	5211A25031G	CW521125031G	R
352118	Tube Assembly,Menifold(Indoor)	5211A21710G	CW521121710G	R
263230-1	Thermistor,NTC	6323AQ3226E	CW6323Q3226E	R
263230-2	Thermistor,NTC	6323A20025D	CW632320025D	R
263230-3	Thermistor Assembly	6323A10002D	CW632310002D	R
249951	Case Assembly,Control(Indoor)	ABQ33690212	CWABQ3BH0212	R
268711-1	PCB Assembly	EBR39319505	CWEBR3X39505	R
268711-3	PCB Assembly,Option	EBR54408002	CWEBR5DR8002	R
268711-2	PCB Assembly,Main	6871A20912C	CW687120912C	R
330870	Pan Assembly,Drain	AGK34943701	CWAGK3FE3701	R
130411-1	Base Assembly,Indoor	AAN35764301	CWAAN3J04301	R
130411-2	Base Assembly,Indoor	AAN35764201	CWAAN3J04201	R
267110	REMOTE CONTROLLER ASSEMBLY	AKB69811902	CWAKB6YN1902	R

17.2 Outdoor Unit CU-L72JD1R8



Service Part List

LOC	DESCRIPTION	CU-L72JD1R8		SVC
		Factory	Customer	
430411	Base Assembly,Weld(Outdoor)	3041A10048W	CW304110048W	R
330870	Pan Assembly,Drain	3087A10023H	CW308710023H	R
549610	Bracket Assembly,Motor(Outdoor)	4960A10022A	CW496010022A	R
435312	Grille Assembly,Rear(Outdoor)	3531A11004H	CW353111004H	R
435310	Grille Assembly,Front(Outdoor)	3531A10266K	CW353110266K	R
435311	Grille Assembly,Discharge(Outdoor)	3531A20211R	CW353120211R	R
447910-2	Barrier Assembly,Outdoor	4791A10022B	CW479110022B	R
447910-1	Barrier Assembly,Outdoor	4791A10022C	CW479110022C	R
552117	Tube Assembly,Suction(Outdoor)	5211A32041C	CW521132041C	R
552204	Valve,Solenoid	5220A90002A	CW522090002A	R
552116	Tube Assembly,Reverse	5211A32040D	CW521132040D	R
552111	Tube Assembly,Capillary	5211A12316L	CW521112316L	R
148390	Separator,Oil	5998A20002E	CW599820002E	R
55211G	Tube Assembly,Expansion	5211A10564E	CW521110564E	R
552113-1	Tube Assembly,Condenser(Out)	5211A10646A	CW521110646A	R
552113-2	Tube Assembly,Condenser(Out)	5211A10647B	CW521110647B	R
552112-1	Tube Assembly,Condenser(In)	5211A10480B	CW521110480B	R
552112-2	Tube Assembly,Condenser(In)	5211A10480J	CW521110480J	R
552112-3	Tube Assembly,Condenser(In)	5211A10480H	CW521110480H	R
552112-4	Tube Assembly,Condenser(In)	5211A10649A	CW521110649A	R
548490	Accumulator	4849A10070D	CW484910070D	R
552203-1	Valve,Service	5220A90012M	CW522090012M	R
552203-2	Valve,Service	5220A90012K	CW522090012K	R
263230-1	Thermistor Assembly	EBG30431201	CWEBG31B1201	R
263230-2	Thermistor Assembly	EBG30431301	CWEBG31B1301	R
649950-1	Case Assembly,Control	ABQ34394721	CWABQ3DQ4721	R
268711-1	PCB Assembly,Main	EBR37055607	CWEBR3P15607	R
649950-2	Case Assembly,Control	ABQ34394906	CWABQ3DQ4906	R
268711-2	PCB Assembly,Inverter	EBR36932806	CWEBR3NN2806	R
268711-3	PCB Assembly,Main	EBR34853103	CWEBR3F53103	R
554030-4	Condenser Assembly,First	ACG72916604	CWACG7936604	R
554030-3	Condenser Assembly,First	ACG72916603	CWACG7936603	R
554030-2	Condenser Assembly,First	ACG72916602	CWACG7936602	R
554030-1	Condenser Assembly,First	ACG72916601	CWACG7936601	R
546810	Motor Assembly,DC,Outdoor	EAU39049401	CWEAU3W89401	R
554160	Compressor	MCB40261101	CWMCB40U1101	R
237204-1	Panel,Side	3720A10169D	CW372010169D	R
237204-2	Panel,Side	3720A10169J	CW372010169J	R
437211	Panel,Front	3721A10107F	CW372110107F	R
559010	Fan Assembly,Propeller	5901A10029C	CW590110029C	R
237205	Panel,Rear	3720A10168D	CW372010168D	R